

$$\text{In[1]:= } c = \text{Quantity}\left[299\,792\,458, \frac{\text{"Meters"}}{\text{"Seconds"}}\right]$$

$$e = \text{Quantity}\left[1\,602\,176\,634 \times 10^{-28}, \text{"Coulombs"}\right]$$

$$h = \text{Quantity}\left[\frac{662\,607\,015}{10^{42}}, \frac{\text{"Joules"}}{\text{"Hertz"}}\right]$$

$$\Omega = \text{Quantity}\left[100 \sqrt[8]{4\pi}, \frac{\text{"Kilograms"} \text{"Seconds"}}{\text{"Meters"} \text{"Coulombs"}}\right]$$

$$\alpha\gamma = 4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi$$

$$\alpha\delta = 1\,972\,044\,687\,500\,000\,000\,000\,000\,000$$

$$\alpha = \frac{\alpha\gamma}{\alpha\delta}$$

$$\Gamma = \frac{\alpha}{2\pi} \frac{h}{c}$$

$$\Phi = \frac{\Gamma}{e^2}$$

$$ZP = \text{UnitConvert}\left[\frac{2\,h}{e^2}, \text{"Ohms"}\right]$$

$$\mu P = \frac{2\,h}{c\,e^2}$$

$$\epsilon P = \frac{1}{2} \frac{e^2}{c\,h}$$

$$Z_0 = ZP\,\alpha$$

$$\mu_0 = \mu P\,\alpha$$

$$\epsilon_0 = \epsilon P \frac{1}{\alpha}$$

$$G = \text{UnitConvert}\left[\frac{4\pi\Phi}{\Omega^2}, \frac{\text{"Meters"}^3}{\text{"Kilograms"} \text{"Seconds"}^2}\right]$$

$$mP = \sqrt{\frac{h\,c}{2\pi\,G}};$$

$$lP = \sqrt{\frac{h\,G}{2\pi\,c^3}};$$

$$(\star me = \frac{mP}{23892177732494625341423} \star)$$

$$me = \text{UnitConvert}\left[\frac{mP}{23\,892\,177\,732\,494\,625\,341\,440}, \text{"Kilograms"}\right];$$

$$mp = \text{UnitConvert}\left[\frac{mP}{167\,262\,192\,595 \times 10^{-38}}, \text{"Kilograms"}\right];$$

$$(\star mp = \frac{mP}{13012086673584404467} \star)$$

$$m\mu = \frac{mP}{115\,550\,496\,527\,972\,941\,829}$$

Out[13]=

$$\frac{149\,896\,229\,\pi}{1\,250\,000}\,\Omega$$

Out[14]=

$$\frac{\pi}{2\,500\,000} \text{ s J / (m Hz C}^2\text{)}$$

Out[15]=

$$\frac{625\,000}{22\,468\,879\,468\,420\,441\,\pi} \text{ Hz s C}^2/(\text{m J})$$

Out[16]=

$$\frac{\pi^{3/4}}{25\,000\,000\,000\,\sqrt{2}} \text{ m}^3/(\text{kg s}^2)$$

Out[21]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{231\,100\,993\,055\,945\,883\,658\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{s} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[26]=

True

Out[27]=

True

Out[28]=

True

$$ln[\bullet] := \frac{\Gamma}{\alpha} == mP \text{ } lP$$

$$e^2 \Phi = \Gamma = \frac{e^2}{4 \pi c^2 \epsilon_0}$$

$$\ln[\bullet] := \frac{\text{me mp}}{\text{me} + \text{mp}} / \frac{\frac{\text{me mp}}{\text{me} + \text{mp}}}{\text{mP}}$$

$$1.0 / \frac{\frac{me \cdot mp}{me + mp}}{mp}$$

Out[•]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{2\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \text{ kg}$$

Out[•]=

$$2.38922 \times 10^{22}$$

$$\frac{\text{Quantity}[1.67262192595 \times 10^{-27}, \text{"Kilograms"}]}{\text{mP}} = 167\,262\,192\,595 \times 10^{-38}$$

`Out[ ]=` 1.30121×10^{19}

$$Out[*]=\frac{35\,000\,000\,000\,000\,000\,000\,000\,000\times 2^{3/4}\sqrt{405\,397\,113\,703\,863}}{33\,452\,438\,519\pi^{7/8}}\sqrt{\text{kg}}\sqrt{\text{s}}\sqrt{\text{J}}\,(\text{m}\sqrt{\text{Hz}})$$

$$ln[\otimes] := \frac{\frac{\frac{\mathbf{mp}}{\mathbf{mP}}}{\mathbf{mP}}}{\mathbf{mp}} \quad \psi \left[\frac{\mathbf{mP}}{\mathbf{mp}} \right] \mathbf{1.0}$$

$$\text{Out}[\ast]=\frac{1}{130\,120\,866\,735\,844\,044\,779}$$

Out[•]=

130 120 866 735 844 044 779

$Out[\bullet]=$ $0.$

$$\begin{aligned} \text{In}[*]:= & \frac{\text{m}\mu \alpha}{\Gamma} \frac{1 \text{ P}}{\Delta \alpha \left[\frac{\text{mP}}{\text{me}^2} \right]} 1.0 \\ & \left(\frac{\text{mP} \alpha}{\Gamma} \frac{1 \text{ P}}{\Delta \alpha \left[\frac{\text{mP}}{\text{me}^2} \right]} \right) == \frac{\text{mP}}{\text{me}} 1.0 \\ & \frac{1 \text{ P mP}}{\text{me}^2} 1.0 \\ & \text{mP} \alpha 1.0 \\ & \frac{1}{2 \pi} \frac{1 \text{ P}}{\Delta \alpha \left[\frac{\text{mP}}{\text{me}^2} \right]} 1.0 \end{aligned}$$

Out[*]=

206.768

Out[*]=

True

Out[*]=

$$1.9308 \times 10^{-13} \sqrt{s} \sqrt{J} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

Out[*]=

$$1.22057 \times 10^{-29} \sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

$$6.14592 \times 10^{-14} \sqrt{s} \sqrt{J} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

$$\begin{aligned} \text{In}[*]:= & \text{me} \left(\frac{\text{mP} \alpha}{\Gamma} \frac{1 \text{ P}}{\Delta \alpha \left[\frac{\text{mP}}{\text{me}^2} \right]} \right) 1.0 \\ & \text{Solve} \left[\left(\frac{\alpha}{\Gamma} \frac{1 \text{ P}}{\Delta \alpha \left[\frac{\text{mP}}{\text{me}^2} \right]} \right) == \frac{1}{X} 1.0, X \right] \\ & \left(\frac{\alpha}{\Gamma} \frac{1 \text{ P}}{\Delta \alpha \left[\frac{\text{mP}}{\text{me}^2} \right]} \right) 1.0 \end{aligned}$$

Out[*]=

$$1.67262 \times 10^{-27} \sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (\text{m} \sqrt{\text{Hz}})$$

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[*]=

$$\left\{ \left\{ X \rightarrow 9.10938 \times 10^{-31} \text{ kg} \right\} \right\}$$

Out[*]=

$$1.09777 \times 10^{30} \text{ m} \sqrt{\text{Hz}} / (\sqrt{\text{kg}} \sqrt{s} \sqrt{J})$$

In[*]:=
$$\frac{23\,892\,177\,732\,494\,625\,341\,423 - 23\,890\,191\,132\,700\,966\,601\,412}{23\,892\,177\,732\,494\,625\,341\,423 - 13\,012\,086\,673\,584\,404\,467}$$

$$\frac{238\,901\,911\,327\,009\,666\,214\,244}{238\,901\,911\,327\,009\,666\,214\,244} \times 1.0$$

Out[*]= 1 986 599 793 658 740 011

Out[*]=
$$\frac{5\,969\,791\,411\,455\,260\,234\,239}{59\,725\,477\,831\,752\,416\,553\,561}$$

Out[*]= 10.0046

In[*]:= 531 mP - me

Out[*]=
$$\frac{22\,201\,806\,157\,920\,630\,598\,517\,321 \sqrt{405\,397\,113\,703\,863}}{11\,946\,088\,866\,247\,312\,670\,711\,500\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

In[*]:= SetPrecision $\left[\frac{\text{me}}{\text{Quantity}[9.1093837139 \times 10^{-31}, \text{"Kilograms"}]}, 28\right]$

SetPrecision $\left[\frac{\text{mP}}{\text{Quantity}[9.1093837139 \times 10^{-31}, \text{"Kilograms"}]}, 28\right]$

$$\frac{2\,389\,217\,773\,249\,462\,534\,144}{10^{43}} \times 1.0$$

UnitConvert $\left[\frac{\text{mP}}{23\,892\,177\,732\,494\,625\,341\,440}, \text{"Kilograms"}\right]$

Out[*]= 1.0000000000000000222044604925

Out[*]= $2.389217773249462534144000000 \times 10^{22}$

Out[*]= 2.38922×10^{-22}

Out[*]=
$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,880\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \text{kg}$$

$$IN[*] := \frac{\text{me mp}}{\text{me} + \text{mp}} \times \frac{1}{23\,892\,177\,732\,494\,625\,341\,440} \times 1.0$$
$$Out[\bullet]=$$

$$\frac{310\,382\,165\,179\,520\,109\,375}{2\,497\,661\,270\,879\,054\,008\,538\,940\,571\,648} \sqrt[2]{\frac{7\sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,880\,000\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8} + \frac{35\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \sqrt{405\,397\,113\,703\,863}}{33\,452\,438\,519 \pi^{7/8}}}} \pi^{7/4} \text{ kg}$$

$$Out[\bullet]=$$

$$4.18547 \times 10^{-23}$$

```

In[*]:= n =  $\frac{m_e}{m_p}$ 

Print[" $\alpha =$ ", SetPrecision[ScientificForm[ $\alpha$ ], 16]];

Print[" $\frac{1}{\alpha} =$ ", SetPrecision[1 /  $\alpha$ , 12]];

re = SetPrecision[UnitConvert[ $\frac{1p}{\Lambda_\alpha[n]}$ , "Meters"], 32];

 $\lambda_c$  = SetPrecision[UnitConvert[ $\frac{1p}{\Lambda_\alpha[n]}$ , "Meters"]  $\frac{2\pi}{\alpha}$ , 32];

 $\alpha\theta$  = SetPrecision[UnitConvert[ $\frac{1p}{\Lambda_\alpha[n]}$ , "Meters"]  $\frac{1}{\alpha^2}$ , 32];

Print[" $\mu = \frac{m_e m_p}{m_e + m_p} =$ ", SetPrecision[UnitConvert[ $\mu$ , "Kilograms"], 32]];

Print[" $n = \frac{m_e}{m_p} =$ ", n];

Print[" $r_e = \frac{1p}{\Lambda_\alpha[n]} =$ ", re];

Print[" $\lambda_c = \frac{1p}{\Lambda_\alpha[n]} \frac{2\pi}{\alpha} =$ ",  $\lambda_c$ ];

Print[" $\alpha_\theta = \frac{1p}{\Lambda_\alpha[n]} \frac{1}{\alpha^2} =$ ",  $\alpha\theta$ ];

```

Out[•]=

$$\frac{1}{23\,892\,177\,732\,494\,625\,341\,440}$$

$$\alpha = 7.29735256530521 \times 10^{-3}$$

$$\frac{1}{\alpha} = 137.035999159$$

$$\mu = \frac{m_e m_p}{m_e + m_p} = 9.1093837138999999367776031165177 \times 10^{-31} \text{ kg}$$

$$n = \frac{m_e}{m_p} = \frac{1}{23\,892\,177\,732\,494\,625\,341\,440}$$

$$r_e = \frac{1P}{\Lambda_\alpha[n]} = 2.817940320834913440078730766476 \times 10^{-15} \text{ m}$$

$$\lambda_c = \frac{1P}{\Lambda_\alpha[n]} \frac{2\pi}{\alpha} = 2.426310235380317284332371107899 \times 10^{-12} \text{ m}$$

$$\alpha_\theta = \frac{1P}{\Lambda_\alpha[n]} \frac{1}{\alpha^2} = 5.291772104738457275940481016826 \times 10^{-11} \text{ m}$$

In[]:= **Clear[n]**

Table[**UnitConvert**[**SetPrecision**[$-\frac{\alpha^2}{2 n^2} \text{me c}^2$, 16], "Electronvolts"], {n, 1, 10}]

Table[**UnitConvert**[**SetPrecision**[$\frac{2 n^2 h}{\alpha^2 \text{me c}}$, 16], "Meters"], {n, 1, 10}]

Table[**UnitConvert**[**SetPrecision**[$\frac{\alpha^2 \text{me c}^2}{2 n^2 h}$, 16], "Hertz"], {n, 1, 10}]

Table[**UnitConvert**[**SetPrecision**[$2 \pi \frac{\alpha^2 \text{me c}^2}{2 n^2 h}$, 16], "Hertz"], {n, 1, 10}]

Out[]:=

{ -13.60569312659197 eV , -3.401423281647992 eV ,
 -1.511743680732441 eV , -0.850355820411998 eV , -0.544227725063679 eV ,
 -0.377935920183110 eV , -0.2776672066651422 eV , -0.2125889551029995 eV ,
 -0.1679715200813823 eV , -0.1360569312659197 eV }

Out[]:=

{ $9.11267050341422 \times 10^{-8} \text{ m}$, $3.64506820136569 \times 10^{-7} \text{ m}$,
 $8.20140345307280 \times 10^{-7} \text{ m}$, $1.458027280546276 \times 10^{-6} \text{ m}$,
 $2.278167625853556 \times 10^{-6} \text{ m}$, $3.280561381229120 \times 10^{-6} \text{ m}$, $4.46520854667297 \times 10^{-6} \text{ m}$,
 $5.83210912218510 \times 10^{-6} \text{ m}$, $7.38126310776552 \times 10^{-6} \text{ m}$, $9.11267050341422 \times 10^{-6} \text{ m}$ }

Out[]:=

{ $3.289841961120809 \times 10^{15} \text{ Hz}$, $8.22460490280202 \times 10^{14} \text{ Hz}$,
 $3.65537995680090 \times 10^{14} \text{ Hz}$, $2.056151225700506 \times 10^{14} \text{ Hz}$,
 $1.315936784448324 \times 10^{14} \text{ Hz}$, $9.13844989200225 \times 10^{13} \text{ Hz}$, $6.71396318596084 \times 10^{13} \text{ Hz}$,
 $5.14037806425126 \times 10^{13} \text{ Hz}$, $4.06153328533433 \times 10^{13} \text{ Hz}$, $3.289841961120809 \times 10^{13} \text{ Hz}$ }

Out[]:=

{ $2.067068667305715 \times 10^{16} \text{ Hz}$, $5.16767166826429 \times 10^{15} \text{ Hz}$,
 $2.296742963673016 \times 10^{15} \text{ Hz}$, $1.291917917066072 \times 10^{15} \text{ Hz}$,
 $8.26827466922286 \times 10^{14} \text{ Hz}$, $5.74185740918254 \times 10^{14} \text{ Hz}$, $4.21850748429738 \times 10^{14} \text{ Hz}$,
 $3.22979479266518 \times 10^{14} \text{ Hz}$, $2.551936626303351 \times 10^{14} \text{ Hz}$, $2.067068667305715 \times 10^{14} \text{ Hz}$ }

In[]:= **me**

Out[]:=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,880\,000\,000\,000\,000\,000} \text{ kg}$$

In[]:= **SetPrecision**[**me c**², 32]

Out[]:=

$8.187105787968449615915667084159 \times 10^{-14} \text{ kg m}^2/\text{s}^2$

```
In[ ]:= UnitSimplify[Quantity[7.72319 × 10-13,  $\frac{\sqrt{\text{"Joules"}} \sqrt{\text{"Seconds"}}}{\sqrt{\text{"Hertz"}} \sqrt{\text{"Kilograms"}}}$ ]]
```

```
Out[ ]:=
```

0.772319 pm

```
In[ ]:= n =  $\frac{mP}{me}$ 

$$\frac{\psi[n] \ 1.0 \text{ mp}}{\frac{1}{\alpha} + \frac{1.0}{\alpha^2}}$$


$$\frac{1}{\alpha} + \frac{1.0}{\alpha^2}$$


$$\frac{\psi[n]}{\frac{1}{\alpha} + \frac{1}{\alpha^2}} \ 1.0 \text{ me}$$

mp 2.162683900875855`*27 × 1.0

$$\frac{\psi[n] \text{ me } 1.0}{1.0} \ 2.162683900875855`*^27$$

```

```
Out[ ]:=
```

23 892 177 732 494 625 341 423

```
Out[ ]:=
```

3.61735 $\sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (m \sqrt{\text{Hz}})$

```
Out[ ]:=
```

18 915.9

```
Out[ ]:=
```

0.00197007 $\sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (m \sqrt{\text{Hz}})$

```
Out[ ]:=
```

3.61735 $\sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (m \sqrt{\text{Hz}})$

```
Out[ ]:=
```

8.05939 × 10²⁸ $\sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (m \sqrt{\text{Hz}})$

```

In[ ]:=
mP
mp
130 120 866 735 844 044 779 * 1836
mP      1
-----
me  13 012 086 673 584 404 467
n = -----
      mP
      me
Solve[ψ[n]  $\frac{X}{mP} == \frac{1.0}{me mP}$ , X] 1.0
238 921 777 324 946 253 414 230 me 1.0
Clear[n]
    
```

```

Out[ ]:=
13 012 086 673 584 404 467
    
```

```

Out[ ]:=
238 901 911 327 009 666 214 244
    
```

```

Out[ ]:=
23 892 177 732 494 625 341 423
-----
13 012 086 673 584 404 467
    
```

```

Out[ ]:=
23 892 177 732 494 625 341 423
    
```

```

Out[ ]:=
{ { 1. (X → 0.0268343 /kg) } }
    
```

```

Out[ ]:=
2.17643 × 10-7 √kg √s √J / (m √Hz )
    
```

```

n = -----
      mP
      me
ψ[n]  $\frac{130 120 866 735 844 044 779.0 me 1836}{mP}$ 
Solve[ψ[n]  $\frac{130 120 866 735 844 044 779.0 me X}{mP} == 10$ , X]
-----
1
2.227980020131375`*^29
    
```

```

Out[ ]:=
23 892 177 732 494 625 341 423
    
```

```

Out[ ]:=
9.99916851455907393523
    
```

```

Out[ ]:=
4.48837 × 10-30
    
```

```

In[ ]:= n =  $\frac{mP}{me}$ 

 $\psi[n] \text{ 1.0 mp}$ 
 $\frac{\psi[n] \text{ 1.0 mp}}{1836.15} \text{ 1.0}$ 
 $\frac{mP}{me} \text{ 1.0}$ 
 $\psi\left[\frac{mP}{me}\right] \text{ 130 120 866 735 844 044 779.0 me 1836}$ 

Solve  $\left[\psi\left[\frac{mP}{me}\right] \text{ 130 120 866 735 844 044 779 me } X == 1.0, X\right]$ 


$$\frac{1}{5.840306715505077 \times 10^{-9}}$$


$$\frac{7 \sqrt{405397113703863}}{200000000000000 \times 2^{1/4} \pi^{7/8}} \text{ kg 1.0}$$

 $\psi\left[\frac{mP}{me}\right] \text{ me 1.0}$ 

```

```

Out[ ]=
23 892 177 732 494 625 341 440

```

```

Out[ ]=
2.66156 × 1050 kg

```

```

Out[ ]=
1.44954 × 1047 kg

```

```

Out[ ]=
1.16806 × 10-9

```

```

Out[ ]=
2.17624918026184363956 × 10-7 kg

```

```

Out[ ]=
 $\left\{ \left\{ X \rightarrow 4.12453 \times 10^{-22} / \text{kg} \right\} \right\}$ 

```

```

Out[ ]=
1.71224 × 108

```

```

Out[ ]=
2.17643 × 10-7 kg

```

```

Out[ ]=
18.6328 kg

```

```

In[ ]:=  $\psi\left[\frac{\text{mP}}{\text{me}}\right] \text{me } 1.0$ 

 $\psi\left[\frac{\text{mp}}{\text{mP}}\right] 1.0$ 

 $n = \frac{\text{mP}}{\text{me}}$ 

 $\frac{\psi[n] \text{me}}{\psi[n] 1.0 \frac{1}{\alpha}} \text{mP}$ 

 $\psi[n] 1.0 \frac{\text{mp}}{\alpha} \text{mP}$ 

 $\frac{\psi[n] 1.0 \text{mp}}{1836.15}$ 

 $\frac{\psi[n] 1.0 \text{mp}}{\alpha \text{mP}}$ 

 $\psi[n] 1.0 \text{mp } \alpha^2$ 

 $\psi[n] 1.0 \text{mP}$ 
Clear[n]
1
me 1.0
    
```

```

Out[ ]=
37.2657  $\sqrt{\text{kg}}$   $\sqrt{\text{s}}$   $\sqrt{\text{J}}$  / (m  $\sqrt{\text{Hz}}$  )
    
```

```

Out[ ]=
1.
    
```

```

Out[ ]=
23 892 177 732 494 625 341 423
    
```

```

Out[ ]=
1.44677  $\times 10^{-40}$  kg s J / (m2Hz)
    
```

```

Out[ ]=
0.0204079 kg s J / (m2Hz)
    
```

```

Out[ ]=
3.72657  $\sqrt{\text{kg}}$   $\sqrt{\text{s}}$   $\sqrt{\text{J}}$  / (m  $\sqrt{\text{Hz}}$  )
    
```

```

Out[ ]=
4.30832  $\times 10^{13}$ 
    
```

```

Out[ ]=
0.364375  $\sqrt{\text{kg}}$   $\sqrt{\text{s}}$   $\sqrt{\text{J}}$  / (m  $\sqrt{\text{Hz}}$  )
    
```

```

Out[ ]=
8.90358  $\times 10^{23}$   $\sqrt{\text{kg}}$   $\sqrt{\text{s}}$   $\sqrt{\text{J}}$  / (m  $\sqrt{\text{Hz}}$  )
    
```

```

Out[ ]=
1.09777  $\times 10^{30}$  m  $\sqrt{\text{Hz}}$  / (  $\sqrt{\text{kg}}$   $\sqrt{\text{s}}$   $\sqrt{\text{J}}$  )
    
```

In[]:= **SetPrecision**[0.0072973525643, 12]

Out[]:=

0.00729735256430

SetPrecision $\left[\frac{\text{Abs}[\alpha - 0.0072973525643]}{0.0072973525643}, 12\right]$

SetPrecision $\left[\text{Abs}\left[\frac{\text{Abs}[\alpha - 0.0072973525643]}{0.0072973525643} - (1.6 * 10^{-10})\right], 3\right]$

SetPrecision $\left[\frac{\text{Abs}[\alpha - 0.0072973525643]}{0.0072973525643}, 12\right] - (1.6 * 10^{-10})$

SetPrecision $\left[\frac{\text{Abs}[\alpha - (0.0072973525643 + 1.6 * 10^{-10})]}{(0.0072973525643 + 1.6 * 10^{-10})}, 12\right]$

SetPrecision $\left[\frac{\text{Abs}[\alpha - (0.0072973525643 - 1.6 * 10^{-10})]}{(0.0072973525643 - 1.6 * 10^{-10})}, 12\right]$

0.0072973525643 + 1.6 * 10⁻¹⁰

0.0072973525643 - 1.6 * 10⁻¹⁰

Out[]:=

1.37750643072 × 10⁻¹⁰

Out[]:=

2.22 × 10⁻¹¹

Out[]:=

-2.22494 × 10⁻¹¹

Out[]:=

2.17880087789 × 10⁻⁸

Out[]:=

2.20635110265 × 10⁻⁸

Out[]:=

0.00729735

Out[]:=

0.00729735

Out[]:=

0.001161409732253

In[]:= **SetPrecision** $\left[\frac{\alpha}{2\pi}, 13\right]$

Out[]:=

0.001161409732253

In[]:= **53 - 43**

Out[]:=

10

```
In[ ]:= SetPrecision[ $\alpha$ , 12]
```

```
Out[ ]:=  
0.0072973525653
```

```
In[ ]:= 23 892 177 732 494 625 341 444
```

```
Out[ ]:=  
23 892 177 732 494 625 341 444
```

```
In[ ]:= 23 892 177 732 494 625 341 475
```

```
Out[ ]:=  
23 892 177 732 494 625 341 475
```

```
In[ ]:= NextPrime[115 550 496 527 972 941 850, -1]
```

```
Out[ ]:=  
115 550 496 527 972 941 829
```

```
In[ ]:= NextPrime[23 892 177 732 494 625 341 444, -1]
```

```
Out[ ]:=  
23 892 177 732 494 625 341 423
```

$$\text{In}[*]:= \text{me} = \frac{\text{mP}}{23\,892\,177\,732\,494\,625\,341\,423}$$

$$\text{mp} = \frac{\text{mP}}{13\,012\,086\,673\,584\,404\,467}$$

$$\text{m}\mu = \frac{\text{mP}}{115\,550\,496\,527\,972\,941\,829}$$

SetPrecision[me, 32]

SetPrecision[mp, 32]

mp

me 1.0

SetPrecision[mμ, 32]

$$\text{Round}\left[\frac{\text{mP}}{167\,262\,192\,595 \times 10^{-38}}\right]$$

mP

13 012 086 673 584 404 467

167 262 192 595 × 10⁻³⁸ × 1.0

13 012 086 673 584 404 461

Out[*]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,846\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{26\,024\,173\,347\,168\,808\,934\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{231\,100\,993\,055\,945\,883\,658\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

$$9.109383713899999367782512764505 \times 10^{-31} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

$$1.6726219259499999992669515508927 \times 10^{-27} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

1836.15

Out[*]=

$$1.8835316270000002567209704690888 \times 10^{-28} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

$$13\,012\,086\,673\,584\,404\,461 \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{26\,024\,173\,347\,168\,808\,934\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$\frac{2 \pi \Gamma}{c^4} \frac{2 n^2}{m_e} 1.0$$

$$\frac{2 \pi \Gamma}{c^4} \quad \frac{2 n^2}{m e}$$

n = 1

$$\frac{\alpha^2 m_e c^2}{2 n^2 h} \quad 1.0$$

$$2 \pi \frac{\alpha^2 m_e c^2}{2 n^2 h}$$

$$\frac{2 n^2 h}{\alpha^2 m_e c}$$

Out[•]=

$$\left(\frac{5\,110\,875\,222\,870\,301\,507\,065\,971\,913\,830\,033\,423\,949}{135\,132\,371\,234\,621} \sqrt{\frac{3}{\pi^{15/8}}} \right) /$$

$$(883\,488\,452\,991\,209\,868\,825\,728\,206\,610\,341\,750\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 : \\ 000\,000 \times 2^{3/4}) \text{ s}^{9/2} \sqrt{\text{J}} / (\sqrt{\text{kg m}^4} \sqrt{\text{Hz}})$$

Out[•]=

1

Out[•]=

$$3.28984 \times 10^{15} \sqrt{\text{kg m}} \sqrt{\text{Hz}} / (\text{s}^{3/2} \sqrt{\text{J}})$$

$$\ln[\bullet] := \text{UnitSimplify}\left[\text{Quantity}\left[4.38388 \times 10^{-48}, \frac{\sqrt{\text{"Joules"}} \text{"Seconds"}^{9/2}}{\sqrt{\text{"Hertz"}} \sqrt{\text{"Kilograms"}} \text{"Meters"}^4}\right]\right]$$

Out[•]=

$$4.38388 \times 10^{-48} \text{ s}^4/\text{m}^3$$

```
In[*]:= Print[{-2813.225802760800611361`9. eV, -703.3064506902001528402`9. eV,
-312.580644751200067929`9. eV, -175.8266126725500382101`9. eV,
-112.5290321104320244544`9. eV, -78.1451611878000169822`9. eV,
-57.4127714849142981911`9. eV, -43.9566531681375095525`9. eV,
-34.731182750133340881`9. eV, -28.1322580276080061136`9. eV}]

-2813.22580 eV -703.306451 eV -312.580645 eV -175.826613 eV -112.529032 eV
-78.1451612 eV -57.4127715 eV -43.9566532 eV -34.7311828 eV -28.1322580 eV
```

```
In[*]:= Clear[n]
```

```
Table[UnitConvert[SetPrecision[- $\frac{\alpha^2}{2 n^2}$  me c2, 16], "Electronvolts"], {n, 1, 10}]
```

```
Table[UnitConvert[SetPrecision[ $\frac{2 n^2 h}{\alpha^2 \text{ me c}}$ , 16], "Meters"], {n, 1, 10}]
```

```
Table[UnitConvert[SetPrecision[ $\frac{\alpha^2 \text{ me c}^2}{2 n^2 h}$ , 16], "Hertz"], {n, 1, 10}]
```

```
Table[UnitConvert[SetPrecision[ $2 \pi \frac{\alpha^2 \text{ me c}^2}{2 n^2 h}$ , 16], "Hertz"], {n, 1, 10}]
```

```
Out[*]=
```

```
{ -13.60569312659197 eV , -3.401423281647992 eV ,  
  -1.511743680732441 eV , -0.850355820411998 eV , -0.544227725063679 eV ,  
  -0.377935920183110 eV , -0.2776672066651422 eV , -0.2125889551029995 eV ,  
  -0.1679715200813823 eV , -0.1360569312659197 eV }
```

```
Out[*]=
```

```
{ 9.11267050341422  $\times 10^{-8}$  m , 3.64506820136569  $\times 10^{-7}$  m ,  
  8.20140345307280  $\times 10^{-7}$  m , 1.458027280546276  $\times 10^{-6}$  m ,  
  2.278167625853556  $\times 10^{-6}$  m , 3.280561381229120  $\times 10^{-6}$  m , 4.46520854667297  $\times 10^{-6}$  m ,  
  5.83210912218510  $\times 10^{-6}$  m , 7.38126310776552  $\times 10^{-6}$  m , 9.11267050341422  $\times 10^{-6}$  m }
```

```
Out[*]=
```

```
{ 3.289841961120809  $\times 10^{15}$  Hz , 8.22460490280202  $\times 10^{14}$  Hz ,  
  3.65537995680090  $\times 10^{14}$  Hz , 2.056151225700506  $\times 10^{14}$  Hz ,  
  1.315936784448324  $\times 10^{14}$  Hz , 9.13844989200225  $\times 10^{13}$  Hz , 6.71396318596084  $\times 10^{13}$  Hz ,  
  5.14037806425126  $\times 10^{13}$  Hz , 4.06153328533433  $\times 10^{13}$  Hz , 3.289841961120809  $\times 10^{13}$  Hz }
```

```
Out[*]=
```

```
{ 2.067068667305715  $\times 10^{16}$  Hz , 5.16767166826429  $\times 10^{15}$  Hz ,  
  2.296742963673016  $\times 10^{15}$  Hz , 1.291917917066072  $\times 10^{15}$  Hz ,  
  8.26827466922286  $\times 10^{14}$  Hz , 5.74185740918254  $\times 10^{14}$  Hz , 4.21850748429738  $\times 10^{14}$  Hz ,  
  3.22979479266518  $\times 10^{14}$  Hz , 2.551936626303351  $\times 10^{14}$  Hz , 2.067068667305715  $\times 10^{14}$  Hz }
```

```
Table[UnitConvert[SetPrecision[- $\frac{\alpha^2}{2 n^2}$  mp c2, 16], "Electronvolts"], {n, 1, 10}]
```

```
Table[UnitConvert[SetPrecision[- $\frac{\alpha^2}{2 n^2}$  mμ c2, 16], "Electronvolts"], {n, 1, 10}]
```

```
In[*]:= NextPrime[13 012 086 673 584 404 470, -1]
```

```
Out[*]=
```

```
13 012 086 673 584 404 467
```

```
In[ ]:= UnitSimplify[Quantity[10,  $\frac{\text{"Joules"}^{3/2} \text{"Seconds"}^{3/2}}{\text{"Coulombs"}^4 \text{"Hertz"}^{3/2} \sqrt{\text{"Kilograms"} \text{"Meters"}}}$ ]]
```

```
Out[ ]:=
```

$$10 \text{ kg m}^2 / (\text{s}^4 \text{ A}^4)$$

```
In[ ]:= UnitSimplify[Quantity[100000000,  $\frac{\sqrt{\text{"Joules"}} \sqrt{\text{"Seconds"}}}{\text{"Coulombs"}^2 \sqrt{\text{"Hertz"}} \sqrt{\text{"Kilograms"}}}$ ]]
```

```
Out[ ]:=
```

$$100000000 / (\text{F N})$$

```
In[ ]:=  $\frac{\text{me c}^2}{\Gamma} 1.0$ 
```

```
Out[ ]:=
```

$$3.1894 \times 10^{31} \sqrt{\text{kg}} \text{ m}^2 \sqrt{\text{Hz}} / (\text{s}^{5/2} \sqrt{\text{J}})$$

```
In[ ]:= Table[rn[n, me], {n, 1.0, 10.0}]
```

```
Out[ ]:=
```

$$\left\{ \begin{array}{l} 5.29177 \times 10^{-11} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), 2.11671 \times 10^{-10} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), \\ 4.76259 \times 10^{-10} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), 8.46684 \times 10^{-10} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), \\ 1.32294 \times 10^{-9} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), 1.90504 \times 10^{-9} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), \\ 2.59297 \times 10^{-9} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), 3.38673 \times 10^{-9} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), \\ 4.28634 \times 10^{-9} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}), 5.29177 \times 10^{-9} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}}) \end{array} \right\}$$

```
In[ ]:= me / mP 1.0  
me 1.0  
mP
```

```
Out[ ]:=
```

$$4.18547 \times 10^{-23}$$

```
Out[ ]:=
```

$$9.10938 \times 10^{-31} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

```
Out[ ]:=
```

$$\frac{7 \sqrt{405397113703863}}{200000000000000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

```
In[ ]:= UnitSimplify[Quantity[ $\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,846\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}}$ ,  

 $\frac{\sqrt{\text{"Joules"}} \text{"Kilograms"}^{3/2} \sqrt{\text{"Seconds"}}}{\sqrt{\text{"Hertz"}} \text{"Meters"}}$ ]]
```

Out[]:=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,846\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \text{ kg}^2$$

```
In[ ]:= UnitSimplify[Quantity[ $\frac{7 \sqrt{405\,397\,113\,703\,863}}{2\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}}$ ,  $\frac{\sqrt{\text{"Joules"}} \sqrt{\text{"Kilograms"}} \sqrt{\text{"Seconds"}}}{\sqrt{\text{"Hertz"}} \text{"Meters"}}$ ]]
```

Out[]:=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{2\,000\,000 \times 2^{1/4} \pi^{7/8}} \mu\text{g}$$

```
In[ ]:= Table[ $\frac{\text{mP lP n}^2}{\alpha \text{ me}}$ , {n, 1.0, 10.0}]  

Table[ $\frac{\text{r n}^2}{\alpha^2 \text{ me}}$ , {n, 1.0, 10.0}]
```

Out[]:=

$$\left\{ 5.29177 \times 10^{-11} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 2.11671 \times 10^{-10} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \right. \\ 4.76259 \times 10^{-10} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 8.46684 \times 10^{-10} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \\ 1.32294 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 1.90504 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \\ 2.59297 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 3.38673 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \\ \left. 4.28634 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 5.29177 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) \right\}$$

Out[]:=

$$\left\{ 5.29177 \times 10^{-11} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 2.11671 \times 10^{-10} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \right. \\ 4.76259 \times 10^{-10} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 8.46684 \times 10^{-10} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \\ 1.32294 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 1.90504 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \\ 2.59297 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 3.38673 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , \\ \left. 4.28634 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) , 5.29177 \times 10^{-9} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz}) \right\}$$

$$\text{In}[*]:= 2 \sqrt{\alpha}$$

$$\frac{\alpha}{e} \sqrt{\frac{h}{c \Phi}}$$

$$\frac{\alpha}{e} \sqrt{\frac{h}{c \Phi}} == \sqrt{2 \pi \alpha}$$

$$\frac{\alpha}{e} \sqrt{\frac{4 h}{c \mu_0}}$$

$$2 \sqrt{\frac{\pi \alpha}{2 \pi}}$$

$$2 \sqrt{\alpha}$$

$$\frac{\alpha}{e} \sqrt{\frac{2 h}{\pi c \Phi}}$$

$$\frac{\alpha}{e} \sqrt{\frac{4 h}{c \mu_0}} == 2 \sqrt{\frac{\pi \alpha}{2 \pi}}$$

$$\frac{2}{\sqrt{2}} \sqrt{\alpha} == \frac{\alpha}{e \sqrt{\pi}} \sqrt{\frac{h}{c \Phi}} == \sqrt{2 \alpha}$$

Out[*]=

$$\frac{267029439 \sqrt{\frac{64241241 \pi}{12621086}}}{625000000}$$

Out[*]=

$$\frac{267029439 \sqrt{\frac{64241241}{6310543}} \pi}{1250000000}$$

Out[*]=

True

Out[*]=

$$\frac{267029439 \sqrt{\frac{64241241 \pi}{6310543}}}{1250000000}$$

Out[*]=

$$\frac{267029439 \sqrt{\frac{64241241 \pi}{6310543}}}{1250000000}$$

Out[*]=

$$\frac{267029439 \sqrt{\frac{64241241 \pi}{12621086}}}{625000000}$$

Out[*]=

$$\frac{267029439 \sqrt{\frac{64241241 \pi}{12621086}}}{625000000}$$

Out[⌘]=

True

Out[⌘]=

True

In[⌘]:=
$$\frac{\alpha}{e \sqrt{\pi}} \sqrt{\frac{h}{c \Phi}} == \sqrt{2 \alpha}$$
$$\frac{\alpha}{\sqrt{2 \alpha} e \sqrt{\pi}} \sqrt{\frac{h}{c \Phi}}$$
$$\frac{\sqrt{\alpha} \sqrt{\frac{h}{c \Phi}}}{e \sqrt{2 \pi}}$$
$$\frac{h \alpha}{2 c e^2} == \pi \Phi$$
$$\pi \Phi$$

Out[⌘]=

True

Out[⌘]=

1

Out[⌘]=

1

Out[⌘]=

True

Out[⌘]=

$$\frac{\pi}{10\,000\,000} \text{ s J / (mHz C}^2\text{)}$$

$$In[\bullet] := \text{UnitSimplify}\left[\frac{n}{c}\right]$$

$$\frac{1}{2} \sqrt{\frac{18\,931\,629}{21\,413\,747}} \frac{c}{h} \Phi$$

$$\frac{1}{e} \sqrt{\frac{h}{c \Phi}}$$

$$\sqrt{2\pi} \frac{\sqrt{\alpha}}{\alpha}$$

$$\frac{\sqrt{\alpha}}{\alpha}$$

$$e^2 \frac{18\,931\,629}{21\,413\,747} \frac{c \cdot \Phi}{h}$$

e

Out[•]=

[illegible]

Out[•]=

100 000 000 000 000 000 /C

Out[•]=

$$\frac{25\,000\,000\,000 \sqrt{\frac{6\,310\,543}{64\,241\,241}}}{267\,029\,439}$$

Out[•]=

$$\frac{25\,000\,000\,000 \sqrt{\frac{6\,310\,543}{64\,241\,241}}}{267\,029\,439}$$

Out[•]=

$$\frac{12\,500\,000\,000 \sqrt{\frac{12\,621\,086}{64\,241\,241 \pi}}}{267\,029\,439}$$

Out[•]=

641 742 491 633 892 489

625 000 000 000 000 000 000

Out[•]=

$$\begin{array}{r} 801\,088\,317 \\ \hline 5\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \end{array} \quad \text{C}$$

$$In[\bullet] := \frac{2 \pi c e^2}{h}$$

Out[•]=

$$\frac{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}{197\,204\,468\,750\,000\,000\,000}\text{ mHz C}^2/(\text{s J})$$

In[]:=

$$\frac{2 \pi c \Gamma}{h}$$
$$\frac{h}{2 \pi c} == 1 \text{ P mP}$$
$$\left(\frac{h}{2 \pi c e} \right) \frac{1}{c \Omega}$$
$$2 \sqrt{\alpha \pi} \frac{\left(\frac{h}{2 \pi c e} \right) \frac{1}{c \Omega}}{1 \text{ P}} == 1$$

Out[]:=

$$\frac{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000}$$

Out[]:=

True

Out[]:=

$$\frac{6\,310\,543}{27\,427\,896\,134\,449\,959\,206\,229\,024\,000\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{9/8}} \text{ s J / (kg m Hz)}$$

Out[]:=

True

$$\ln[\bullet] := 2 \sqrt{\alpha \pi} \frac{\left(\frac{h}{2 \pi c e}\right) \frac{1}{c \Omega}}{1 p} = 1$$

$$2 \sqrt{\alpha \pi} \frac{\left(\frac{h}{2 \pi c e}\right) \frac{1}{c \Omega}}{\sqrt{\frac{h G}{2 \pi c^3}}}$$

$$\frac{\sqrt{2} \hbar \sqrt{\alpha}}{c^2 e \sqrt{\frac{\hbar 4 \pi \Phi}{c^3}}}$$

$$\frac{G}{4\pi\epsilon}$$

$$\sqrt{\frac{h 4 \pi \Phi}{c^3}}$$

Out[•]=

True

Out[•]=

1

Out[•]=

1

Out[•]=

$$\frac{1}{10\,000\sqrt{2}\pi^{1/4}}\text{ m}^4\text{Hz C}^2/(\text{kg s}^3\text{J})$$

Out[•]=

$$\frac{\sqrt{\frac{18\,931\,629\,\pi}{21\,413\,747}}}{299\,792\,458\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000} \text{ s}^2\text{J} / (\text{m}^2\text{Hz C})$$

$$\ln[\bullet] :=$$

$$ln[\bullet] := \frac{h \sqrt{2\alpha}}{e \sqrt{c h 4 \pi \Phi}}$$

Out[•]=

1

$$\ln[*]:= \text{UnitSimplify}\left[\text{Quantity}\left[1.60009 \times 10^{-26}, \frac{\text{"Joules"}}{\text{"Hertz" "Kilograms"}}\right]\right]$$

Out[•]=

$$1.60009 \times 10^{-26} \text{ m}^2/\text{s}$$

$$ln[\bullet] := \mu P \alpha \in P \frac{1}{\alpha} == \mu P \in P == \mu 0 \in 0 == \frac{1}{c^2}$$

Out[•]=

True

In[]:= 1.0

Out[]:=
1.

In[]:= $\frac{\alpha}{2\pi} \frac{c \hbar m p^2}{\Gamma^2} 1.0$

Out[]:=
979 526. kg m/s²

In[]:= $\text{Table}\left[\frac{mP \ 1P \ me}{\Gamma^2} \left(\frac{1P}{\Delta\alpha[n]} - \frac{2 \ G \ me}{c^2}\right), \{n, 1.0, 10.0\}\right]$

$\frac{\Gamma}{me \ 1.0 \ \alpha^2}$
 $\frac{\Gamma}{me \ 1.0}$

Out[]:=
{1.5777 × 10⁻¹⁸, 3.1468 × 10⁻¹⁸, 4.71781 × 10⁻¹⁸, 6.2893 × 10⁻¹⁸, 7.86099 × 10⁻¹⁸,
9.43276 × 10⁻¹⁸, 1.10046 × 10⁻¹⁷, 1.25765 × 10⁻¹⁷, 1.41483 × 10⁻¹⁷, 1.57202 × 10⁻¹⁷}

Out[]:=
5.29177 × 10⁻¹¹ √s √J / (√kg √Hz)

Out[]:=
2.81794 × 10⁻¹⁵ √s √J / (√kg √Hz)

In[]:= $\Delta\alpha\left[\frac{12}{\sqrt{\alpha} \ 2}\right] 1.0$
 $\frac{me}{mP} \left(\frac{1P}{\Delta\alpha[2.0]} - \frac{2 \ G \ me}{c^2}\right)$
2 √2 α 1.0

Out[]:=
0.00503369

Out[]:=
2.70839 × 10⁻⁵⁷ √s √J / (√kg √Hz)

Out[]:=
0.241617

In[]:=

$$M = mP$$

$$Q = e$$

$$R_s = \frac{2 G M}{c^2}$$

$$\frac{\frac{\alpha}{2 \pi} \frac{c h M}{\Gamma Q}}{\frac{\alpha}{2 \pi} \frac{c h M^2}{\Gamma^2}} \frac{e}{1P}$$

$$\frac{\frac{\alpha}{2 \pi} \frac{c h M}{\Gamma Q}}{\frac{\alpha}{2 \pi} \frac{c h M^2}{\Gamma^2}} \frac{e}{1P} == \frac{mP}{M} \frac{e}{Q}$$

$$\frac{\frac{\alpha}{2 \pi} \frac{c h M}{\Gamma Q}}{\frac{\alpha}{2 \pi} \frac{c h M^2}{\Gamma^2}} \frac{e}{1P} == \frac{mP}{M} \frac{e}{Q}$$

$$\frac{\frac{\alpha}{2 \pi} \frac{c h M}{\Gamma Q}}{\frac{\alpha}{2 \pi} \frac{c h M^2}{\Gamma^2}} \frac{e}{1P} == \frac{mP}{M} \frac{e}{Q}$$

$$\frac{\Gamma}{M Q} \frac{e}{1P} == \frac{e}{Q} \frac{mP}{M}$$

$$\frac{\Gamma e^2}{M Q^2} == \frac{mP 1P e^2}{M Q^2}$$

$$\lambda c = \frac{\Gamma e^2}{M Q^2} \frac{2 \pi}{\alpha}$$

$$\lambda c = \frac{mP 1P e^2}{M Q^2} \frac{2 \pi}{\alpha}$$

$$\frac{\Gamma e^2}{M Q^2} \frac{2 \pi}{\alpha} == \frac{mP 1P e^2}{M Q^2} \frac{2 \pi}{\alpha}$$

SetPrecision[UnitConvert[λc, "Meters"], 32]

SetPrecision[λc / (2 G mP / c^2), 32]

$$\lambda c / \left(\frac{2 G mP}{c^2} \right) == \pi$$

$$\frac{\frac{\Gamma}{\alpha M}}{R_s} == \frac{1}{2}$$

$$\frac{\lambda c}{R_s} == \pi$$

$$\frac{\lambda c}{2 \pi R_s} == \frac{1}{2}$$

Out[]:=

$$\frac{7 \sqrt{405397113703863}}{2000000000000000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (m \sqrt{\text{Hz}})$$

Out[]:=

$$\frac{801088317}{5000000000000000000000000000000} C$$

Out[]:=

$$\frac{\sqrt{\frac{18931629}{21413747}}}{149896229000 \times 2^{3/4} \pi^{1/8}} \sqrt{s} \sqrt{J} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

```
Out[ ]=  
4 580 703 784 999 263 461 548 761  $\pi$   
1 972 044 687 500 000 000 000 000 000  
  
Out[ ]=  
True  
  
Out[ ]=  
True  
  
Out[ ]=  
True  
  
Out[ ]=  

$$\frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}} \pi^{7/8}}{14\,989\,622\,900\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}} \sqrt{s} \sqrt{J} / ( \sqrt{kg} \sqrt{Hz} )$$
  
  
Out[ ]=  

$$\frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}} \pi^{7/8}}{14\,989\,622\,900\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}} \sqrt{s} \sqrt{J} / ( \sqrt{kg} \sqrt{Hz} )$$
  
  
Out[ ]=  
True  
  
Out[ ]=  
 $1.0155249398134214613345260392937 \times 10^{-34} \text{ m}$   
  
Out[ ]=  
3.1415926535897932384626433832795  
  
Out[ ]=  
True  
  
Out[ ]=  
True  
  
Out[ ]=  
True  
  
Out[ ]=  
True
```

$$\text{In}[*]:= \text{Solve}\left[\frac{1\text{P mP}}{2\text{ }\Gamma}\frac{2^{3/4}}{100\sqrt{\pi}}X\equiv\frac{1}{\alpha\Omega}, X\right]$$

$$\frac{1\text{P mP}}{2\text{ }\Gamma}\frac{2^{3/4}}{100\sqrt{\pi}}$$

$$\frac{1}{100\times 2^{1/4}\pi^{1/8}}$$

$$\frac{9\,860\,223\,437\,500\,000\,000\,000\,000\times 2^{3/4}}{4\,580\,703\,784\,999\,263\,461\,548\,761\pi^{9/8}}\bigg/\frac{1}{100\times 2^{1/4}\pi^{1/8}}$$

$$\frac{2^{3/4}}{\pi^{9/8}}1.0$$

$$\text{Solve}\left[\frac{1}{100\alpha X2^{1/4}\pi^{1/8}}\equiv 1, X\right]$$

$$\frac{1}{\alpha\Omega}$$

$$\text{Out}[*]=\left\{\left\{X\rightarrow 1\text{ m A/kg}\right\}\right\}$$

$$\text{Out}[*]=\frac{986\,022\,343\,750\,000\,000\,000\,000\,000}{4\,580\,703\,784\,999\,263\,461\,548\,761\pi}$$

$$\text{Out}[*]=\frac{1}{50\times 2^{1/4}\pi^{1/8}}$$

$$\text{Out}[*]=\frac{1}{100\times 2^{1/4}\pi^{1/8}}$$

$$\text{Out}[*]=\frac{1\,972\,044\,687\,500\,000\,000\,000\,000\,000}{4\,580\,703\,784\,999\,263\,461\,548\,761\pi}$$

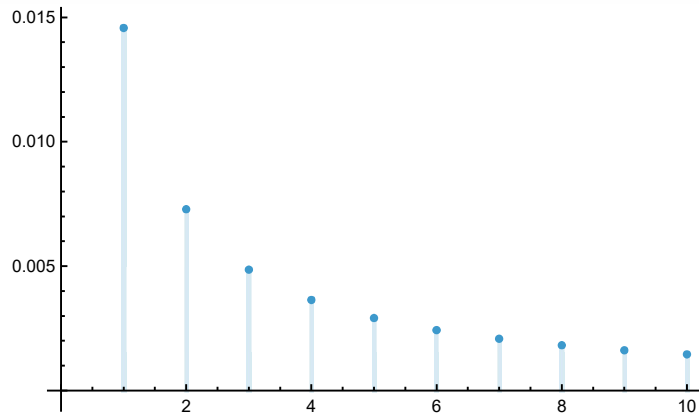
$$\text{Out}[*]=0.463958$$

$$\text{Out}[*]=\left\{\left\{X\rightarrow\frac{9\,860\,223\,437\,500\,000\,000\,000\,000\times 2^{3/4}}{4\,580\,703\,784\,999\,263\,461\,548\,761\pi^{9/8}}\right\}\right\}$$

$$\text{Out}[*]=\frac{9\,860\,223\,437\,500\,000\,000\,000\,000\times 2^{3/4}}{4\,580\,703\,784\,999\,263\,461\,548\,761\pi^{9/8}}\text{ m C/ (kg s)}$$

```
In[ ]:= DiscretePlot[ $\frac{1}{50 n 2^{1/4} \pi^{1/8}}$ , {n, 0, 10}]
```

```
Out[ ]:=
```



```
In[ ]:= FactorInteger[446 726 628 773 276 015 625 000 000 000 000 000]
```

```
Out[ ]:=
```

```
{{2, 15}, {3, 1}, {5, 22}, {19, 1}, {73, 1}, {389, 1}, {12 043, 1}, {293 339, 1}}
```

```
In[ ]:= FactorInteger[1 786 906 515 093 104 062 500 000 000 000 000 000]
```

```
Out[ ]:=
```

```
{{2, 17}, {3, 1}, {5, 22}, {19, 1}, {73, 1}, {389, 1}, {12 043, 1}, {293 339, 1}}
```

```
In[ ]:=
e
α
6 310 543
91 489 613 572 766 928 000 000 π
e
α
αγ
αδ
```

```
Out[ ]:=
          6 310 543
          ----- C
91 489 613 572 766 928 000 000 π
```

```
Out[ ]:=
6 310 543
```

```
Out[ ]:=
91 489 613 572 766 928 000 000 π
```

```
Out[ ]:=
          801 088 317
          ----- C
5 000 000 000 000 000 000 000 000 000
```

```
Out[ ]:=
          4 580 703 784 999 263 461 548 761 π
          -----
1 972 044 687 500 000 000 000 000 000
```

```
Out[ ]:=
4 580 703 784 999 263 461 548 761 π
```

```
Out[ ]:=
1 972 044 687 500 000 000 000 000 000
```

```
In[ ]:=
  c² Γ
  --- == G
m P² α
```

```
Out[ ]:=
True
```

```
In[ ]:=
  α   c h M
  ---  --- Q   c²  M
  2 π  r q
  α   c h M²
  ---  --- m P  α  m P
  2 π  r²
```

```
Out[ ]:=
          π³/⁴
          ----- m³/ (kg s²)
25 000 000 000 √2
```

```
In[ ]:= n = 1.0
```

$$\frac{1}{\alpha} \sqrt{\frac{e^4 Z_0^2 c^2}{8 h^2}} == \frac{c}{\sqrt{2}}$$

$$\frac{1}{\alpha^2} \left(\frac{e^4 Z_0^2 c^2}{8 h^2} \right) == \frac{c^2}{2}$$

$$\left(\frac{e^4 Z_0^2 c^2}{8 h^2} \right) == \frac{1}{2} \alpha^2 c^2$$

$$\text{UnitConvert}\left[\frac{e^4 Z_0^2 m_e c^2}{8 h^2}, \text{"Joules"}\right] \frac{1}{n^2}$$

```
Clear[n]
```

```
Out[ ]:=
```

```
1.
```

```
Out[ ]:=
```

```
True
```

```
Out[ ]:=
```

```
True
```

```
Out[ ]:=
```

```
True
```

```
Out[ ]:=
```

```
2.17987 × 10-18 J
```

Clear [x]

$$In[\ast] := \text{En}[m_ , n_] := \text{UnitConvert}\left[\frac{1}{2 (n^2)} \alpha^2 c^2 m, \text{"Electronvolts"}\right]$$

$$\ln[\bullet] :=$$
$$\text{En}[m_, n_] := \text{UnitConvert}\left[-\frac{\alpha}{2 n^2} \frac{m c^2 e^2 Z_0}{2 h}, \text{"Electronvolts"}\right];$$
$$\text{En}[m_, n_] := \text{UnitConvert}\left[\frac{\alpha^2}{2 n^2} c^2 m, \text{"Electronvolts"}\right];$$

Clear[x, n]

$$ln[\bullet] := \text{Solve}\left[\frac{\alpha}{2 \times 1^2} c^2 \frac{me \, mp}{\Gamma} == G \frac{me \, mp}{R^2}, R\right]$$

Out[•]=

$$\left\{ \left\{ \mathbf{R} \rightarrow - \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{1/8}} \mathbf{m} \right\}, \right. \\ \left. \left\{ \mathbf{R} \rightarrow \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{1/8}} \mathbf{m} \right\} \right\}$$

```
In[•]:= Ezn[ me, 1.0, 2]
```

$Out[\bullet]=$

54.4228 eV

```
In[*]:= SetPrecision[ $\frac{\text{Ezn}[\text{me}, 1, 1]}{h}$ , 64]
```

$$\frac{E_{zn}[me, 1, 1]}{h} \quad 1.0$$

$$\frac{Z^2 \alpha^2}{2 n^2} c^2 \frac{m}{h}$$

$$\frac{Z^2 \alpha^2}{2 n^2} c^2 m$$

$$\text{UnitConvert}\left[\frac{\text{Ezn}[\text{me}, 1, 1]}{c^2}, \text{"Kilograms"}\right] 1.0$$

$$\frac{\Gamma}{\alpha \frac{E_{zn} [me, 1, 1]}{c^2}} \quad 1.0$$

$$1.45032655577 \times 10^{-8}$$

$$\frac{\Gamma}{\alpha \frac{E_{zn} [me, 1, 1]}{c^2}}$$

$$\frac{\Gamma}{\frac{z^2 \alpha^3}{2 n^2}} m$$

Out[•]=

$$3.289841961120809456555756458522755998205077664069207774817678628 \times 10^{15} \text{ Hz}$$

Out[•]=

$$3.28984 \times 10^{15} \text{ Hz}$$

$$\frac{Z^2}{n^2} \left(\frac{2\,757\,082\,244\,883\,288\,657\,435\,505\,237\,673\,473\,158\,969\,848\,435\,997\,957\,647\,430\,636\,891}{8\,150\,502\,585\,577\,922\,157\,045\,550\,820\,446\,886\,230\,468\,750\,000\,000} \cdot 2^{1/4} \sqrt{\frac{64\,243\,243}{6\,310\,543}} \cdot \pi^{9/8} \right) \text{ kg m}^2 \text{ Hz} / (\text{s}^2 \text{ J})$$

$$\frac{Z^2 \left(\frac{57\,898\,727\,142\,549\,061\,806\,145\,609\,991\,142\,936\,338\,366\,817\,155\,957\,110\,596\,043\,374\,711}{258\,313\,827\,687\,345\,515\,498\,287\,574\,316\,406\,250\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000} \sqrt{\frac{64\,241\,241}{6\,310\,543}} r^{9/8} \right)}{n^2} \text{ kg m}^2/\text{s}^2$$

$$2.42544 \times 10^{-35} \text{ kg}$$

$$1.45033 \times 10^{-8} \text{ m/ (Hz s)}$$

$$1.45033 \times 10^{-8}$$

$$\frac{464\,578\,647\,377\,941\,562\,951\,285\,279\,945\,512\,757\,324\,218\,750\,000\,000 \sqrt{\frac{6\,310\,543}{64\,241\,241}} 2^{1/4}}{1\,048\,416\,554\,617\,577\,827\,617\,156\,389\,887\,486\,562\,862\,641\,199\,945\,620\,952\,903 \pi^{17/8}} \text{ m / (Hz s)}$$

$$\frac{n^2 \left(\frac{8\,150\,502\,585\,577\,922\,157\,045\,550\,820\,446\,886\,230\,468\,750\,000\,000 \sqrt{\frac{6\,310\,543}{64\,241\,241}} 2^{1/4}}{18\,393\,272\,888\,027\,681\,186\,265\,901\,576\,973\,448\,471\,274\,407\,016\,589\,841\,279 \pi^{17/8}} \right) \text{ s J / (kg m Hz)}}{Z^2}$$

```

In[ ]:= n = 1
        z = 1

        SetPrecision[UnitConvert[ $\left(\frac{c^2 m_e z^2 \alpha^2}{2 n^2}\right)$ , "Electronvolts"], 6]

        SetPrecision[UnitConvert[ $\left(\frac{c^2 m_e z^2 \alpha^2}{2 (n + z)^2}\right)$ , "Electronvolts"], 6]

        n = 3
        z = 1

        SetPrecision[UnitConvert[ $\frac{c^2 z^2 m_e \alpha^2}{2 (n)^2}$ , "Electronvolts"], 6]

        Clear[n, z]

Out[ ]:=
1

Out[ ]:=
1

Out[ ]:=
13.6057 eV

Out[ ]:=
3.40142 eV

Out[ ]:=
3

Out[ ]:=
1

Out[ ]:=
1.51174 eV

In[ ]:=

```

```
In[ ]:= 1 Δα [ 1 / 4 α ] 1.0
1.7 / 0.18234473881797703`
1.7
Δα [ 4^2 ]
1.7003238951071893723` 5.541362150974352 eV
Δα [ α / 4 ] 1.0
```

```
9.323 / Δα [ 2 α / 1 ]
```

```
Clear[Z]
```

```
Z = 4;
```

```
Z^2 α γ
(- 2 Z-1 / 2 α δ + Z^2 α γ) 2 1.0
```

```
Clear[z, Z]
```

```
Out[ ]:= 0.0145947
```

```
Out[ ]:= 9.323
```

```
Out[ ]:= 54.4008
```

```
Out[ ]:= 6.8075 eV
```

```
Out[ ]:= 4.93363
```

```
Out[ ]:= -0.0172553
```

```
In[ ]:= 4 π 1.0
π 2 √ π 1.0
√ 2 π
401 / 2000 1.0
5 √ 2 π - √ α 1.0
```

```
Out[ ]:= 12.5664
```

```
Out[ ]:= 11.1367
```

```
Out[ ]:= 12.5019
```

```
Out[ ]:= 12.4477
```

```
In[ ]:= Table[UnitConvert[ $\frac{c^2 (4) m e \alpha^2}{2 \left(\frac{1}{\sqrt{2}} z + 2 n\right)^2}$ , "Electronvolts"] 1.0, {n, 1, 4}, {z, 4, 4}]
```

```
Out[ ]:= { { 2.33437 eV }, { 1.16718 eV }, { 0.698255 eV }, { 0.464141 eV } }
```

```
In[ ]:= z = 4.0;
```

```
UnitConvert[ $\frac{c^2 z^2 m e \alpha^2}{2 (n)^2}$ , "Electronvolts"]
```

```
UnitConvert[ $\frac{c^2 z^2 m e \alpha^2}{2 (n)^2}$ , "Electronvolts"]
```

```
UnitConvert[ $\frac{c^2 z^2 m p \alpha^2}{2 (n)^2}$  1.0, "Electronvolts"]
```

```
UnitConvert[ $\frac{c^2 m e \alpha^2}{2}$  1.0, "Electronvolts"]
```

```
Clear[z, n];
```

```
Out[ ]:= UnitConvert[ $\frac{3.4878 \times 10^{-17} \sqrt{\text{kg m}} \sqrt{\text{J}} / (\sqrt{\text{Hz}} \text{ s}^{3/2})}{n^2}$ , Electronvolts]
```

```
Out[ ]:= UnitConvert[ $\frac{3.4878 \times 10^{-17} \sqrt{\text{kg m}} \sqrt{\text{J}} / (\sqrt{\text{Hz}} \text{ s}^{3/2})}{n^2}$ , Electronvolts]
```

```
UnitConvert[ $\frac{6.40413 \times 10^{-15} \sqrt{\text{kg m}} \sqrt{\text{J}} / (\sqrt{\text{Hz}} \text{ s}^{3/2})}{n^2}$ , Electronvolts]
```

```
Out[ ]:= 13.6057 eV
```

```
In[ ]:= .0
```

```
Out[ ]:= 0.
```

In[]:= **z = 8**

$$\text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (1)^2}, \text{"Electronvolts"}\right] 1.0$$

$$\frac{\text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (1)^2}, \text{"Electronvolts"}\right] - \left(\sum_{n=2}^z \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (n)^2}, \text{"Electronvolts"}\right]\right)}{4 \pi} 1.0$$

$$x1 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (1)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x2 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (2)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x3 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (3)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x4 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (4)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x5 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (5)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x6 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (6)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x7 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (7)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x8 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (8)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x9 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (9)^2}, \text{"Electronvolts"}\right] 1.0$$

$$\frac{x1 - x2 - x3 - x4 - x5 - x6 - x7 - x8 - x9}{4 \pi}$$

Clear[z, n]

Out[]:=

8

Out[]:=

870.764 eV

Out[]:=

32.7465 eV

Out[]:=

870.764 eV

Out[]:=

217.691 eV

Out[*]=	96.7516 eV
Out[*]=	54.4228 eV
Out[*]=	34.8306 eV
Out[*]=	24.1879 eV
Out[*]=	17.7707 eV
Out[*]=	13.6057 eV
Out[*]=	10.7502 eV
Out[*]=	31.891 eV
In[*]:=	$\frac{x1 - x2 - x3 - x4 - x5 - x6 - x7 - x8 - x9}{4 \pi}$ $\frac{x1 - x2 - x3 - x4 - x5 - x6 - x7 - x8 - x9}{\frac{\sqrt{2 \pi}}{\frac{192}{1000}}}$
Out[*]=	31.891 eV
Out[*]=	30.6965 eV

In[]:= **z = 4**

$$x1 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (1)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x2 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (2)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x3 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (3)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x4 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (4)^2}, \text{"Electronvolts"}\right] 1.0$$

$$x5 = \text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 (5)^2}, \text{"Electronvolts"}\right] 1.0$$

$$\frac{x1 - x2 - x3 - x4 - x5}{4 \pi}$$

$$\frac{217 - 54 - 24 - 13 - 8.0}{4 \pi}$$

Clear[z, n]

Out[]:=

4

Out[]:=

217.691 eV

Out[]:=

54.4228 eV

Out[]:=

24.1879 eV

Out[]:=

13.6057 eV

Out[]:=

8.70764 eV

Out[]:=

9.29203 eV

Out[]:=

9.39014

In[]:= **UnitSimplify** $\left[\text{Quantity}\left[4.96526 \times 10^{-17}, \frac{\sqrt{\text{"Joules"}} \sqrt{\text{"Kilograms"}} \text{"Meters"}}{\sqrt{\text{"Hertz"}} \text{"Seconds"}^{3/2}}\right]\right]$

Out[]:=

0.309907 keV

$$\text{In}[*]:= \left(\frac{z^2 \alpha \gamma}{\left(-\frac{2z-1}{2} \alpha \delta + z^2 \alpha \gamma\right) 2} \right)^2 \sqrt{\frac{1 + \frac{\pi}{3}}{2}} \sqrt{\frac{3}{2}}$$

```
Table[SetPrecision[UnitConvert[ $\left(\frac{c^2 z^2 m_e \alpha^2}{2 n^2}\right)$ , "Electronvolts"], 6],
  {n, 1, 5}, {z, 4, 4}]
```

```
Table[SetPrecision[UnitConvert[ $\left(\frac{c^2 z^2 m_e \alpha^2}{2 n^2}\right)$ , "Electronvolts"], 6], {n, 1, 5}, {z, 5, 5}]
```

```
z = 5
```

$$\left(\frac{z^2 \alpha \gamma}{\left(-\frac{2z-1}{2} \alpha \delta + z^2 \alpha \gamma\right) 2} \right)^2 \sqrt{\frac{1 + \frac{\pi}{3}}{2}} \sqrt{\frac{3}{2}} 1.0$$

217 - 54 - 24 - 13 - 8.0

4 π

```
Clear[n, z]
```

```
Out[*]=
```

$$\frac{4580703784999263461548761 \sqrt{3 \left(1 + \frac{\pi}{3}\right)} \pi z^2}{2 \left(986022343750000000000000 (1 - 2z) + 4580703784999263461548761 \pi z^2\right)}$$

```
Out[*]=
```

```
{ { 217.691 eV }, { 54.423 eV }, { 24.1879 eV }, { 13.6057 eV }, { 8.7076 eV } }
```

```
Out[*]=
```

```
{ { 340.142 eV }, { 85.036 eV }, { 37.794 eV }, { 21.2589 eV }, { 13.6057 eV } }
```

```
Out[*]=
```

```
5
```

```
Out[*]=
```

```
-0.0523573
```

```
Out[*]=
```

```
9.39014
```

```
In[*]:= Grid[Table[SetPrecision[Ezn[me, n, z], 6], {n, 1, 4}, {z, 2, 4}]]
```

$$\text{Table}\left[\text{SetPrecision}\left[\frac{\text{UnitConvert}\left[\frac{c^2 (z-1)^2 m_e \alpha^2}{2 (n)^2}, \text{"Electronvolts"}\right]}{\text{UnitConvert}\left[\frac{c^2 (z+1)^2 m_e \alpha^2}{2 (z)^2}, \text{"Electronvolts"}\right]}, 6\right], \{n, 1, 1\}, \{z, 4, 4\}\right]$$

$$\text{Table}\left[\left(\frac{z^2 \alpha \gamma}{\left(-\frac{2z-1}{2} \alpha \delta + z^2 \alpha \gamma\right) 2}\right)^2 \sqrt{\frac{1 + \frac{\pi}{3}}{2}} \sqrt{\frac{3}{2}}\right]$$

$$\text{SetPrecision}\left[\text{UnitConvert}\left[\frac{c^2 z^2 m_e \alpha^2}{2 n^2}, \text{"Electronvolts"}\right], 6\right], \{n, 1, 5\}, \{z, 1, 5\}\right]$$

1.7

9.323

Clear[n, z]

Out[*]=

```
54.423 eV  122.451 eV  217.691 eV
13.6057 eV  30.6128 eV  54.423 eV
6.0470 eV  13.6057 eV  24.1879 eV
3.40142 eV  7.6532 eV  13.6057 eV
```

Out[*]=

```
{ {5.76000} }
```

Out[*]=

```
{ { -0.249696 eV , -1.33832 eV , -4.0936 eV , -9.3090 eV , -17.8089 eV },
  { -0.062424 eV , -0.334580 eV , -1.02339 eV , -2.32725 eV , -4.4522 eV },
  { -0.0277440 eV , -0.148702 eV , -0.45484 eV , -1.03433 eV , -1.97877 eV },
  { -0.0156060 eV , -0.083645 eV , -0.255849 eV , -0.58181 eV , -1.11306 eV },
  { -0.0099878 eV , -0.053533 eV , -0.163743 eV , -0.37236 eV , -0.71236 eV } }
```

Out[*]=

0.182345

```
In[*]:= Rationalize[1.7778, 0]
```

Out[*]=

$\frac{16}{9}$

```
In[*]:= Grid[Table[SetPrecision[Ezn[me, n, z], 6], {n, 1, 5}, {z, 1, 10}]]
```

```
Out[*]=
```

13.6057 eV	54.423 eV	122.451 eV	217.691 eV	340.142 eV	489.80 eV	666.68 eV	870.76 eV	1102.06 eV	1360.57 eV
3.40142 eV	13.6057 eV	30.6128 eV	54.423 eV	85.036 eV	122.451 eV	166.670 eV	217.691 eV	275.515 eV	340.142 eV
1.51174 eV	6.0470 eV	13.6057 eV	24.1879 eV	37.794 eV	54.423 eV	74.075 eV	96.752 eV	122.451 eV	151.174 eV
0.85036 eV	3.40142 eV	7.6532 eV	13.6057 eV	21.2589 eV	30.6128 eV	41.667 eV	54.423 eV	68.879 eV	85.036 eV
0.54423 eV	2.17691 eV	4.8980 eV	8.7076 eV	13.6057 eV	19.5922 eV	26.6672 eV	34.831 eV	44.082 eV	54.423 eV

```
In[*]:= Grid[%1332]
```

```
Out[*]=
```

13.605693126591\	54.422772506367\	122.45123813932\	217.69109002547\	340.14232816479\
9685631286154\	8742525144618\	7717068157539\	1497010057847\	9214078215386\
5257513827577\	1030055310308\	0731762444819\	2412022124123\	3143784568942\
0740716372812\	2962865491249\	3666644735531\	3185146196499\	6851790932030\
39736481113 eV	5894592445 eV	15762833002 eV	83578369781 eV	99341202784 eV
3.4014232816479\	13.605693126591\	30.612809534831\	54.422772506367\	85.035582041199\
9214078215386\	9685631286154\	9292670393847\	8742525144618\	8035195538465\
3143784568942\	5257513827577\	6829406112048\	1030055310308\	7859461422356\
6851790932030\	0740716372812\	4166611838827\	2962865491249\	7129477330077\
99341202784 eV	39736481113 eV	89407082505 eV	5894592445 eV	4835300696 eV
1.5117436807324\	6.0469747229297\	13.605693126591\	24.187898891719\	37.793592018311\
4095145873505\	6380583494020\	9685631286154\	0552233397608\	0237864683762\
0286126475085\	1144505900342\	5257513827577\	0457802360137\	5715316187714\
6378573747569\	5514294990277\	0740716372812\	0205717996110\	0946434368923\
33040534570 eV	3216213828 eV	39736481113 eV	92864855313 eV	3260133643 eV
0.8503558204119\	3.4014232816479\	7.6532023837079\	13.605693126591\	21.258895510299\
9803519553846\	9214078215386\	8231675984619\	9685631286154\	9508798884616\
5785946142235\	3143784568942\	2073515280121\	5257513827577\	4464865355589\
6712947733007\	6851790932030\	0416529597069\	0740716372812\	1782369332519\
74835300696 eV	99341202784 eV	7351770626 eV	39736481113 eV	37088251740 eV
0.5442277250636\	2.1769109002547\	4.8980495255731\	8.7076436010188\	13.605693126591\
7874252514461\	1497010057847\	0868272630156\	5988040231388\	9685631286154\
8103005531030\	2412022124123\	2927049779277\	9648088496493\	5257513827577\
8296286549124\	3185146196499\	4666578942124\	2740584785999\	0740716372812\
95894592445 eV	83578369781 eV	6305133201 eV	3431347913 eV	39736481113 eV

Out[•]=

$$\ln[\bullet] := \mathbf{z} = 3.0$$

```
Clear[z, n]
```

Out[•]=

3.

Out[•]=

2

Out[•]=

$$2.92664 \times 10^{-32} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$Out[\bullet]=$$

30.6128 eV

$$\frac{z \propto m e 1.0}{\left(\frac{2 \sqrt{\frac{\pi}{3}}}{3} - \frac{\alpha}{z^2} \right)}$$

Out[•]=

$$\frac{2 \sqrt{\frac{\pi}{3}}}{3} - \frac{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000\,z^2} = 0.170392\,z$$

Out[•]=

$$\frac{z \left(6.64744 \times 10^{-33} \sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (\text{m} \sqrt{\text{Hz}}) \right)}{\frac{2}{3} \frac{\sqrt{\pi}}{z^2} - \frac{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000\,000\,z^2}}$$

$$\ln[\bullet] := \frac{z \propto me \, 1.0}{3.901252198918781 \cdot 10^{-32} \text{ kg}}$$

$$\left(\frac{2.0 \sqrt{\pi}}{\sqrt{3}} / \frac{801\,088\,317}{267\,029\,439} \right)$$

$$\left(\frac{1.0 \sqrt{\pi}}{\sqrt{3}} / \frac{801\,088\,317}{267\,029\,439} \right)$$

$$\left(\frac{2 \sqrt{\pi}}{\sqrt{3}} \middle/ \frac{801\,088\,317}{267\,029\,439} \right) - \frac{\alpha}{4^2} \mathbf{1.0}$$

$$\frac{2\sqrt{\frac{\pi}{3}}}{3} - \frac{\alpha}{z^2}$$

`Out[•]=`

0.170392 z

$Out[\bullet]=$ 0.682218

$Out[\bullet]=$

0.341109

$Out[\bullet]=$

0.681762

$$\frac{2 \sqrt{\frac{\pi}{3}}}{3} - \frac{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000\,z^2}$$

$$\frac{\text{me}}{\text{mP}} = \frac{\text{mP}}{23\,892\,177\,732\,494\,625\,341\,424}$$

$$\text{Out}[\ast]=\frac{1}{23\,892\,177\,732\,494\,625\,341\,423}$$

$$\frac{\sqrt{\frac{135\,132\,371\,234\,621}{3}}}{2\,275\,445\,498\,332\,821\,461\,088\,000\,000\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$In[\bullet] := \text{QuantityMagnitude}[G]$$

$$\frac{\pi^{3/4}}{25\,000\,000\,000\,\sqrt{2}}$$

```
In[ ]:= SetPrecision[G, 7]
SetPrecision[ $\frac{\text{Abs}[\text{QuantityMagnitude}[G] - 6.67430 \times 10^{-11}]}{6.67430 \times 10^{-11}}$ , 7]
ScientificForm[SetPrecision[Abs[ $\frac{\text{Abs}[\text{QuantityMagnitude}[G] - 6.67430 \times 10^{-11}]}{6.67430 \times 10^{-11}} - 2.2 \times 10^{-5}$ ], 3]]
```

```
Out[ ]:=
 $6.67433 \times 10^{-11} \text{ m}^3 / (\text{kg s}^2)$ 
```

```
Out[ ]:=
 $3.855361 \times 10^{-6}$ 
```

```
Out[ ]//ScientificForm=
 $1.81 \times 10^{-5}$ 
```

```
In[ ]:= M =  $\frac{\text{mP}}{23\,892\,177\,732\,494\,625\,341\,300}$  1.0
```

```
SetPrecision[ $\frac{\alpha^2 \text{M c}}{2 \text{h}}$ , 13]
```

```
M = me
```

```
SetPrecision[ $\frac{\alpha^2 \text{me c}}{2 \text{h}}$ , 13]
```

```
Abs[ $\frac{\alpha^2 \text{me c}}{2 \text{h}} - \text{Quantity}[10\,973\,731.568157, \text{"Meters"}^{-1}]$ ]
```

```
SetPrecision[Abs[ $\frac{\text{Abs}[\frac{\alpha^2 \text{me c}}{2 \text{h}} - \text{Quantity}[10\,973\,731.568157, \text{"Meters"}^{-1}]]}{\text{Quantity}[10\,973\,731.568157, \text{"Meters"}^{-1}]} - (1.1 \times 10^{-12})$ ], 3]
```

```
Out[ ]:=
 $9.10938 \times 10^{-31} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$ 
```

```
Out[ ]:=
 $1.097373157106 \times 10^7 \sqrt{\text{kg}} \sqrt{\text{Hz}} / (\sqrt{\text{s}} \sqrt{\text{J}})$ 
```

```
Out[ ]:=
 $\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,846\,000\,000\,000\,000\,000} \times 2^{1/4} \pi^{7/8} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$ 
```

```
Out[ ]:=
 $1.097373157106 \times 10^7 \sqrt{\text{kg}} \sqrt{\text{Hz}} / (\sqrt{\text{s}} \sqrt{\text{J}})$ 
```

```
Out[ ]:=
 $2.64702 \times 10^{-10}$ 
```

```
Out[ ]:=
 $2.64 \times 10^{-10}$ 
```


0.0428267

12 799.1

0.68157

$$\frac{267\,029\,439 \sqrt{\frac{64\,241\,241 \pi}{12\,621\,086}}}{12\,500\,000\,000}$$

267 029 439

0.682218

2.04665

$$5.34104 \times 10^{-30} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$
$$7.4575 \times 10^{-30} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$
$$\begin{aligned} \ln[\circ] &:= \frac{\Gamma}{\frac{\alpha}{c^2} \text{Ezn}[\text{me}, 1, 4]} \\ \text{Ezn}\left[\frac{\alpha \text{me}^2 \lambda}{\Gamma 1.0}, 1, 4\right] \\ \text{Ezn}\left[\frac{\text{me}}{10.0}, 10, 10\right] \\ k &= \text{SetPrecision}\left[\sqrt[4]{1 - \frac{4 \text{G me}}{c^2 \lambda}}, 64\right] \end{aligned}$$
$$\frac{29\,036\,165\,461\,121\,347\,684\,455\,329\,996\,594\,547\,332\,763\,671\,875\,000 \sqrt{\frac{6\,310\,543}{64\,241\,241}} 2^{1/4}}{1\,048\,416\,554\,617\,577\,827\,617\,156\,389\,887\,486\,562\,862\,641\,199\,945\,620\,952\,903 \pi^{17/8}} \text{ m/ (Hz s)}$$

510 999. eV

1.36057 eV

[illegible]
$$\frac{me \text{ } mp}{me + mp}$$
$$9.06004 \times 10^{-31} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

```
Table[SetPrecision[Ezn[me, n, 1] - Ezn[me, n, 2], 64], {n, 2, 2}]
```

```
In[ ]:= En[m_, n_] := UnitConvert[- $\frac{\alpha^2}{2 n^2} c^2 m$ , "Electronvolts"];
```

```
Ezn[m_, n_, Z_] := UnitConvert[- $\frac{Z^2 \alpha^2}{2 n^2} c^2 m$ , "Electronvolts"];
```

```
In[ ]:= SetPrecision[Ezn[me, 2, 1] - Ezn[me, 1, 1], 32]
```

```
Out[ ]:=
```

```
10.204269844943976422346034492749 eV
```

```
In[ ]:= 
$$\frac{\text{Abs}\left[\frac{-4 \pi^2 - 2 \pi \alpha + \alpha^2}{4 \pi^2} - (-1.00115965218046)\right]}{1.00115965218046}$$

```

```
Out[ ]:=
```

```
4.08206  $\times 10^{-7}$ 
```

```
In[ ]:= SetPrecision[ $\left(\left(\frac{\alpha}{2 \pi} - 1\right) \frac{\alpha}{2 \pi} - 1.000005\right)$ , 64]
```

```
Out[ ]:=
```

```
-1.00116506085968648420703175361268222332000732421875000000000000
```

```

$$\frac{\alpha}{2 \pi}$$

```

```
In[ ]:= 
$$\left(\frac{\alpha}{2 \pi} - 1\right) \frac{\alpha 1.0}{2 \pi} - 1 == \frac{-4 \pi^2 - 2 \pi \alpha + \alpha^2}{4 \pi^2}$$

```

```
SetPrecision[ $\frac{-4 \pi^2 - 2 \pi \alpha + \alpha^2}{4 \pi^2}$ , 64]
```

```
SetPrecision[ $\left(\frac{\alpha}{2 \pi} - 1\right) \frac{\alpha}{2 \pi} - 1$ , 64]
```

```
e 1.0
```

```
Out[ ]:=
```

```
True
```

```
Out[ ]:=
```

```
-1.001160060859686476748789721591264765695231148761543517076005171
```

```
Out[ ]:=
```

```
-1.001160060859686476748789721591264765695231148761543517076005171
```

```
Out[ ]:=
```

```
1.60218  $\times 10^{-19}$  C
```

In[]:= $\Omega == 100 \times 2^{1/4} \pi^{1/8} \text{ kg s / (m C)}$

$$\frac{\Omega}{100 \sqrt[4]{4 \pi}}$$

Out[]:=

True

Out[]:=

1 kg s / (m C)

In[]:= $\text{SetPrecision}\left[\left(\frac{\alpha}{2 \pi} - 1\right) \frac{\alpha}{2 \pi} - 1, 64\right] / 1838.281$

$$\frac{1 - \frac{\alpha}{2 \pi} 1.0}{-0.0005446175317373604} \cdot \frac{\alpha}{2 \pi} 1.0$$

Out[]:=

-0.000544618

Out[]:=

-1834.02

Out[]:=

0.998839

$$\frac{\frac{\alpha}{2 \pi} \Lambda \alpha[1] - \frac{\alpha}{2 \pi} 1.0}{\text{SetPrecision}\left[\left(\frac{\alpha}{2 \pi} - 1\right) \frac{\alpha}{2 \pi} - 1, 64\right]} - 1838.281$$

$$\frac{\text{SetPrecision}\left[\left(\frac{\alpha}{2 \pi} - 1\right) \frac{\alpha}{2 \pi} - 1, 64\right]}{1.0 - \frac{\alpha}{2 \pi}}$$

$$\frac{1}{\alpha}$$

In[]:= $\frac{1838.281}{\frac{1}{\alpha \pi^2}} \alpha$

$$\frac{\frac{\alpha}{\frac{1}{\alpha \pi^2}} 1.0}{\frac{1}{\alpha \pi^2}}$$

$$\frac{1}{\frac{\alpha}{\frac{1}{\alpha \pi^2}}} 1.0 - \frac{1}{2 \alpha}$$

Out[]:=

0.966145

Out[]:=

0.00052557

Out[]:=

1834.18

$$\left(\left(\frac{1}{\pi^2 \alpha^2} - \frac{1}{2\alpha} \right) 1.0 \right)$$

Out[•]=

1838.18

$$ln[\bullet] := \frac{2 - \pi^2 \alpha}{2 \pi^2 \alpha^2}$$

Out[•]=

$$\frac{1\,944\,480\,124\,748\,486\,328\,125\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000}{20\,982\,847\,165\,906\,578\,496\,057\,223\,492\,009\,660\,153\,544\,780\,635\,121}\pi^4 \left(2 - \frac{4\,580\,703\,784\,999\,263\,461\,548\,761}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000}\pi^3\right)$$

$$\begin{aligned} \ln[\otimes] &:= \frac{1}{\pi^2 \alpha^2} - \frac{1}{2 \alpha} = \frac{2 - \pi^2 \alpha}{2 \pi^2 \alpha^2} \\ 1. \theta \left(\frac{1}{\pi^2 \alpha^2} - \frac{1}{2 \alpha} \right) &+ \left(3 + \left(\frac{\pi}{3} \right) \right) \\ \frac{2 - \pi^2 \alpha}{2 \pi^2 \alpha^2} 1. \theta & \end{aligned}$$

Out[•]=

True

Out[•]=

1838.23

Out[•]=

1834.18

$$3 + \frac{\pi}{3} + \frac{1}{\pi^2 \alpha^2} - \frac{1}{2\alpha}$$

Out[•]=

1838.23

$$\ln[\bullet] := \mathbf{m} = \left(\mathbf{1} - \frac{\mathbf{1}}{\frac{\mathbf{me} + \mathbf{mp}}{\mathbf{me}} + \mathbf{1}} \right) \mathbf{me}$$

$$\ln[\bullet] := \Lambda\alpha[2.0] \ 4$$
 $\Delta\alpha[1.0] \ 2$

Out[•]=

0.999089

Out[•]=

0.996365

$$\text{SetPrecision}\left[\left(\frac{\alpha}{2\pi} - 1\right) \frac{\alpha}{2\pi} - 1, 32\right]$$

Out[•]=

-1.0000004082058497889562431737431

```
In[*]:= SetPrecision[ $\frac{\alpha}{2\pi} 1.00115965218046$ , 32]

SetPrecision[ $\frac{\frac{\alpha}{2\pi} - 1 - \Lambda\alpha[1] - \Lambda\alpha[1]}{2} - 1$ , 32]

 $\frac{\alpha}{2\pi} \Lambda\alpha[1] - \frac{\alpha}{2\pi} 1.0$ 
 $\frac{\alpha}{2\pi} \Lambda\alpha[1] 1.0$ 

Solve[ $\frac{\alpha}{2\pi} - X == -1.00115965218046$ , X]

 $\left(\frac{\alpha}{2\pi} \Lambda\alpha[1] - \frac{\alpha}{2\pi}\right)$ 
1.00115965218046
128  $\Lambda\alpha[64] 1.0$ 
```

```
Out[*]=
0.0011627565635810622763496313680776
```

```
Out[*]=
-1.9976015892130295933442034404389
```

```
Out[*]=
-0.000582816
```

```
Out[*]=
0.000578594
```

```
Out[*]=
{ {X  $\rightarrow$  1.00232} }
```

```
Out[*]=
-0.000582141
```

```
Out[*]=
0.999999
```

```
In[*]:= c
```

```
Out[*]=
299 792 458 m/s
```

```

me
In[ ]:= -----
mp

                mP
                ----- 1.0
23 892 177 732 494 625 341 422
2.3892211741104968`*^22 - 23 892 177 732 494 625 341 423
23 892 177 732 494 625 341 423 / 3.400861034217472`*^16
Λα[2.3892211741104968`*^22] 1.0
23 892 177 732 494 625 341 423 - 2.3892211741104968`*^22
Λα[23 892 177 732 494 625 341 423] 1.0 × 2.3892211741104968`*^22
Solve[Λα[23 892 177 732 494 625 341 423] X ==  $\frac{1}{2} \times 1.0$ , X]

Solve[Λα[ $\frac{mP}{me}$ ] X ==  $\frac{1}{2} \times 1.0$ , X]
Solve[Λα[ $\frac{mP}{mp}$ ] X ==  $\frac{1}{2} \times 1.0$ , X]

Λα[ $\frac{mP}{me}$ ] 1.0
Λα[ $\frac{mP}{mp}$ ] 1.0
2.3892211741104968`*^22
Out[ ]=
          1
          -----
23 892 177 732 494 625 341 423

Out[ ]=
9.10938 × 10-31  $\sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$ 

Out[ ]=
3.40086 × 1016

Out[ ]=
702 533.

Out[ ]=
2.09273 × 10-23

Out[ ]=
-3.40086 × 1016

Out[ ]=
0.500001

Out[ ]=
{ {X → 2.38922 × 1022 } }

Out[ ]=
{ {X → 2.38922 × 1022 } }

Out[ ]=
{ {X → 1.30121 × 1019 } }

Out[ ]=
2.09274 × 10-23

```

$$\text{Out}[\ast]= 3.84258 \times 10^{-20}$$

$$\text{Out}[\ast]= 2.38922 \times 10^{22}$$

$$\text{In}[\ast]:= \text{Solve}\left[\left(\text{SetPrecision}\left[\text{Ezn}\left[\text{mP}, 2, 1\right] \frac{1}{X}, 64\right] - \text{SetPrecision}\left[\text{Ezn}\left[\frac{\text{mP}}{X}, 2, 2\right], 64\right]\right) == \text{Quantity}[10.20425532, \text{"Electronvolts"}], X\right]$$

... Solve: Solve was unable to determine the units of quantities that appear in the input.

$$\text{Out}[\ast]= \text{Solve}\left[\frac{-8.12674095885789522405700875672448666033729689405809941345637295 \times 10^{22} \text{ eV}}{X} - \text{UnitConvert}\left[\frac{-52081.8978994115002176173364558294516113484465695226412747556237 \sqrt{\text{kg m}} \sqrt{\text{J}} / (\sqrt{\text{Hz}} \text{ s}^{3/2})}{X}, \text{Electronvolts}\right] == 10.2043 \text{ eV}, X\right]$$

$$\text{In}[\ast]:= \frac{\text{e } 3 \text{ } 343 \text{ } 006 \text{ } 498 \text{ } 094}{1} \frac{11 \text{ } 259 \text{ } 524 \text{ } 195 \text{ } 272}{\text{mP}} 1.0$$

$$\text{Out}[\ast]= \frac{1 \text{ } 339 \text{ } 021 \text{ } 724 \text{ } 639 \text{ } 093 \text{ } 083 \text{ } 899}{2 \text{ } 500 \text{ } 000 \text{ } 000 \text{ } 000 \text{ } 000 \text{ } 000 \text{ } 000 \text{ } 000} \text{ C}$$

$$\text{Out}[\ast]= 5.17339 \times 10^{20} \text{ m} \sqrt{\text{Hz}} / (\sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}})$$

$$\text{In}[\ast]:= 0.007407071436459126 \alpha$$

$$\text{Out}[\ast]= 0.000054052$$

$$\text{In}[\ast]:= \frac{0.00005325147164796948}{\alpha^2} \alpha^2 1.0$$

$$\text{Out}[\ast]= 1.$$

$$\text{Out}[\ast]= 0.0000532514$$

$$\text{In}[\ast]:= \text{Log}\left[\frac{c^2 \Gamma}{G \text{ mP mP}}\right]$$

$$\text{Out}[\ast]= \text{Log}\left[\frac{4 \text{ } 580 \text{ } 703 \text{ } 784 \text{ } 999 \text{ } 263 \text{ } 461 \text{ } 548 \text{ } 761 \pi}{1 \text{ } 972 \text{ } 044 \text{ } 687 \text{ } 500 \text{ } 000 \text{ } 000 \text{ } 000 \text{ } 000 \text{ } 000}\right]$$

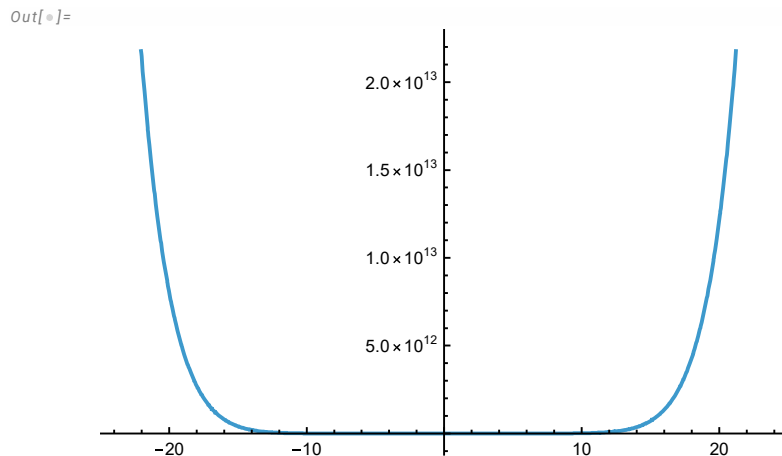
In[]:= **RootApproximant** $\left[\text{Log} \left[\frac{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000} \right] \right]$

$$13 + 2 \#1 + 6 \#1^2 + 3 \#1^4 + 5 \#1^5 + 7 \#1^6 - 6 \#1^7 - 6 \#1^8 + 4 \#1^9 + \#1^{10} \&$$

Out[]:= $10 \sqrt{-4.92\dots}$

Out[]:= $13 + 2 \#1 + 6 \#1^2 + 3 \#1^4 + 5 \#1^5 + 7 \#1^6 - 6 \#1^7 - 6 \#1^8 + 4 \#1^9 + \#1^{10} \&$

In[]:= **Plot** $\left[\left(13 + 2 \#1 + 6 \#1^2 + 3 \#1^4 + 5 \#1^5 + 7 \#1^6 - 6 \#1^7 - 6 \#1^8 + 4 \#1^9 + \#1^{10} \& \right) [x], \{x, -24., 24.\} \right]$



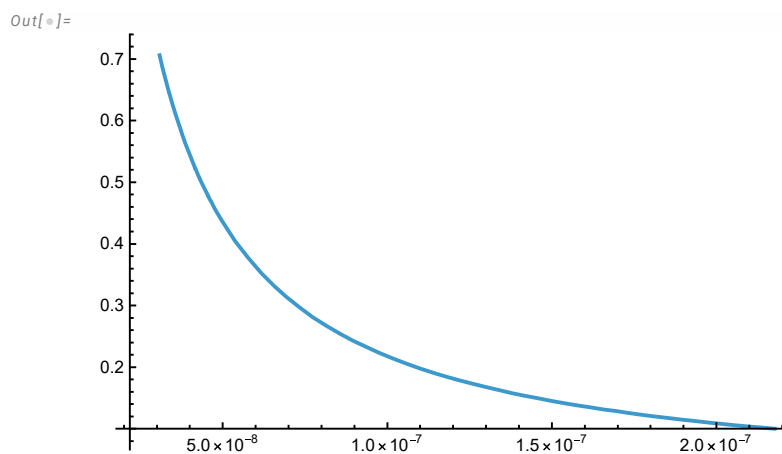
In[]:= **RootApproximant** $[-\text{Log}[2]]$

Out[]:= $19 \sqrt{-0.693\dots}$

In[]:= **RootApproximant** $[\text{Log}[23\,892\,177\,732\,494\,625\,341\,423]]$

Out[]:= $3 \sqrt{51.5\dots}$

In[]:= **Plot** $\left[\frac{c^2 \Gamma}{G m P \alpha}, \{m, 1 \text{ mP}, 10 \text{ mP}\} \right]$



```
In[ ]:= 
$$\frac{c^2 \Gamma}{G \text{ me mP } \alpha} == \frac{\text{mP}}{\text{me}}$$

Clear[m]
Table[
$$\frac{c^2 \Gamma}{G \text{ m mP } \alpha}, \{m, \text{Quantity}[1, \text{"Kilograms"}], \text{Quantity}[10, \text{"Kilograms"}]\}]$$

```

Out[]=

True

Out[]=

$$\left\{ \frac{7 \sqrt{405397113703863}}{2000000000000000 \times 2^{1/4} \pi^{7/8}}, \frac{7 \sqrt{405397113703863}}{4000000000000000 \times 2^{1/4} \pi^{7/8}}, \frac{7 \sqrt{\frac{135132371234621}{3}}}{2000000000000000 \times 2^{1/4} \pi^{7/8}}, \frac{7 \sqrt{405397113703863}}{8000000000000000 \times 2^{1/4} \pi^{7/8}}, \frac{7 \sqrt{405397113703863}}{10000000000000000 \times 2^{1/4} \pi^{7/8}}, \right.$$

$$\frac{7 \sqrt{\frac{135132371234621}{3}}}{4000000000000000 \times 2^{1/4} \pi^{7/8}}, \frac{\sqrt{405397113703863}}{2000000000000000 \times 2^{1/4} \pi^{7/8}}, \frac{7 \sqrt{405397113703863}}{16000000000000000 \times 2^{1/4} \pi^{7/8}},$$

$$\left. \frac{7 \sqrt{\frac{135132371234621}{3}}}{6000000000000000 \times 2^{1/4} \pi^{7/8}}, \frac{7 \sqrt{405397113703863}}{2000000000000000 \times 2^{1/4} \pi^{7/8}} \right\}$$

```
In[ ]:= 
$$\frac{\text{me}}{\Gamma 502963.2282568433} \text{ mP } \frac{G}{c^2}$$


$$\frac{1}{\frac{\text{me}}{\Gamma} \alpha \text{ mP } \frac{G}{c^2}}$$


$$\frac{\text{me}}{\Gamma} \alpha \text{ mP } \frac{G}{c^2} 1.0$$


$$\frac{1}{\frac{\text{me}}{\Gamma} \alpha \text{ mP } \frac{G}{c^2}}$$

23892177732494625341423
```

Out[]=

$$1.14036 \times 10^{-26}$$

Out[]=

23892177732494625341423

Out[]=

$$4.18547 \times 10^{-23}$$

Out[]=

23892177732494625341423

Out[]=

23892177732494625341423

```
In[*]:= n = 1836.15;
      mP ( 1P
      Γ ( Δα[n] - 2 G mP ) 1.0
           c²
```

```
Out[*]=
502963.
```

```
In[*]:= FactorInteger[20549487]
```

```
Out[*]=
{{3, 1}, {7, 1}, {29, 1}, {41, 1}, {823, 1}}
```

```
In[*]:= FactorInteger[15402726]
```

```
Out[*]=
{{2, 1}, {3, 2}, {149, 1}, {5743, 1}}
```

```
In[*]:= FactorInteger[275]
```

```
Out[*]=
{{5, 2}, {11, 1}}
```

```
In[*]:= UnitSimplify[Quantity[2.31184×10-18,
      "Joules" "Seconds"
      "Hertz" "Meters"]]
```

```
Out[*]=
2.31184×10-18 kg m
```

```
In[*]:= SetPrecision[(Ezn[ me , 2, 1] - Ezn[ me , 2, 2] ), 32] - Quantity[
      α²
      √(27/2), "Electronvolts"]
```

```
10.2042698449439764223532951363498102054957874365308768309934`31.54136215097435 -
      α²
      √(27/2)
```

```
Out[*]=
10.204255351761026991441378142195 eV
```

```
Out[*]=
10.204255351761026991441378142195
```

```
In[ ]:= Solve[Ezn[ me , 2, 1] X Δα[X] - Ezn[ me , 2, 2] X Δα[X] ==
Quantity[10.20425532, "Electronvolts"], X]

Solve[Ezn[ me , 2, 1]  $\frac{\Gamma}{\Delta\alpha[X] \text{ 1P mP}}$  - Ezn[ me , 2, 2]  $\frac{\Gamma}{\Delta\alpha[X] \text{ 1P mP}}$  ==
Quantity[10.20425532, "Electronvolts"], X]
```

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

```
Out[ ]:=
{{X -> 0. - 0.0854246 I}, {X -> 0. + 0.0854246 I}}
```

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

```
Out[ ]:=
{{X -> 0.0000532515}, {X -> 68.5178}}
```

```
In[ ]:= 0. - 0.0854246 I
```

```
Out[ ]:=
0. - 0.0854246 I
```

```
In[ ]:=  convert to exponential 0. - 0.0854246 I
```

```
Out[ ]:=
0.0854246 ei (-1.5708)
```

```
In[ ]:= Abs[0. - 0.0854246 I]
```

```
Out[ ]:=
0.0854246
```

```

SetPrecision[
  
$$\frac{\text{Ezn[ me , 2, 1]}}{\sqrt{23\,892\,177\,732\,494\,625\,341\,423} \times 23\,892\,177\,732\,494\,625\,341\,387 \text{ me}} - \frac{\text{Ezn[ me , 2, 2]}}{\sqrt{23\,892\,177\,732\,494\,625\,341\,423} \times 23\,892\,177\,732\,494\,625\,341\,387 \text{ me}},$$

  64] == Quantity[10.20425532, "Electronvolts"]
Solve[ $\frac{\text{Ezn[ me , 2, 1] - Ezn[ me , 2, 2]}}{X} == \text{Quantity[10.20425532, "Electronvolts"]}, X]$ 
-Quantity[10.20425532, "Electronvolts"]
1.000001423420281
121 538 416 - 121 538 243
mP
me

$$\sqrt{23\,892\,177\,732\,494\,625\,341\,423} \times 23\,892\,177\,732\,494\,625\,341\,387 \text{ me}$$

mP

```

Out[8]=

False

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numerizing the result.

Out[8]=

{ {X → 1.} }

Out[8]=

-10.2043 eV

Out[8]=

1.

Out[8]=

173

Out[8]=

23 892 177 732 494 625 341 423

Out[8]=

$\sqrt{\frac{23\,892\,177\,732\,494\,625\,341\,387}{23\,892\,177\,732\,494\,625\,341\,423}}$

In[8]:= **FactorInteger[162]**

Out[8]=

{ {2, 1}, {3, 4} }

In[8]:= **23 892 177 732 494 625 341 423**

In[8]:= **23 892 177 732 494 625 341 422**

Out[8]=

23 892 177 732 494 625 341 422

In[8]:= **NextPrime[23 892 177 732 494 625 341 600, -1]**

Out[8]=

23 892 177 732 494 625 341 549

```
In[ ]:= NumberForm[1., 16]
```

```
Out[ ]//NumberForm=
1.000001423420281
```

```
In[ ]:= Rationalize[1., 0]
```

```
Out[ ]:=
121 538 416
-----
121 538 243
```

```
In[ ]:= Solve[ ( (Ezn[ me , 2, 1] - Ezn[ me , 2, 2] ) / x ) == Quantity[10.20425532, "Electronvolts"], x]
```

$$\frac{1}{2 \Delta \alpha \left[\frac{m_P}{m_e} \right]} X$$

$$\frac{1}{2 \Delta \alpha \left[\frac{m_P}{m_e} \right]} X$$

$$\frac{1}{2 \Delta \alpha \left[\frac{m_P}{m_e} \right]} X$$

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

```
Out[ ]:=
{ {X -> 1.} }
```

```
Out[ ]:=
( 2 251 428 820 905 099 894 277 953 422 003 736 982 349 360 175 029 029 375 000 000 000 000 000 000 +
  4 580 703 784 999 263 461 548 761 π ) / 94 232 884 340 343 644 053 712 201 680 625 000 000 000 000 000 000
```

```
Out[ ]:=
( ( 2 251 428 820 905 099 894 277 953 422 003 736 982 349 360 175 029 029 375 000 000 000 000 000 000 +
  4 580 703 784 999 263 461 548 761 π ) X ) /
94 232 884 340 343 644 053 712 201 680 625 000 000 000 000 000 000
```

```
Out[ ]:=
( ( 2 251 428 820 905 099 894 277 953 422 003 736 982 349 360 175 029 029 375 000 000 000 000 000 000 +
  4 580 703 784 999 263 461 548 761 π ) X ) /
94 232 884 340 343 644 053 712 201 680 625 000 000 000 000 000 000
```

$$\text{In}[*]:= \text{m} = \frac{\text{mP}}{23\,892\,177\,732\,494\,625\,341\,423 - \text{X}}$$

$$\left(\frac{\text{me mp}}{\text{me} + \text{mp}} \right)$$

$$\Delta\alpha[0.999998576581745`]$$

$$\Delta\alpha[0.49818299804601934`]$$

$$\Delta\alpha\left[\frac{1.0}{299\,792\,458}\right]$$

$$\sqrt[8]{\Delta\alpha\left[\frac{1.0}{2} - \alpha\right]}$$

$$\frac{\Delta\alpha\left[\frac{1.0}{2} - \alpha\right]^2 + \Delta\alpha\left[\frac{1.0}{2} - \alpha\right]^2}{2}$$

$$\text{m} \frac{\Delta\alpha\left[\frac{1.0}{2} - \alpha\right]^2 + \Delta\alpha\left[\frac{1.0}{2} - \alpha\right]^2}{2}$$

$$\text{SetPrecision}[\text{Ezn}[\text{m}, 2, 2], 64]$$

$$\text{SetPrecision}[\text{Ezn}[\text{m}, 2, 1], 64]$$

$$\text{SetPrecision}[\text{Ezn}[\text{m}, 2, 2], 64] - \text{SetPrecision}[\text{Ezn}[\text{m}, 2, 1], 64]$$

$$\text{Solve}\left[\left(\text{SetPrecision}\left[\text{Ezn}\left[\frac{\text{mP}}{\text{X}}, 2, 1\right], 64\right] - \text{SetPrecision}\left[\text{Ezn}\left[\frac{\text{mP}}{\text{X}}, 2, 2\right], 64\right]\right) == \text{Quantity}[10.20425532, \text{"Electronvolts"}], \text{X}\right]$$

$$\text{Out}[*]= \frac{\frac{7 \sqrt{405\,397\,113\,703\,863}}{2\,000\,000\,000\,000\,000 \cdot 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})}{23\,892\,177\,732\,494\,625\,341\,423 - \text{X}}$$

$$\text{Out}[*]= \frac{7 \sqrt{\frac{135\,132\,371\,234\,621}{3}}}{15\,936\,793\,212\,778\,806\,497\,260\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$\text{Out}[*]= 0.498183$$

$$\text{Out}[*]= 0.989106$$

$$\text{Out}[*]= 4.57103 \times 10^{-7}$$

$$\text{Out}[*]= 0.999973$$

$$\text{Out}[*]= 0.999568$$

$$\text{Out}[*]= \frac{2.17549 \times 10^{-8} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})}{23\,892\,177\,732\,494\,625\,341\,423 - \text{X}}$$

10.20426984494397642235329513634981020549578743653087683099397050 eV

```
In[ ]:= SetPrecision[ $\frac{\alpha}{2\pi}$ , 16]
```

$$\frac{\alpha}{2\pi}$$

```
Λα[431.25] 1.0
```

$$\left(\frac{\sqrt{\Lambda\alpha[431]^2 + \Lambda\alpha[432]^2}}{\sqrt{2}} \right) 1.0$$

```
Out[ ]=
```

0.001161409732252648

```
Out[ ]=
```

$$\frac{4\,580\,703\,784\,999\,263\,461\,548\,761}{3\,944\,089\,375\,000\,000\,000\,000\,000\,000}$$

```
Out[ ]=
```

0.00115942

```
Out[ ]=
```

0.00115875

```
In[ ]:= SetPrecision[ $\frac{\alpha}{2\pi}$ , 16]
```

$$\frac{\alpha}{2\pi}$$

```
Λα[531] 1.0
```

$$\text{Solve}\left[\Lambda\alpha[X] == \frac{\alpha 1.0}{2\pi}, X\right]$$

```
Out[ ]=
```

0.001161409732252648

```
Out[ ]=
```

0.00094162

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

```
Out[ ]=
```

$$\left\{ \left\{ X \rightarrow 8.47522 \times 10^{-6} \right\}, \left\{ X \rightarrow 430.511 \right\} \right\}$$

```

In[ ]:= (* Rydberg Energy *)
Table[SetPrecision[Ezn[me, n, 1], 64], {n, 1, 5}]
(* Rydberg Frequency *)
Table[SetPrecision[ $\frac{Ezn[me, n, 1]}{h}$ , 64], {n, 1, 5}]
(* Rydberg wavelength *)
Table[SetPrecision[ $\frac{2 \pi \Gamma}{\frac{\alpha}{c^2} Ezn[me, n, 1]}$ , 64], {n, 1, 5}]
(* Rydberg Angular Wavelength *)
Table[SetPrecision[ $\frac{\Gamma}{\frac{\alpha}{c^2} Ezn[me, n, 1]}$ , 64], {n, 1, 5}]
    
```

```

Out[ ]:=
{
  -13.60569312659196856313772684846641360732771658204116910799196067 eV ,
  -3.401423281647992140784431712116603401831929145510292276997990168 eV ,
  -1.511743680732440951459747427607379289703079620226796567554662297 eV ,
  -0.850355820411998035196107928029150850457982286377573069249497542 eV ,
  -0.544227725063678742525509073938656544293108663281646764319678427 eV }
    
```

```

Out[ ]:=
{
  -3.289841961120809456555756458522755998205077664069207774817678628  $\times 10^{15}$  Hz ,
  -8.22460490280202364138939114630688999551269416017301943704419657  $\times 10^{14}$  Hz ,
  -3.65537995680089939617306273169195110911675296007689752757519848  $\times 10^{14}$  Hz ,
  -2.056151225700505910347347786576722498878173540043254859261049142  $\times 10^{14}$  Hz ,
  -1.315936784448323782622302583409102399282031065627683109927071451  $\times 10^{14}$  Hz }
    
```

```

Out[ ]:=
{
  -9.11267050341422258633257676416997343700435694858834583795169500  $\times 10^{-8}$  m/ (Hz s) ,
  -3.64506820136568903453303070566798937480174277943533833518067800  $\times 10^{-7}$  m/ (Hz s) ,
  -8.20140345307280032769931908775297609330392125372951125415652550  $\times 10^{-7}$  m/ (Hz s) ,
  -1.458027280546275613813212282267195749920697111774135334072271199  $\times 10^{-6}$  m/ (Hz s) ,
  -2.278167625853555646583144191042493359251089237147086459487923749  $\times 10^{-6}$  m/ (Hz s) }
    
```

```

Out[ ]:=
{
  -1.450326555386083816779526853319027576743129564531052429370064333  $\times 10^{-8}$  m/ (Hz s) ,
  -5.80130622154433526711810741327611030697251825812420971748025733  $\times 10^{-8}$  m/ (Hz s) ,
  -1.305293899847475435101574167987124819068816608077947186433057900  $\times 10^{-7}$  m/ (Hz s) ,
  -2.320522488617734106847242965310444122789007303249683886992102933  $\times 10^{-7}$  m/ (Hz s) ,
  -3.62581638846520954194881713329756894185782391132763107342516083  $\times 10^{-7}$  m/ (Hz s) }
    
```

```
In[ ]:= n = 1011
```

$$\Gamma / \left(\frac{2 n^2}{\alpha n + 2 n^2} \text{1P} \right) \text{1.0}$$

$$\text{Ezn} \left[\frac{\Gamma / \left(\frac{2 n^2}{\alpha n + 2 n^2} \text{1P} \right)}{\text{mP}} \text{me, 1, 1} \right] \text{1.0}$$

```
Clear[n]
```

```
Out[ ]=
```

1011

```
Out[ ]=
```

$$1.58822 \times 10^{-10} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

```
Out[ ]=
```

0.0992859 eV

```
In[ ]:= 5 / 11.0
```

```
Out[ ]=
```

0.454545

```
In[ ]:= Table[SetPrecision[Ezn[me, n, 4], 64], {n, 1, 5}]
Ezn[me, 4, 4]
```

```
Out[ ]=
```

{ 217.6910900254714970102036295754626177172434653126587057278713707 eV ,
54.4227725063678742525509073938656544293108663281646764319678427 eV ,
24.18789889171905522335595884171806863524927392362874508087459675 eV ,
13.60569312659196856313772684846641360732771658204116910799196067 eV ,
8.70764360101885988040814518301850470868973861250634822911485483 eV }

```
Out[ ]=
```

$$\frac{4\,119\,679\,911\,803\,402\,973\,545\,424\,666\,947\,361\,524\,543\,252\,798\,292\,345\,217\,131 \sqrt{\frac{64\,241\,241}{6\,310\,543}} \pi^{9/8}}{2\,944\,777\,635\,635\,738\,876\,678\,506\,302\,519\,531\,250\,000\,000\,000\,000\,000\,000\,000 \times 2^{1/4}} \text{eV}$$

```
In[ ]:= Table[SetPrecision[Ezn[mp, n, 1], 64], {n, 1, 10}]
```

```
Out[ ]=
```

{ 2498.21298081447340229318729050455963406223422232715880025407274 eV ,
624.553245203618350573296822626139908515558555558178970006351818 eV ,
277.5792200904970446992430322782844037846926913591906533361563638 eV ,
156.1383113009045876433242056565349771288896388895447425015879546 eV ,
99.9285192325789360917274916201823853624893688893086352010162910 eV ,
69.3948050226242611748107580695711009461731728397976633340390909 eV ,
50.9839383839688449447589242960114211033109024945452220413348423 eV ,
39.034577825226146910831051414133744282224097223861856253969887 eV ,
30.84213556561078274436033691980937819829918792879896148179515153 eV ,
24.9821298081447340229318729050455963406223422232715880025407274 eV }

In[]:=

Table[SetPrecision[Ezn[me, 1, z], 64], {z, 1, 103}]

Out[]:=

```
{ 13.60569312659196856313772684846641360732771658204116910799196067 eV ,
  54.4227725063678742525509073938656544293108663281646764319678427 eV ,
  122.4512381393277170682395416361977224659494492383705219719276460 eV ,
  217.6910900254714970102036295754626177172434653126587057278713707 eV ,
  340.1423281647992140784431712116603401831929145510292276997990168 eV ,
  489.804952557310868272958166544790889863797796953482087887710584 eV ,
  666.678963203006459593748615574854266759058112520017286291606073 eV ,
  870.764360101885988040814518301850470868973861250634822911485483 eV ,
  1102.061143253949453614155874725779502193545043145334697747348814 eV ,
  1360.569312659196856313772684846641360732771658204116910799196067 eV ,
  1646.288868317628196139664948664436046486653706426981462067027241 eV ,
  1959.219810229243473091832666179163559455191187813928351550842337 eV ,
  2299.362138394042687170275837390823899638384102364957579250641354 eV ,
  2666.715852812025838374994462299417067036232450080069145166424292 eV ,
  3061.280953483192926705988540904943061648736230959263049298191151 eV ,
  3483.05744040754395216325807320740188347589544500253929164594193 eV ,
  3932.04531358507891474680305920679353251771009220989787220967663 eV ,
  4408.24457301579781445662349890311800877418017258133879098939526 eV ,
  4911.65521869970065129271939229637531224530568611686204798509780 eV ,
  5442.27725063678742525509073938656544293108663281646764319678427 eV ,
  6000.11066882705813634373754017368840083152301268015557662445466 eV ,
  6585.15547327051278455865979465774418594661482570792584826810897 eV ,
  7197.41166396715136989985750283873279827636207189977845812774720 eV ,
  7836.87924091697389236733066471665423782076475125571340620336935 eV ,
  8503.55820411998035196107928029150850457982286377573069249497542 eV ,
  9197.44855357617074868110334956329559855353640945983031700256541 eV ,
  9918.55028928554508252740287253201551974190538830801227972613933 eV ,
  10666.86341124810335349997784919766826814492980032027658066569717 eV ,
  11442.38791946384556159882827956025384376260964549662321982123892 eV ,
  12245.12381393277170682395416361977224659494492383705219719276460 eV ,
  13075.07109465488178917535550137622347664193563534156351278027421 eV ,
  13932.22976163017580865303229282960753390358178001015716658376773 eV ,
  14816.59981485865376525698453797992441837988335784283315860324517 eV ,
  15728.18125434031565898721223682717413007084036883959148883870654 eV ,
  16666.97408007516148984371538937135666897645281300043215729015182 eV ,
  17632.97829206319125782649399561247203509672069032535516395758103 eV ,
  18626.19389030440496293554805555052022843164400081436050884099416 eV ,
```

19 646.62087479880260517087756918550124898122274446744819194039121 eV ,
 20 694.25924554638418453248253651741509674545692128461821325577218 eV ,
 21 769.10900254714970102036295754626177172434653126587057278713707 eV ,
 22 871.17014580109915463451883227204127391789157441120527053448589 eV ,
 24 000.44267530823254537495016069475360332609205072062230649781862 eV ,
 25 156.92659106854987324165694281439875994894796019412168067713528 eV ,
 26 340.62189308205113823463917863097674378645930283170339307243586 eV ,
 27 551.52858134873634035389686814448755483862607863336744368372036 eV ,
 28 789.64665586860547959943001135493119310544828759911383251098878 eV ,
 30 054.97611664165855597123860826230765858692592972894255955424112 eV ,
 31 347.51696366789556946932265886661695128305900502285362481347739 eV ,
 32 667.26919694731652009368216316785907119384751348084702828869757 eV ,
 34 014.23281647992140784431712116603401831929145510292276997990168 eV ,
 35 388.4078222657102327212275328611417926593908298890808498870897 eV ,
 36 789.7942143046829947244133982531823942141456378393212680102617 eV ,
 38 218.3919925968396938538747173421558229835558789536440243494175 eV ,
 39 674.2011571421803301096114901280620789676215532320491189045573 eV ,
 41 157.2217079407049034916237166109011621663426606745365516756810 eV ,
 42 667.4536449924134139999113967906730725797192012811063226627887 eV ,
 44 204.8969682973058616344745306673778102077511750517584318658802 eV ,
 45 769.5516778553822463953131182410153750504385819864928792849557 eV ,
 47 361.4177736666425682824271595115857671077814220853096649200151 eV ,
 48 980.4952557310868272958166544790889863797796953482087887710584 eV ,
 50 626.7841240487150234354816031435250328664334017751902508380857 eV ,
 52 300.2843786195271567014220055048939065677425413662540511210968 eV ,
 54 000.9960194435232270936378615631956074837071141214001896200919 eV ,
 55 728.9190465207032346121291713184301356143271200406286663350709 eV ,
 57 484.0534598510671792568959347705974909596025591239394812660338 eV ,
 59 266.3992594346150610279381519196976735195334313713326344129807 eV ,
 61 075.9564452713468799252558227657306832941197367828081257759115 eV ,
 62 912.7250173612626359488489473086965202833614753583659553548261 eV ,
 64 776.7049757043623290987175255485951844872586470980061231497248 eV ,
 66 667.8963203006459593748615574854266759058112520017286291606073 eV ,
 68 586.299051150113526772810431191909945390192900695334733874737 eV ,
 70 531.9131682527650313059759824498881403868827613014206558303241 eV ,
 72 504.7386716086004729609463754775181134494016656973901764891584 eV ,
 74 504.7755612176198517421922222020809137265760032574420353639766 eV ,
 76 532.0238370798231676497135226235765412184057739815762324547788 eV ,
 78 586.4834991952104206835102767420049959248909778697927677615648 eV ,

```

80 668.1545475637816108435824845573662778460316149220916412843348 eV ,
82 777.0369821855367381299301460696603869818276851384728530230887 eV ,
84 913.1308030604758025425532612788873233322791885189364029778266 eV ,
87 076.4360101885988040814518301850470868973861250634822911485483 eV ,
89 266.9526035699057427466258527881396776771484947721105175352540 eV ,
91 484.6805832043966185380753290881650956715662976448210821379436 eV ,
93 729.6199490920714314558002590851233408806395336816139849566171 eV ,
96 001.7707012329301814998006427790144133043682028824892259912745 eV ,
98 301.1328396269728686700764801698383129427523052474468052419159 eV ,
100 627.7063642741994929666277712575950397957918407764867227085411 eV ,
102 981.4912751746100543894545160422845938634868094696089783911503 eV ,
105 362.4875723282045529385567145239069751458372113268135722897434 eV ,
107 770.6952557349829886139343667024621836428430463481005044043205 eV ,
110 206.1143253949453614155874725779502193545043145334697747348814 eV ,
112 668.7447813080916713435160321503710822808210158829213832814263 eV ,
115 158.5866234744219183977200454197247724217931503964553300439551 eV ,
117 675.6398518939361025781995123860112897774207180740716150224678 eV ,
120 219.9044665666342238849544330492306343477037189157702382169645 eV ,
122 791.3804674925162823179848074093828061326421529215511996274451 eV ,
125 390.0678546715822778772906354664678051322360200914144992539095 eV ,
128 015.9666281038322105628719172204856313464853204253601370963580 eV ,
130 669.0767877892660803747286526714362847753900539233881131547903 eV ,
133 349.3983337278838873128608418193197654189502205854984274292065 eV ,
136 056.9312659196856313772684846641360732771658204116910799196067 eV ,
138 791.6755843646713125679515812058852083500368534019660706259908 eV ,
141 553.6312890628409308849101314445671706375633195563233995483588 eV ,
144 342.7983800141944863281441353801819601397452188747630666867108 eV }
    
```

```
In[ ]:= SetPrecision[ $\alpha$ , 64]
```

```
SetPrecision[ $\frac{\alpha\gamma}{\alpha\delta}$ , 64]
```

```
Out[ ]:=
```

```
0.00729735256530521488043396038932213551321844064341505620784677965
```

```
Out[ ]:=
```

```
0.00729735256530521488043396038932213551321844064341505620784677965
```

```
In[ ]:=
```

```
Table[SetPrecision[Ezn[me, 1, z], 64], {z, 1, 103}]
```

```
Out[ ]:=
```

```
{ 13.60569312659196856313772684846641360732771658204116910799196067 eV ,
  54.4227725063678742525509073938656544293108663281646764319678427 eV ,
  122.4512381393277170682395416361977224659494492383705219719276460 eV ,
```

217.6910900254714970102036295754626177172434653126587057278713707 eV ,
340.1423281647992140784431712116603401831929145510292276997990168 eV ,
489.804952557310868272958166544790889863797796953482087887710584 eV ,
666.678963203006459593748615574854266759058112520017286291606073 eV ,
870.764360101885988040814518301850470868973861250634822911485483 eV ,
1102.061143253949453614155874725779502193545043145334697747348814 eV ,
1360.569312659196856313772684846641360732771658204116910799196067 eV ,
1646.288868317628196139664948664436046486653706426981462067027241 eV ,
1959.219810229243473091832666179163559455191187813928351550842337 eV ,
2299.362138394042687170275837390823899638384102364957579250641354 eV ,
2666.715852812025838374994462299417067036232450080069145166424292 eV ,
3061.280953483192926705988540904943061648736230959263049298191151 eV ,
3483.05744040754395216325807320740188347589544500253929164594193 eV ,
3932.04531358507891474680305920679353251771009220989787220967663 eV ,
4408.24457301579781445662349890311800877418017258133879098939526 eV ,
4911.65521869970065129271939229637531224530568611686204798509780 eV ,
5442.27725063678742525509073938656544293108663281646764319678427 eV ,
6000.11066882705813634373754017368840083152301268015557662445466 eV ,
6585.15547327051278455865979465774418594661482570792584826810897 eV ,
7197.41166396715136989985750283873279827636207189977845812774720 eV ,
7836.87924091697389236733066471665423782076475125571340620336935 eV ,
8503.55820411998035196107928029150850457982286377573069249497542 eV ,
9197.44855357617074868110334956329559855353640945983031700256541 eV ,
9918.55028928554508252740287253201551974190538830801227972613933 eV ,
10 666.86341124810335349997784919766826814492980032027658066569717 eV ,
11 442.38791946384556159882827956025384376260964549662321982123892 eV ,
12 245.12381393277170682395416361977224659494492383705219719276460 eV ,
13 075.07109465488178917535550137622347664193563534156351278027421 eV ,
13 932.22976163017580865303229282960753390358178001015716658376773 eV ,
14 816.59981485865376525698453797992441837988335784283315860324517 eV ,
15 728.18125434031565898721223682717413007084036883959148883870654 eV ,
16 666.97408007516148984371538937135666897645281300043215729015182 eV ,
17 632.97829206319125782649399561247203509672069032535516395758103 eV ,
18 626.19389030440496293554805555052022843164400081436050884099416 eV ,
19 646.62087479880260517087756918550124898122274446744819194039121 eV ,
20 694.25924554638418453248253651741509674545692128461821325577218 eV ,
21 769.10900254714970102036295754626177172434653126587057278713707 eV ,
22 871.17014580109915463451883227204127391789157441120527053448589 eV ,
24 000.44267530823254537495016069475360332609205072062230649781862 eV ,

25 156.92659106854987324165694281439875994894796019412168067713528 eV ,
 26 340.62189308205113823463917863097674378645930283170339307243586 eV ,
 27 551.52858134873634035389686814448755483862607863336744368372036 eV ,
 28 789.64665586860547959943001135493119310544828759911383251098878 eV ,
 30 054.97611664165855597123860826230765858692592972894255955424112 eV ,
 31 347.51696366789556946932265886661695128305900502285362481347739 eV ,
 32 667.26919694731652009368216316785907119384751348084702828869757 eV ,
 34 014.23281647992140784431712116603401831929145510292276997990168 eV ,
 35 388.4078222657102327212275328611417926593908298890808498870897 eV ,
 36 789.7942143046829947244133982531823942141456378393212680102617 eV ,
 38 218.3919925968396938538747173421558229835558789536440243494175 eV ,
 39 674.2011571421803301096114901280620789676215532320491189045573 eV ,
 41 157.2217079407049034916237166109011621663426606745365516756810 eV ,
 42 667.4536449924134139999113967906730725797192012811063226627887 eV ,
 44 204.8969682973058616344745306673778102077511750517584318658802 eV ,
 45 769.5516778553822463953131182410153750504385819864928792849557 eV ,
 47 361.4177736666425682824271595115857671077814220853096649200151 eV ,
 48 980.4952557310868272958166544790889863797796953482087887710584 eV ,
 50 626.7841240487150234354816031435250328664334017751902508380857 eV ,
 52 300.2843786195271567014220055048939065677425413662540511210968 eV ,
 54 000.9960194435232270936378615631956074837071141214001896200919 eV ,
 55 728.9190465207032346121291713184301356143271200406286663350709 eV ,
 57 484.0534598510671792568959347705974909596025591239394812660338 eV ,
 59 266.3992594346150610279381519196976735195334313713326344129807 eV ,
 61 075.9564452713468799252558227657306832941197367828081257759115 eV ,
 62 912.7250173612626359488489473086965202833614753583659553548261 eV ,
 64 776.7049757043623290987175255485951844872586470980061231497248 eV ,
 66 667.8963203006459593748615574854266759058112520017286291606073 eV ,
 68 586.2990511501135267772810431191909945390192900695334733874737 eV ,
 70 531.9131682527650313059759824498881403868827613014206558303241 eV ,
 72 504.7386716086004729609463754775181134494016656973901764891584 eV ,
 74 504.7755612176198517421922222020809137265760032574420353639766 eV ,
 76 532.0238370798231676497135226235765412184057739815762324547788 eV ,
 78 586.4834991952104206835102767420049959248909778697927677615648 eV ,
 80 668.1545475637816108435824845573662778460316149220916412843348 eV ,
 82 777.0369821855367381299301460696603869818276851384728530230887 eV ,
 84 913.1308030604758025425532612788873233322791885189364029778266 eV ,
 87 076.4360101885988040814518301850470868973861250634822911485483 eV ,
 89 266.9526035699057427466258527881396776771484947721105175352540 eV ,

```

91 484.6805832043966185380753290881650956715662976448210821379436 eV ,
93 729.6199490920714314558002590851233408806395336816139849566171 eV ,
96 001.7707012329301814998006427790144133043682028824892259912745 eV ,
98 301.1328396269728686700764801698383129427523052474468052419159 eV ,
100 627.7063642741994929666277712575950397957918407764867227085411 eV ,
102 981.4912751746100543894545160422845938634868094696089783911503 eV ,
105 362.4875723282045529385567145239069751458372113268135722897434 eV ,
107 770.6952557349829886139343667024621836428430463481005044043205 eV ,
110 206.1143253949453614155874725779502193545043145334697747348814 eV ,
112 668.7447813080916713435160321503710822808210158829213832814263 eV ,
115 158.5866234744219183977200454197247724217931503964553300439551 eV ,
117 675.6398518939361025781995123860112897774207180740716150224678 eV ,
120 219.9044665666342238849544330492306343477037189157702382169645 eV ,
122 791.3804674925162823179848074093828061326421529215511996274451 eV ,
125 390.0678546715822778772906354664678051322360200914144992539095 eV ,
128 015.9666281038322105628719172204856313464853204253601370963580 eV ,
130 669.0767877892660803747286526714362847753900539233881131547903 eV ,
133 349.3983337278838873128608418193197654189502205854984274292065 eV ,
136 056.9312659196856313772684846641360732771658204116910799196067 eV ,
138 791.6755843646713125679515812058852083500368534019660706259908 eV ,
141 553.6312890628409308849101314445671706375633195563233995483588 eV ,
144 342.7983800141944863281441353801819601397452188747630666867108 eV }

```

```
In[ ]:= Ezn[ me, 1.0, 2]
```

```
M = me;
```

```
n = 1.0
```

```
Z = 1
```

```
UnitConvert[ $\frac{Z^2 \alpha^2}{2 n^2} c^2 M$ , "Electronvolts"]
```

```
Out[ ]:=
```

```
54.4228 eV
```

```
Out[ ]:=
```

```
1.
```

```
Out[ ]:=
```

```
1
```

```
Out[ ]:=
```

```
13.6057 eV
```

In[*]:=

```
Table[SetPrecision[En[ $\frac{mP}{me}$  mP, n], 32], {n,  $\frac{mP}{me} - 5$ ,  $\frac{mP}{me} + 5$ }]
```

$$\frac{\alpha}{e} \sqrt{\frac{h}{c \Phi}} == \sqrt{2 \pi \alpha}$$

$$\frac{\alpha}{e} \sqrt{\frac{h}{c \Phi}}$$

Out[*]=

```
{ 13.605693126591968563143421470898 eV ,  
 13.605693126591968563142282546412 eV , 13.605693126591968563141143621926 eV ,  
 13.605693126591968563140004697439 eV , 13.605693126591968563138865772953 eV ,  
 13.605693126591968563137726848466 eV , 13.605693126591968563136587923980 eV ,  
 13.605693126591968563135448999494 eV , 13.605693126591968563134310075007 eV ,  
 13.605693126591968563133171150521 eV , 13.605693126591968563132032226034 eV }
```

Out[*]=

True

Out[*]=

$$\frac{267029439 \sqrt{\frac{64241241}{6310543}} \pi}{12500000000}$$

In[*]:=

M = 1.883531627*^-28 kg

```
Table[SetPrecision[En[M, n], 32], {n, 1, 10}]
```

```
Clear[n, x, m, M]
```

Out[*]=

1.88353×10^{-28} kg

Out[*]=

```
{ 2813.2258027607999792962800711393 eV , 703.30645069019999482407001778483 eV ,  
 312.58064475120005454300553537905 eV , 175.82661267254999870601750444621 eV ,  
 112.52903211043201281427172943950 eV , 78.145161187800013635751383844763 eV ,  
 57.412771484914287611900363117456 eV , 43.956653168137499676504376111552 eV ,  
 34.731182750133328340780280996114 eV , 28.132258027608003203567932359874 eV }
```

```
In[ ]:= M = 3.16754*^-27 kg
```

```
Table[SetPrecision[En[M, n], 32], {n, 1, 10}]
```

```
Clear[n, x, m, M]
```

```
Out[ ]:=
```

$3.16754 \times 10^{-27} \text{ kg}$

```
Out[ ]:=
```

{ 47 310.090956476116843987256288528 eV , 11 827.522739119029210996814072132 eV ,
5256.6767729417915688827633857727 eV , 2956.8806847797573027492035180330 eV ,
1892.4036382590447828988544642925 eV , 1314.1691932354478922206908464432 eV ,
965.51206033624714564211899414659 eV , 739.22017119493932568730087950826 eV ,
584.07519699353235864691669121385 eV , 473.10090956476119572471361607313 eV }

```
In[ ]:= n = 1
```

$$\frac{\alpha^2}{2 n^2} c^2 \frac{1}{\Gamma} 1.0$$

$$\frac{\alpha}{2 n^2} c^2 m_e 1.0$$

$$\frac{\alpha}{2 n^2} c^2 \frac{1}{\Gamma} == G \frac{1}{\left(\frac{2 n 1P}{\sqrt{2}}\right)^2}$$

$$\left(\frac{2 n 1P}{\sqrt{2}}\right)^2 \frac{\alpha}{2 n^2} c^2 \frac{1}{\Gamma}$$

```
Clear[n]
```

```
Out[ ]:=
```

1

```
Out[ ]:=
```

$9.32226 \times 10^{56} \text{ m}^3 \text{ Hz} / (\text{s}^3 \text{ J})$

```
Out[ ]:=
```

$2.98721 \times 10^{-16} \sqrt{\text{kg m}} \sqrt{\text{J}} / (\sqrt{\text{Hz}} \text{ s}^{3/2})$

```
Out[ ]:=
```

True

```
Out[ ]:=
```

$$\frac{\pi^{3/4}}{25\,000\,000\,000 \sqrt{2}} \text{ m}^3 / (\text{kg s}^2)$$

```
In[ ]:= UnitSimplify[Quantity[8.49201*10^26,  $\frac{\text{"Hertz" "Kilograms" "Meters"}^3}{\text{"Joules" "Seconds"}^3}$ ]]
```

```
Out[ ]:=
```

$8.49201 \times 10^{26} \text{ m/s}^2$

$$\ln[\bullet] :=$$

Clear[x, n]

Out[•]=

$$\frac{3\,300\,227\,447\,125\,296\,522\,950\,299\,769\,495\,147\,371\,286\,908\,577\,889\,555\,303\,974\,472\,358\,527}{14\,723\,888\,178\,178\,694\,383\,392\,531\,512\,597\,656\,250}\sqrt{\frac{64\,241\,241}{6\,310\,543}}\pi^{9/8}$$

 $\text{kg}^{3/2}\text{m}^3 \sqrt{\text{Hz}} / (\text{s}^{5/2} \sqrt{\text{J}})$

 n^2

$$\ln[\bullet] :=$$

En [me, x]

$$\text{UnitConvert}\left[\frac{e^4 Z_0^2 m_e c^2}{8 h^2} \frac{1}{x^2}, \text{"Electronvolts"}\right]$$
$$\text{UnitConvert}\left[-\frac{1}{2x^2} \frac{m e^2 c \pi \Gamma e^2 Z_0}{h^2}, \text{"Electronvolts"}\right]$$

En [me, x]

Clear[x, n]

Out[•]=

1.

Out[•]=

13.6057 eV

Out[•]=

13.6057 eV

Out[•]=

-13.6057 eV

Out[•]=

13.6057 eV

$$\ln[\ast]:= \sqrt{\frac{e^4 Z_0^2 c^2}{8 h^2}}$$

$$x = 4.0$$

$$\text{UnitConvert}\left[\frac{e^4 Z_0^2 c^2}{8 h^2} \frac{\text{me}}{x^2}, \text{"Electronvolts"}\right]$$

$$\left(\frac{c \alpha}{\sqrt{2}}\right)^2$$

$$\text{UnitConvert}\left[\left(\frac{c \alpha}{\sqrt{2}}\right)^2 \frac{\text{me}}{x^2}, \text{"Electronvolts"}\right]$$

Out[⌘]=

$$\frac{686\,630\,223\,537\,416\,360\,663\,645\,773\,522\,269\,\pi}{986\,022\,343\,750\,000\,000\,000\,000\,000\,\sqrt{2}} \text{ mHz C}^2\Omega / (\text{s J})$$

Out[⌘]=

$$4.$$

Out[⌘]=

$$0.850356 \text{ eV}$$

Out[⌘]=

$$\frac{471\,461\,063\,875\,042\,360\,421\,471\,395\,642\,163\,910\,183\,844\,082\,555\,650\,757\,710\,638\,908\,361\,\pi^2}{1\,944\,480\,124\,748\,486\,328\,125\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000} \text{ m}^2/\text{s}^2$$

Out[⌘]=

$$0.850356 \text{ eV}$$

```

In[ ]:= n = 1.0
        
$$\frac{\alpha}{2\pi} \frac{c h m_e}{\Gamma} 1.0$$

        
$$\frac{c h m_e}{\Gamma} 1.0$$

        
$$\frac{e^4 Z_0^2 m_e c^2}{8 h^2} 1.0$$

        UnitConvert[ $\frac{e^4 Z_0^2 m_e c^2}{8 h^2}$ , "Joules"]  $\frac{1}{n^2}$ 
        UnitConvert[ $\frac{e^4 Z_0^2 m_e c^2}{8 h^2}$ , "Electronvolts"]  $\frac{1}{n^2}$ 
        Clear[n]
        
$$\frac{1}{\alpha} \sqrt{\frac{e^4 Z_0^2 c^2}{8 h^2}} == \frac{c}{\sqrt{2}}$$

        
$$\frac{e^2 Z_0 c}{\alpha h \sqrt{8}} == \frac{c}{\sqrt{2}}$$

        
$$\sqrt{2} \frac{e^2 Z_0}{\alpha h \sqrt{8}}$$

        
$$\frac{1}{\alpha} \left( \frac{e^2 Z_0 c m_P}{2 h} \right)^2 1.0$$

Out[ ]:=
1.

Out[ ]:=
 $8.18711 \times 10^{-14} \sqrt{\text{kg m}} \sqrt{\text{J}} / (\sqrt{\text{Hz}} \text{ s}^{3/2})$ 

Out[ ]:=
 $7.04928 \times 10^{-11} \sqrt{\text{kg m}} \sqrt{\text{J}} / (\sqrt{\text{Hz}} \text{ s}^{3/2})$ 

Out[ ]:=
 $2.17987 \times 10^{-18} \sqrt{\text{kg m Hz}^{3/2} \text{C}^4 \Omega^2} / (\text{s}^{3/2} \text{J}^{3/2})$ 

Out[ ]:=
 $2.17987 \times 10^{-18} \text{ J}$ 

Out[ ]:=
13.6057 eV

Out[ ]:=
True

Out[ ]:=
True

Out[ ]:=
1
    
```

Out[8]=

$$0.310668 \text{ kg Hz C}^4 \Omega^2 / (\text{s J})$$

$$\text{In[8]} := \text{UnitSimplify}\left[\text{Quantity}\left[0.310668, \frac{\text{"Coulombs"}^4 \text{"Hertz"} \text{"Kilograms"} \text{"Ohms"}^2}{\text{"Joules"} \text{"Seconds"}}\right]\right]$$

Out[8]=

$$0.310668 \text{ kg J}$$

$$\text{In[9]} := \text{UnitSimplify}\left[\text{Quantity}\left[149896229 \sqrt{2}, \frac{\text{"Coulombs"}^2 \text{"Hertz"} \text{"Meters"} \text{"Ohms"}}{\text{"Joules"} \text{"Seconds"}}\right]\right]$$

$$\text{In[9]} := \frac{\left(\frac{\alpha}{2\pi} \frac{c h M}{r Q}\right)}{\frac{\alpha}{2\pi} \frac{c h M^2}{r^2}} \frac{Q}{m P^2} \frac{M c^2}{\alpha G}$$

Out[9]=

$$\frac{c^2 \Gamma}{G m P^2 \alpha}$$

$$\frac{V}{a} \Rightarrow \frac{\left(\frac{\alpha}{2\pi} \frac{c h M}{r Q}\right)}{\frac{\alpha}{2\pi} \frac{c h M^2}{r^2}}$$

$$\frac{V}{a} \frac{Q}{m P^2} \frac{M c^2}{\alpha G} == 1$$

$$\frac{V}{a} \Rightarrow \frac{\left(\frac{\alpha}{2\pi} \frac{c h M}{r Q}\right)}{\frac{\alpha}{2\pi} \frac{c h M^2}{r^2}}$$

$$\frac{\left(\frac{\alpha}{2\pi} \frac{c h M}{r Q}\right)}{\frac{\alpha}{2\pi} \frac{c h M^2}{r^2}} \frac{Q}{m P^2} \frac{M c^2}{\alpha G} \Rightarrow \frac{c^2 \Gamma}{G m P^2 \alpha} = 1$$

$$\text{In[9]} := \frac{\left(\frac{\alpha}{2\pi} \frac{c h M}{r Q}\right)}{\frac{\alpha}{2\pi} \frac{c h M^2}{r^2}} \frac{Q}{m P^2} \frac{M c^2}{\alpha G}$$

Out[9]=

$$\frac{c^2 \Gamma}{G m P^2 \alpha}$$

$$\frac{V}{a} \frac{Q}{m P^2} \frac{M c^2}{\alpha G} == 1$$

$$\frac{\left(\frac{\alpha}{2\pi} \frac{c h M}{r Q}\right)}{\frac{\alpha}{2\pi} \frac{c h M^2}{r^2}} \frac{Q}{m P^2} \frac{M c^2}{\alpha G}$$

M = me

$$Q = -e$$
$$\text{UnitConvert}\left[\frac{\alpha}{2\pi} \frac{\text{c h M}}{\Gamma}, \text{"Joules"}\right]$$

```
UnitConvert[ $\frac{\alpha}{2\pi} \frac{c h M}{\Gamma Q}$ , "Volts"]
```

$$V = \frac{\alpha}{2\pi} \frac{chM}{\Gamma Q}$$

$$a = Q \frac{\alpha}{2\pi} \frac{c \hbar M^2}{\Gamma^2 Q}$$

$$a = \frac{\alpha}{2\pi} \frac{c h M^2}{\Gamma^2}$$

$$\frac{V}{a} = \frac{\Gamma}{M} = \frac{Q}{\Gamma c^2}$$

$$\frac{V}{a} \frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{c^2 \alpha \delta}{G \alpha \gamma} = \frac{V}{a} \frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{c^2}{G \alpha}$$

$$\frac{\frac{V}{a} \frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{c^2}{G \alpha}}{\frac{mP}{M}} = \frac{mP}{M}$$

$$\frac{V}{a} \frac{\Gamma}{mP} \frac{Q}{\Gamma} \frac{c^2}{G\alpha} = \frac{mP}{M}$$

$$\frac{V}{a} \frac{Q}{m_P} \frac{c^2}{G \alpha} = \frac{m_P}{M}$$

$$\frac{V}{a} \frac{Q}{m_P} \frac{c^2}{\alpha} \frac{M}{m_P} \equiv G$$

$$\frac{V}{a} \frac{Q}{m_P} \frac{c^2}{\alpha G} \frac{M}{m_P} = 1$$

$$\frac{V}{a} \frac{Q}{mP^2} \frac{Mc^2}{\alpha G} == 1$$

$$\frac{\frac{V}{a}}{\frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2}} = \frac{1}{4 \pi \epsilon_0}$$

Out[•]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,880\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \text{ kg}$$

Out[]=

$$-\frac{801\,088\,317}{5\,000\,000\,000\,000\,000\,000\,000\,000\,000}\text{ C}$$

Out[]=

$$\frac{157\,282\,156\,278\,943\,087\sqrt{405\,397\,113\,703\,863}}{11\,946\,088\,866\,247\,312\,670\,720\,000\,000\,000\,000\,000}\times 2^{1/4}\pi^{7/8}\text{ J}$$

Out[]=

$$-\frac{38\,398\,963\,935\,288\,839\,599\,609\,375\sqrt{\frac{135\,132\,371\,234\,621}{3}}}{155\,759\,639\,072\,176\,071\,177\,732\,096}\times 2^{1/4}\pi^{7/8}\text{ V}$$

Out[]=

$$-\frac{38\,398\,963\,935\,288\,839\,599\,609\,375\sqrt{\frac{135\,132\,371\,234\,621}{3}}}{155\,759\,639\,072\,176\,071\,177\,732\,096}\times 2^{1/4}\pi^{7/8}\text{ kg m}^2/(\text{s}^2\text{C})$$

Out[]=

$$\frac{692\,798\,640\,188\,177\,556\,548\,445\,863\,207\,750\,953\,733\,921\,051\,025\,390\,625}{2\,274\,474\,859\,115\,114\,867\,566\,227\,888\,187\,475\,417\,937\,968\,432\,676\,864\sqrt{2}\pi^{7/4}}\text{ kg}^2\text{m}^3\text{Hz}/(\text{s}^3\text{J})$$

Out[]=

$$\frac{692\,798\,640\,188\,177\,556\,548\,445\,863\,207\,750\,953\,733\,921\,051\,025\,390\,625}{2\,274\,474\,859\,115\,114\,867\,566\,227\,888\,187\,475\,417\,937\,968\,432\,676\,864\sqrt{2}\pi^{7/4}}\text{ kg}^2\text{m}^3\text{Hz}/(\text{s}^3\text{J})$$

Out[]=

$$\frac{2\,274\,474\,859\,115\,114\,867\,566\,227\,888\,187\,475\,417\,937\,968\,432\,676\,864\sqrt{2}\pi^{7/4}}{692\,798\,640\,188\,177\,556\,548\,445\,863\,207\,750\,953\,733\,921\,051\,025\,390\,625}\text{ s}^3\text{J}/(\text{kg}^2\text{m}^3\text{Hz})$$

Out[]=

True

Out[]=

True

Out[]=

True

Out[]=

True

Out[]=

True

Out[]=

True

Out[]=

True

Out[]=

True

$$\text{In}[*]:= \frac{\frac{V}{a}}{\frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2}} == \frac{1}{4 \pi \epsilon_0}$$

$$\frac{G}{\frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2} V} 1.0$$

$$\frac{1}{\frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2} V}$$

$$\frac{V}{c^2} \frac{Q}{M}$$

$$\frac{a}{V c^2} \frac{Q}{M} 1.0$$

$$\frac{1}{\alpha} \sqrt{\frac{a}{V}} 1.0$$

$$\frac{a}{\frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2}} 1.0$$

$$\frac{a \frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2}}{V} / G^2 1.0$$

$$\frac{1}{\frac{a \frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2}}{V}} 1.0$$

$$\frac{c^2 V}{a} \frac{M}{Q} == \frac{1}{4 \pi \epsilon_0}$$

Out[*]=

True

Out[*]=

$$6.67433 \times 10^{-11} \text{ m}^3 / (\text{kg s}^2)$$

Out[*]=

1

Out[*]=

1

Out[*]=

$$1.11265 \times 10^{-10} \text{ Hz s C}^2 / (\text{m J})$$

Out[*]=

$$(0. + 1.03329 \text{ i}) \sqrt[4]{\text{kg}} \sqrt[4]{\text{Hz}} \sqrt{C} / (\sqrt[4]{\text{s}} \sqrt[4]{J})$$

Out[*]=

$$-1.48463 \times 10^7 \text{ kg}^{3/2} \text{ m}^2 \sqrt{J} / (\sqrt{\text{Hz}} \text{ s}^{7/2} C)$$

Out[]:=

$$2.49772 \times 10^{10} \text{ kg}^2 \text{ Hz s}^5 \text{ C}^2 / (\text{m}^7 \text{ J})$$

Out[]:=

$$8.98755 \times 10^9 \text{ m J} / (\text{Hz s C}^2)$$

Out[]:=

True

In[]:= **c**

Out[]:=

$$299\,792\,458 \text{ m/s}$$

In[]:=

$$\frac{\text{Quantity}\left[299\,792\,458, \frac{\text{"Henries"}}{\text{"Kilograms"}}\right]}{\frac{149\,896\,229 \pi^{3/4}}{125\,000\,000\,000\,000\,000 \sqrt{2}}} \text{ m}^4 / \text{s}^4$$

Out[]:=

$$1 \text{ kg m} / (\text{s H})$$

Out[]:=

$$\frac{504\,850\,544\,566\,405\,639\,328\,987\,546\,634\,481 \pi^3}{976\,562\,500\,000} \text{ m}^{16} / \text{s}^{16}$$

In[]:= **lp**

Out[]:=

$$\frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000} \times 2^{3/4} \pi^{1/8} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

In[]:=

$$\sqrt{\left(\frac{\frac{e^2}{4 \pi \epsilon_0}}{h^2 \alpha c \sqrt{\pi}}\right)} \left(\frac{\frac{G}{4 \pi \epsilon_0}}{h^2 \sqrt{\pi}}\right)$$

Out[]:=

$$\frac{1}{2 \pi^{3/4}}$$

Out[]:=

$$\frac{2\,567\,871\,939\,248\,050\,400\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000}{18\,931\,629} \text{ m}^4 / (\text{kg s}^3 \text{ C}^2)$$

$$\text{In}[*]:= \frac{G}{\sqrt{\left(\frac{\frac{e e}{4 \pi \epsilon_0}}{h^2 \alpha c \sqrt{\pi}}\right)}}$$

$$\frac{1}{G \sqrt{\left(\frac{\frac{e e}{4 \pi \epsilon_0}}{h^2 \alpha c \sqrt{\pi}}\right)}} G$$

Out[*]=

$$\frac{1}{50\,000\,000\,000 \sqrt{2}} \text{ m}^3 / (\text{kg s}^2)$$

Out[*]=

$$2 \pi^{3/4}$$

$$\text{In}[*]:= \left(\frac{c}{h}\right) \left(\frac{G \frac{1 p^2}{4 \pi \epsilon_0}}{h^2 \sqrt{\pi}}\right)$$

$$\frac{e \frac{G^2}{4 \pi \epsilon_0}}{h^2 \alpha c \sqrt{\pi}}$$

$$\frac{1}{50 c}$$

$$\frac{\frac{G}{4 \pi \epsilon_0}}{h^2 \sqrt{\pi}}$$

Out[*]=

$$\frac{4\,000\,000\,000\,000}{132\,521\,403} \text{ m}^5 / (\text{kg}^2 \text{ s}^3 \text{ C}^2)$$

Out[*]=

$$\frac{1\,000\,000}{801\,088\,317} \text{ m}^6 / (\text{kg}^2 \text{ s}^4 \text{ C})$$

Out[*]=

$$\frac{1}{14\,989\,622\,900} \text{ s/m}$$

Out[*]=

$$\frac{2\,567\,871\,939\,248\,050\,400\,000\,000\,000\,000\,000\,000\,000\,000 \sqrt{2} \pi^{1/4}}{18\,931\,629} \text{ m}^4 / (\text{kg s}^3 \text{ C}^2)$$

In[]:= $m\mu = \text{Quantity}\left[\frac{1883\,531\,651 + 1883\,531\,603}{2} * 10^{-28}, \text{"Kilograms"}\right]$

$$m\mu = \frac{\text{mP}}{115\,550\,496\,528}$$

me

mp

$$\frac{115\,550\,496\,528}{801\,088\,317}$$

Out[]:=

$$\frac{1883\,531\,627}{10\,000\,000\,000\,000\,000\,000\,000\,000\,000} \text{ kg}$$

Out[]:=

$$\frac{7 \sqrt{\frac{135\,132\,371\,234\,621}{3}}}{77\,033\,664\,352\,000\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]:=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,846\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]:=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{260\,241\,733\,471\,688\,089\,558\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]:=

$$\frac{38\,516\,832\,176}{267\,029\,439}$$

In[]:= $\frac{115\,550\,496\,528}{801\,088\,317}$

Out[]:=

$$\frac{38\,516\,832\,176}{267\,029\,439}$$

$$\begin{aligned}
In[*] &:= \frac{1\,883\,531\,651 + 1\,883\,531\,603}{2} \\
&\frac{1 + \sqrt{5}}{2} \bigg/ \frac{1\,883\,531\,651 \times 1\,883\,531\,603}{801\,088\,317 \times 1\,883\,531\,627} + 1.0 \\
&\frac{1 + \sqrt{5}}{2} \frac{1\,883\,531\,651 \times 1\,883\,531\,603}{801\,088\,317 \times 1\,883\,531\,627} \bigg/ \frac{2 + \sqrt{5}}{2} \\
&\frac{1 + \sqrt{5}}{2} \bigg/ \frac{1\,883\,531\,651 \times 1\,883\,531\,603}{801\,088\,317 \times 1\,883\,531\,627} 1.0 \\
&\alpha \sqrt{\frac{1\,883\,531\,651 \times 1\,883\,531\,603}{2 \alpha 801\,088\,317 \times 1\,883\,531\,627}} \\
&1\,883\,531\,627 \alpha \\
&1 \bigg/ \frac{\sqrt{1\,883\,531\,651^2 + 1\,883\,531\,603^2}}{801\,088\,317 \times 1\,883\,531\,627 \sqrt{2}} \times 1.0 \\
&\frac{801\,088\,317}{1\,883\,531\,628} \\
&\frac{89\,009\,813}{209\,281\,292} \times 1\,883\,531\,627.0 \\
&801\,088\,317 \sqrt{1\,883\,531\,651^2 + 1\,883\,531\,603^2} \alpha 1.0 \\
&\frac{c \sqrt{\frac{c h}{g}} \sqrt{2 \pi} \Gamma}{h m \mu \alpha} 1.0
\end{aligned}$$

$$Out[*] = 1\,883\,531\,627$$

$$Out[*] = 1.68817$$

$$\begin{aligned}
Out[*] &= \frac{3\,547\,691\,389\,909\,266\,553 \left(1 + \sqrt{5}\right)}{1\,508\,875\,181\,089\,701\,759 \left(2 + \sqrt{5}\right)} \\
Out[*] &= 0.688169
\end{aligned}$$

$$\begin{aligned}
Out[*] &= \frac{\sqrt{\frac{20\,286\,057\,146\,139\,450\,058\,637\,174\,455\,934\,949 \pi}{11\,886\,107\,324\,043\,461}}}{25\,000\,000\,000} \\
Out[*] &= \frac{8\,627\,900\,452\,964\,720\,901\,532\,589\,746\,164\,147 \pi}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000\,000}
\end{aligned}$$

$$Out[*] = 8.01088 \times 10^8$$

$$\begin{aligned}
Out[*] &= \frac{89\,009\,813}{209\,281\,292}
\end{aligned}$$

Out[*]=

$$8.01088 \times 10^8$$

Out[*]=

$$1.51437 \times 10^{18}$$

Out[*]=

$$1.1555 \times 10^{11}$$

$$\begin{aligned} \text{In[*]} := & \frac{801\,088\,317}{\frac{1\,508\,875\,181\,089\,701\,759}{\sqrt{3\,547\,691\,389\,909\,267\,705}}} == \frac{\sqrt{3\,547\,691\,389\,909\,267\,705}}{1\,883\,531\,627} \\ & 1\,883\,531\,627 == \frac{\sqrt{3\,547\,691\,389\,909\,267\,705}}{801\,088\,317} \frac{1\,508\,875\,181\,089\,701\,759}{\sqrt{3\,547\,691\,389\,909\,267\,705}} \\ & \frac{1\,508\,875\,181\,089\,701\,759}{801\,088\,317} \end{aligned}$$

Out[*]=

True

Out[*]=

True

Out[*]=

$$1\,883\,531\,627$$

$$\begin{aligned} \text{In[*]} := & \frac{801\,088\,317}{\frac{1\,508\,875\,181\,089\,701\,759}{\sqrt{3\,547\,691\,389\,909\,267\,705}}} 1.0 \\ & 1\,883\,531\,627 \sqrt{2} \\ & \frac{167\,652\,797\,898\,855\,751}{209\,281\,292} \frac{1\,883\,531\,628}{801\,088\,317} \end{aligned}$$

Out[*]=

1.

Out[*]=

$$1\,883\,531\,627 \sqrt{2}$$

Out[*]=

$$1\,883\,531\,627$$

$$\text{In[*]} := \text{Solve} \left[\frac{801\,088\,317 \times 1\,883\,531\,627 \sqrt{2}}{x} == \sqrt{1\,883\,531\,651^2 + 1\,883\,531\,603^2}, x \right]$$

Out[*]=

$$\left\{ \left\{ x \rightarrow \frac{1\,508\,875\,181\,089\,701\,759}{\sqrt{3\,547\,691\,389\,909\,267\,705}} \right\} \right\}$$

$$\text{In}[*]:= \frac{801\,088\,317 \times 1\,883\,531\,627 \sqrt{2}}{89\,009\,813 \times 1\,883\,531\,627} \times 1.0 == \sqrt{1\,883\,531\,651^2 + 1\,883\,531\,603^2}$$

$$\frac{209\,281\,292}{209\,281\,292}$$

$$\frac{801\,088\,317}{1\,883\,531\,628} \times 1\,883\,531\,627 \times 1.0$$

$$1\,883\,531\,627 == \frac{167\,652\,797\,898\,855\,751}{209\,281\,292} \frac{1\,883\,531\,628}{801\,088\,317}$$

$$\frac{89\,009\,813}{209\,281\,292}$$

$$\text{Out}[*]= \frac{2.13387 \times 10^{18}}{X} == \sqrt{7\,095\,382\,779\,818\,535\,410}$$

$$\text{Out}[*]= \frac{167\,652\,797\,898\,855\,751}{209\,281\,292}$$

$$\text{Out}[*]= 8.01088 \times 10^8$$

$$\text{Out}[*]= \text{True}$$

$$\text{Out}[*]= \frac{89\,009\,813}{209\,281\,292}$$

$$\text{In}[*]:= \text{FactorInteger}[1\,883\,531\,628]$$

$$\text{Out}[*]= \{\{2, 2\}, \{3, 2\}, \{11, 1\}, \{101, 1\}, \{47\,093, 1\}\}$$

$$\text{In}[*]:= \text{FactorInteger}[1\,883\,531\,627]$$

$$\text{Out}[*]= \{\{47, 1\}, \{2789, 1\}, \{14\,369, 1\}\}$$

$$\text{In}[*]:= \frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h \text{Quantity}\left[\frac{1\,883\,531\,651 + 1\,883\,531\,603}{2} * 10^{-28}, \text{"Kilograms"}\right] \alpha}$$

$$\text{Out}[*]= \frac{17\,500\,000\,000\,000 \times 2^{3/4} \sqrt{405\,397\,113\,703\,863}}{1\,883\,531\,627 \pi^{7/8}}$$

$$\text{In}[*]:= \frac{17\,500\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \sqrt{405\,397\,113\,703\,863}}{1\,883\,531\,627 \pi^{7/8}}$$

$$\text{Out}[*]= \frac{17\,500\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \sqrt{405\,397\,113\,703\,863}}{1\,883\,531\,627 \pi^{7/8}}$$

$$\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h \text{ Quantity} [1883\,531\,627 * 10^{-37}, \text{"Kilograms"}] \alpha}$$

$$\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h \text{ Quantity} [\frac{1883\,531\,651 + 1883\,531\,603}{2} * 10^{-28}, \text{"Kilograms"}] \alpha} 1.0$$

$$\frac{115\,550\,496\,527\,972\,940\,541}{1883\,531\,627}$$

$$\frac{\sqrt{1883\,531\,651^2 + 1883\,531\,603^2}}{\sqrt{2}} \text{ mP } 1.0$$

$$\frac{17\,500\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \sqrt{405\,397\,113\,703\,863}}{1\,883\,531\,627 \pi^{7/8}}$$

$$Out[\bullet]=1.1555 \times 10^{11}$$

$$\begin{array}{r} 115\,550\,496\,527\,972\,940\,541 \\ \hline 1\,883\,531\,627 \end{array}$$

$$Out[\bullet]=40.9938 \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

801 088 317	C
5 000 000 000 000 000 000 000 000 000	

`Out[•]=`

1 883 531 655

$$\text{In}[*]:= \frac{\text{NextPrime}[1883\,531\,651, -1] \cdot 1883\,531\,651 + 1883\,531\,603}{2}$$

`Out[•]=`

1 883 531 603

`Out[•]=`

1 883 531 627

```
In[ ]:= NextPrime[1883531627, -1]
```

`Out[•]=`

1 883 531 603

$$\text{In}[*]:= \frac{1\,883\,531\,628}{10\,000\,000\,000\,000\,000\,000\,000\,000\,000} \text{ kg } 1.0$$

Out[*]=

$$1.88353 \times 10^{-19} \text{ kg}$$

$$\text{In}[*]:= 801\,088\,317 \left(\frac{1}{2} (-1 - \sqrt{5}) + \frac{4\,568\,151\,571 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634} \right)$$

Out[*]=

$$801\,088\,317 \left(\frac{1}{2} (-1 - \sqrt{5}) + \frac{4\,568\,151\,571 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634} \right)$$

$$\text{In}[*]:= 801\,088\,317 \left(\frac{4\,568\,151\,571 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634} - \frac{1 + \sqrt{5}}{2} \right) == 1\,883\,531\,627$$

$$\frac{1\,883\,531\,627 + \frac{1}{2} (-2\,965\,974\,937 + 801\,088\,317 \sqrt{5})}{801\,088\,317}$$

Out[*]=

True

Out[*]=

$$\frac{1\,883\,531\,627 + \frac{1}{2} (-2\,965\,974\,937 + 801\,088\,317 \sqrt{5})}{801\,088\,317}$$

$$\text{In}[*]:= \frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h \frac{1\,883\,531\,628}{10\,000\,000\,000\,000\,000\,000\,000\,000\,000} \text{ kg } \alpha} 801\,088\,317$$

$$\text{Solve}\left[x - \frac{1\,883\,531\,627}{801\,088\,317} == \frac{1 + \sqrt{5}}{2}, x\right]$$

$$\frac{4\,568\,151\,571 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634} - \frac{1\,883\,531\,627}{801\,088\,317} == \frac{1 + \sqrt{5}}{2}$$

$$1\,883\,531\,627 * 2$$

$$\frac{-2\,965\,974\,937 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634}$$

$$1\,602\,176\,634$$

$$\frac{-2\,965\,974\,937 + 801\,088\,317 \sqrt{5}}{2 \times 801\,088\,317}$$

$$2 \times 801\,088\,317$$

$$1\,602\,176\,634 / 2$$

$$1\,883\,531\,627 * 2$$

$$c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma$$

$$\frac{\text{mP } 801\,088\,317}{h \text{ m} \mu \alpha}$$

$$1\,883\,531\,630$$

```

Out[ ]:=

$$\frac{389\,417\,931\,875\,000\,000\,000 \times 2^{3/4} \sqrt{405\,397\,113\,703\,863}}{52\,320\,323 \pi^{7/8}}$$


Out[ ]:=

$$\left\{ \left\{ x \rightarrow \frac{4\,568\,151\,571 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634} \right\} \right\}$$


Out[ ]:=
True

Out[ ]:=
3\,767\,063\,254

Out[ ]:=

$$\frac{-2\,965\,974\,937 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634}$$


Out[ ]:=

$$\frac{-2\,965\,974\,937 + 801\,088\,317 \sqrt{5}}{1\,602\,176\,634}$$


Out[ ]:=
801\,088\,317

Out[ ]:=
3\,767\,063\,254

Out[ ]:=

$$\frac{40\,497\,691\,846\,556\,815\,227 \sqrt{405\,397\,113\,703\,863}}{125\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$


Out[ ]:=
1\,883\,531\,630

In[ ]:= 11\,555\,049\,657\,311\,669\,381.0

Out[ ]:=

$$1.1555049657311669381 \times 10^{19}$$


In[ ]:= FactorInteger[11\,555\,049\,657\,311\,669\,383]

Out[ ]:=
{{7, 1}, {193, 1}, {68\,227, 1}, {125\,360\,348\,779, 1}}

In[ ]:=

$$\frac{4\,005\,441\,585\,000\,000\,000}{563\,047\,631 \times 801\,088\,317} 13\,012\,086\,673\,584\,404\,481 \times 563\,047\,631$$


Out[ ]:=
65\,060\,433\,367\,922\,022\,405\,000\,000\,000

In[ ]:=

$$\frac{21\,686\,811\,122\,640\,674\,135\,000\,000\,000}{1\,876\,825\,437}$$


Out[ ]:=

$$\frac{21\,686\,811\,122\,640\,674\,135\,000\,000\,000}{1\,876\,825\,437}$$


```

$$In[*]:= \text{Round}\left[\frac{21\,686\,811\,122\,640\,674\,135\,000\,000\,000}{1\,876\,825\,437}\right]$$

$$Out[*]= 11\,555\,049\,657\,311\,669\,383$$

$$In[*]:= \frac{13\,012\,086\,673\,584\,404\,481}{11\,260\,952\,622 \times 10^{-10}} \\ \frac{65\,060\,433\,367\,922\,022\,405\,000\,000\,000}{11\,260\,952\,622 \times 10^{-10}} \\ \frac{801\,088\,317}{1\,126\,095\,262 \times 10^{-10}} \\ \frac{42\,320\,845\,102\,020\,848\,356\,787\,507}{2\,500\,000\,000\,000\,000\,000\,000\,000\,000} \times 1.0$$

$$Out[*]= \frac{21\,686\,811\,122\,640\,674\,135\,000\,000\,000}{1\,876\,825\,437}$$

$$Out[*]= \frac{108\,434\,055\,613\,203\,370\,675\,000\,000\,000\,000\,000\,000}{1\,876\,825\,437}$$

$$Out[*]= \frac{4\,005\,441\,585\,000\,000\,000}{563\,047\,631}$$

$$Out[*]= 0.0169283$$

$$In[*]:= \text{NextPrime}[1883\,531\,659, -1]$$

$$Out[*]= 1883\,531\,651$$

$$In[*]:= 115\,550\,496\,528 \text{ mP } 1.0$$

$$Out[*]= 2514.88 \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$In[*]:= \frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h \alpha} \text{ mP } 1.0$$

$$\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h \alpha \text{ mp}}$$

$$Out[*]= 4.73685 \times 10^{-16} \text{ kg s J} / (\text{m}^2 \text{Hz})$$

$$Out[*]= 130\,120\,866\,735\,844\,044\,779$$

$$\text{In}[*]:= \text{Round}\left[\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h m \mu \alpha}\right]$$

Out[*]=
115 550 496 528

$$\text{In}[*]:= m = \text{Round}\left[\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h m \mu \alpha}\right]$$

$$m = m \text{ mP}$$

$$m = m e$$

$$\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h m \alpha} \text{ mP} - \frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h m \alpha} \frac{\text{mP}}{\left(\frac{1+\sqrt{5}}{2}\right)} 1.0$$

$$\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h m \alpha} \frac{\text{mP}}{\left(\frac{1+\sqrt{5}}{2}\right)} 1.0$$

$$\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h m \alpha} \text{ mP } 1.0$$

$$\text{UnitConvert}\left[\frac{m}{\text{mP } \alpha} \left(\frac{1 \text{ P}}{\Delta \alpha [1]} - \frac{2 G m}{c^2}\right), \text{"Meters"}\right] 1.0$$

$$\frac{\text{mP } 1 \text{ P } \alpha}{m} 1.0$$

$$\frac{\text{mP } 1 \text{ P}}{m} 2 \pi 1.0$$

$$\frac{\text{mP } 1 \text{ P}}{m \alpha} 1.0$$

$$\frac{\Gamma}{m \alpha} 2 \pi == \frac{\text{mP } 1 \text{ P}}{m} 2 \pi$$

$$\frac{\text{mP } 1 \text{ P}}{m \alpha} / \frac{\text{mP } 1 \text{ P } 2 \pi}{m}$$

$$\frac{\frac{\text{mP } 1 \text{ P } 2 \pi}{m}}{\frac{\text{mP } 1 \text{ P}}{m \alpha}} 1.0$$

Out[*]=
115 550 496 528

$$\frac{50\,553\,342\,231 \sqrt{405\,397\,113\,703\,863}}{125\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{s} \sqrt{\text{J}} / (m \sqrt{\text{Hz}})$$

Out[]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,846\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]=

$$1.98621 \times 10^{14} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]=

$$3.21376 \times 10^{14} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]=

$$5.19997 \times 10^{14} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]=

$$1.86081 \times 10^{-55} \text{ m}$$

Out[]=

$$2.81794 \times 10^{-15} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

Out[]=

$$2.42631 \times 10^{-12} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

Out[]=

$$5.29177 \times 10^{-11} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

Out[]=

True

Out[]=

$$\frac{986\,022\,343\,750\,000\,000\,000\,000\,000}{4\,580\,703\,784\,999\,263\,461\,548\,761 \pi^2}$$

Out[]=

$$0.0458506$$

$$\text{In}[*]:= \frac{c^2 \Gamma}{m_e G \alpha m_P} == \frac{m_P}{m_e}$$

$$\frac{c^2 \Gamma}{m_e G \alpha} \frac{1}{\sqrt{\frac{h c}{2 \pi G}}}$$

$$\frac{c^2 \sqrt{2 \pi} \Gamma}{G \sqrt{\frac{c h}{G}} m_e \alpha} == \frac{\sqrt{\frac{h c}{2 \pi G}}}{m_e}$$

$$\frac{c \sqrt{\frac{c h}{G}} \sqrt{2 \pi} \Gamma}{h m_e \alpha}$$

$$\frac{c^2 \Gamma}{m_e G \alpha m_P}$$

Out[*]=

True

Out[*]=

23 892 177 732 494 625 341 423

Out[*]=

True

Out[*]=

23 892 177 732 494 625 341 423

Out[*]=

23 892 177 732 494 625 341 423

$$\begin{aligned}
 \text{In}[*]:= & \text{mP}^2 \\
 & \frac{h \, c}{2 \, \pi \, G} \\
 & \frac{c^2 \, \Gamma}{m_e \, G} == \alpha \frac{\frac{h \, c}{2 \, \pi \, G}}{m_e} \\
 & \frac{c^2 \, \Gamma}{G \, m_e} == \frac{c \, h \, \alpha}{2 \, G \, m_e \, \pi} \\
 & \frac{c \, h \, \alpha}{2 \, G \, m_e \, \pi} \quad 1.0 \\
 & \frac{c^2 \, \Gamma}{m_e \, G \, \alpha \, \text{mP}} == \frac{\text{mP}}{m_e} \\
 & \frac{c^2 \, \Gamma}{m_e \, G \, \alpha} \frac{1}{\frac{h \, c}{2 \, \pi \, G}} \\
 & \frac{2 \, c \, \pi \, \Gamma}{h \, m_e \, \alpha}
 \end{aligned}$$

Out[*]=

$$\frac{19 \, 864 \, 458 \, 571 \, 489 \, 287}{4 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, \sqrt{2} \, \pi^{7/4}} \text{ kg s J / (m}^2\text{Hz)}$$

Out[*]=

$$\frac{19 \, 864 \, 458 \, 571 \, 489 \, 287}{4 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, 000 \, \sqrt{2} \, \pi^{7/4}} \text{ kg s J / (m}^2\text{Hz)}$$

Out[*]=

True

Out[*]=

True

Out[*]=

$$3.7946 \times 10^{12} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[*]=

True

Out[*]=

$$\frac{47 \, 784 \, 355 \, 464 \, 989 \, 250 \, 682 \, 846 \, 000 \, 000 \, 000 \, 000 \, 000 \times 2^{1/4} \pi^{7/8}}{7 \sqrt{405 \, 397 \, 113 \, 703 \, 863}} \text{ m} \sqrt{\text{Hz}} / (\sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}})$$

Out[*]=

$$\frac{47 \, 784 \, 355 \, 464 \, 989 \, 250 \, 682 \, 846 \, 000 \, 000 \, 000 \, 000 \, 000 \times 2^{1/4} \pi^{7/8}}{7 \sqrt{405 \, 397 \, 113 \, 703 \, 863}} \text{ m} \sqrt{\text{Hz}} / (\sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}})$$

$$\text{In}[*]:= \frac{c^2 \Gamma}{m_e G} == \alpha \frac{m_P^2}{m_e}$$

$$\alpha \frac{m_P^2}{m_e}$$

Out[*]=

True

Out[*]=

$$\frac{766\,100\,922\,797\,792\,752\,002\,206\,044\,103\,034\,957\,193\,913\,388\,321 \sqrt{\frac{64\,241\,241}{6\,310\,543}} \pi^{1/8}}{625\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{1/4}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$ln[\ast]:= \text{UnitSimplify}\left[\text{me} \sqrt{\text{mP}} \sqrt[4]{\frac{\text{G c}^2}{\Gamma}}\right] 1.0$$

me

$Out[\bullet]=$

$$9.34369 \times 10^{-22} \text{ s N}$$

Out[●]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{47\,784\,355\,464\,989\,250\,682\,846\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$\ln[*]:= \text{UnitSimplify}\left[\text{Quantity}\left[4.81957 \times 10^{-26}, \frac{\text{"Joules"}^{7/4} \text{"Kilograms"}^2 \text{"Seconds"}^{3/4}}{\text{"Hertz"}^{7/4} \text{"Meters"}^{5/2}}\right]\right]$$

Out[•]=

$$4.81957 \times 10^{-26} \text{ kg}^{15/4} \text{ m/s}$$

$$ln[\psi] := \text{Solve}\left[\frac{X \frac{G}{4\pi\epsilon_0}}{2h\sqrt{\pi}} = me, X\right]$$

$$\frac{2\,387\,564\,896\,465\,810\,067\,633\,029\,022\,354}{125} \times 2^{3/4} \sqrt{\frac{64\,241\,241}{6\,310\,543}} \pi^{13/8} \text{ m}^6 / (\text{kg s}^6 \text{ A}^2) \quad \mathbf{1.0}$$

$$\frac{G m e c^2}{\Gamma 2 h \sqrt{\alpha}}$$

Out[•]=

[illegible]

Out[•]=

$$6.58506 \times 10^{29} \text{ m}^6 / (\text{kg s}^6 \text{ A}^2)$$

$$Out[\bullet]=$$

$$\frac{1\,572\,821\,562\,789\,430\,870\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{1/4} \sqrt{\frac{135\,132\,371\,234\,621}{3}}}{5\,110\,875\,222\,870\,301\,507\,065\,971\,913\,830\,033\,423\,949 \pi^{1/8}}$$

Out[•]=

[illegible]

$$\text{In}[*]:= \frac{\frac{\frac{G}{4 \pi \epsilon_0} \frac{1}{m_e}}{\text{Quantity}\left[(5+\alpha) \cdot 10^{-12}, \frac{\text{"Meters"}^6}{\text{"Seconds"}^5}\right]}}{2 \sqrt{\pi} h} \quad 1.0$$

$$\text{Solve}\left[\frac{\frac{G \mu_0}{4 \pi \epsilon_0}}{x} == \text{Quantity}\left[299\,792\,458, \frac{\text{"Henries"}}{\text{"Kilograms"}}\right], x\right]$$

$$\frac{7\,964\,059\,244\,164\,875\,113\,813}{125} \times 2^{3/4} \sqrt{\frac{64\,241\,241}{6\,310\,543}} \pi^{13/8} \text{m}^4 / (\text{kg s}^4)$$

$$\frac{\sqrt{2} \alpha}{125} \frac{7\,964\,059\,244\,164\,875\,113\,813}{125} \times 1.0$$

mP

Out[*]=

$$3.089 \times 10^8 \text{ s}^{3/2} \text{J}^{3/2} / (\text{kg}^{3/2} \text{mHz}^{3/2} \text{C}^2)$$

Equal: Invalid comparison of quantities $3.91751 \times 10^{-71} \text{m}^4 \text{J}^2 / (\text{kg}^2 \text{Hz}^2 \text{s}^2 \text{C}^2)$ and 299792458H/kg with incompatible units.

Solve: 4.` is not a valid variable. [i](#)

Out[*]=

$$\text{Solve}\left[3.91751 \times 10^{-71} \text{m}^4 \text{J}^2 / (\text{kg}^2 \text{Hz}^2 \text{s}^2 \text{C}^2) == 299\,792\,458 \text{H/kg}, 4.\right]$$

Out[*]=

$$\frac{7\,964\,059\,244\,164\,875\,113\,813}{125} \times 2^{3/4} \sqrt{\frac{64\,241\,241}{6\,310\,543}} \pi^{13/8} \text{m}^4 / (\text{kg s}^4)$$

Out[*]=

$$\frac{267\,029\,439 \sqrt{\frac{64\,241\,241 \pi}{6\,310\,543}}}{12\,500\,000\,000}$$

Out[*]=

$$6.37125 \times 10^{19}$$

Out[*]=

$$\frac{7 \sqrt{405\,397\,113\,703\,863}}{2\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

$$\text{In}[*]:= \text{UnitSimplify}\left[\text{Quantity}\left[9.97496 \times 10^{-22}, \frac{\text{"Joules"}^2 \text{"Seconds"}^2}{\text{"Coulombs"}^2 \text{"Hertz"}^2 \text{"Kilograms"} \text{"Meters"}^2}\right]\right]$$

Out[*]=

$$9.97496 \times 10^{-22} \text{H}$$

$$\text{In}[*]:= \text{UnitSimplify}\left[\text{Quantity}\left[0.599858, \frac{\text{"Joules"} \text{"Meters"}^4}{\text{"Coulombs"}^2 \text{"Hertz"} \text{"Kilograms"} \text{"Seconds"}^3}\right]\right]$$

Out[*]=

$$0.599858 \text{m}^6 / (\text{s}^6 \text{A}^2)$$

```
In[ ]:= UnitSimplify[Quantity[1.66706,  $\frac{\text{"Coulombs"}^2 \text{"Hertz" "Kilograms" "Seconds"}^3}{\text{"Joules" "Meters"}^4}$ ]]
```

```
Out[ ]:=
```

$1.66706 \text{ s}^6 \text{A}^2/\text{m}^6$

```
In[ ]:= mμ 1.0
```

```
Round[SetPrecision[ $\frac{\text{mP}}{\text{m}\mu}$ , 64]]
```

mP

115 550 496 528

```
Out[ ]:=
```

$1.88353 \times 10^{-19} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$

```
Out[ ]:=
```

115 550 496 528

```
Out[ ]:=
```

$\frac{7 \sqrt{\frac{135 132 371 234 621}{3}}}{77 033 664 352 000 000 000 000 000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$

```
In[ ]:=  $\frac{\frac{\mathbf{V}}{\mathbf{a}} \frac{\mathbf{\Gamma}}{\mathbf{me}} \frac{(-\mathbf{e})}{\mathbf{\Gamma}} \frac{\mathbf{c}^2}{\mathbf{G}} \frac{\mathbf{\alpha}}{\mathbf{\alpha}} \frac{\mathbf{\delta}}{\mathbf{\gamma}}}{\frac{\mathbf{V}}{\mathbf{a}} \frac{\mathbf{\Gamma}}{\mathbf{me}} \frac{(-\mathbf{e})}{\mathbf{\Gamma}} \frac{\mathbf{c}^2}{\mathbf{G}} \frac{\mathbf{\alpha}}{\mathbf{\alpha}}}}{1} == \frac{\mathbf{V}}{\mathbf{a}} \frac{\mathbf{\Gamma}}{\mathbf{me}} \frac{(-\mathbf{e})}{\mathbf{\Gamma}} \frac{\mathbf{c}^2}{\mathbf{G}} \frac{\mathbf{\alpha}}{\mathbf{\alpha}}}$ 
```

$\frac{\frac{\mathbf{V}}{\mathbf{a}} \frac{\mathbf{\Gamma}}{\mathbf{me}} \frac{(-\mathbf{e})}{\mathbf{\Gamma}} \frac{\mathbf{c}^2}{\mathbf{G}} \frac{\mathbf{\alpha}}{\mathbf{\alpha}}}{\frac{\mathbf{V}}{\mathbf{a}} \frac{\mathbf{\Gamma}}{\mathbf{me}} \frac{(-\mathbf{e})}{\mathbf{\Gamma}} \frac{\mathbf{c}^2}{\mathbf{G}} \frac{\mathbf{\alpha}}{\mathbf{\alpha}}}} == - \frac{\mathbf{mP}}{\mathbf{me}}$

$\frac{\frac{\mathbf{V}}{\mathbf{a}} \frac{\mathbf{\Gamma}}{\mathbf{me}} \frac{(-\mathbf{e})}{\mathbf{\Gamma}} \frac{\mathbf{c}^2}{\mathbf{G}} \frac{\mathbf{\alpha}}{\mathbf{\alpha}}}{\frac{\mathbf{V}}{\mathbf{a}} \frac{\mathbf{\Gamma}}{\mathbf{me}} \frac{(-\mathbf{e})}{\mathbf{\Gamma}} \frac{\mathbf{c}^2}{\mathbf{G}} \frac{\mathbf{\alpha}}{\mathbf{\alpha}}}} == - \frac{\mathbf{mP}}{\mathbf{me}}$

```
Out[ ]:=
```

True

```
Out[ ]:=
```

False

```
Out[ ]:=
```

False

```

In[*]:= 
$$\frac{v}{a} \frac{\Gamma}{me} \frac{(-e)}{\Gamma} \frac{c^2}{\alpha \gamma} \frac{\alpha \delta}{G}$$

570 836 156 801 112 016 961 068 199 501 089 647 330 590 721  $\alpha$ 
 $\alpha \gamma$ 
 $\alpha \delta$ 
2 614 831 344 073 286 865 523 707 785 096 251 002 480 532 475 129 241 214 999 727 675 646 681  $\pi$ 


---


 $\alpha \gamma$ 
570 836 156 801 112 016 961 068 199 501 089 647 330 590 721

```

```

Out[*]=

$$v - \left( \left( 3 \, 641 \, 886 \, 054 \, 510 \, 154 \, 286 \, 871 \, 986 \, 578 \, 744 \, 051 \, 281 \, 159 \, 486 \, 710 \, 102 \, 664 \, 136 \, 351 \, 293 \, 077 \, 479 \, 153 \, 734 \, 513 \right. \right. \\ \left. \left. \sqrt{\frac{3}{135 \, 132 \, 371 \, 234 \, 621}} \pi^{7/8} \right) / \right. \\ \left. \left( 786 \, 410 \, 781 \, 394 \, 715 \, 435 \, 000 \, 000 \, 000 \, 000 \times 2^{3/4} \right) \right) \sqrt{\text{Hz s}^{3/2} \text{C}} / \left( \sqrt{\text{kg m}} \sqrt{\text{J}} \right)$$


```

```

Out[*]=
2 614 831 344 073 286 865 523 707 785 096 251 002 480 532 475 129 241 214 999 727 675 646 681  $\pi$ 


---


1 972 044 687 500 000 000 000 000 000

```

```

Out[*]=
4 580 703 784 999 263 461 548 761  $\pi$ 

```

```

Out[*]=
1 972 044 687 500 000 000 000 000 000

```

```

Out[*]=
570 836 156 801 112 016 961 068 199 501 089 647 330 590 721

```

```

Out[*]=
570 836 156 801 112 016 961 068 199 501 089 647 330 590 721

```

```

In[*]:= FactorInteger[570 836 156 801 112 016 961 068 199 501 089 647 330 590 721]

```

```

Out[*]=
{{3, 2}, {19, 2}, {63 221 999, 2}, {6 629 986 705 273, 2}}

```

```

In[*]:= 
$$\frac{\alpha h}{2 \pi \Gamma} == c$$


```

```

Out[*]=
True

```


$$\begin{array}{r} 91\,489\,613\,572\,766\,928\,000\,000\,\pi \\ \hline \alpha\gamma \\ 91\,489\,613\,572\,766\,928\,000\,000\,\pi \\ \hline \alpha\gamma \end{array}$$

$$\begin{array}{r}
 6\,310\,543\,\alpha\gamma \\
 801\,088\,317\,\alpha\gamma \\
 801\,088\,317 \times 6\,310\,543 \\
 \hline
 \alpha\gamma \\
 801\,088\,317 \times 6\,310\,543 \\
 \hline
 \alpha\delta \\
 801\,088\,317 \times 6\,310\,543 \quad \frac{4\,\pi}{\alpha} \\
 \hline
 \alpha\delta\,\alpha\gamma \qquad \alpha \\
 \qquad \qquad \frac{4\,\pi\,\alpha}{\alpha\delta\,\alpha\gamma} \\
 801\,088\,317 \times 6\,310\,543 \\
 4\,\pi\,801\,088\,317 \\
 \hline
 \alpha\delta\,\alpha\gamma\,\text{mP}\,\text{LP} \\
 \frac{4\,\pi\,801\,088\,317}{\alpha\delta\,\alpha\gamma} - 3.5176729417
 \end{array}$$

$$\begin{array}{r}
\Gamma \text{ 1.0} \\
\frac{4 \pi \text{ 801 088 317}}{\alpha \delta \alpha \gamma \text{ mP lP}} / \frac{1}{\Gamma} \\
\frac{4 \pi \text{ 801 088 317}}{\alpha \delta \alpha \gamma} \\
\frac{4 \pi \text{ 801 088 317} \times 6 \text{ 310 543}}{\alpha \delta \alpha \gamma} \\
\frac{4 \pi \text{ 801 088 317}}{\alpha \delta \alpha \gamma \text{ mP lP}} / \frac{1}{\Gamma} \\
\frac{4 \pi}{\alpha \delta \alpha \gamma \text{ mP lP}} / \frac{1}{\Gamma} \\
972 \text{ 240 062 374 243 164 062 500 000 000 000 000 000 000 000 000 000 000} \\
\hline
801 \text{ 088 317 } \alpha \delta \text{ 6 310 543}
\end{array}$$

$$\frac{\frac{1}{\alpha\delta\alpha\gamma\text{ mP lP}} \bigg/ \frac{1}{\Gamma}}{\frac{\pi}{\Gamma} \bigg/ \frac{4\pi}{\alpha\delta\alpha\gamma\text{ mP lP}}}$$

```
Out[•]=
```

16 000 000
801 088 317

$Out[\bullet]=$

16 000 000
801 088 317

Out[*]=

28 906 728 205 500 607 042 432 302 887 223 π

Out[*]=

3 669 548 285 800 589 812 651 691 162 925 237 π

Out[*]=

$$\frac{6\,310\,543}{5\,718\,100\,848\,297\,933\,\pi}$$

Out[*]=

$$\frac{801\,088\,317}{312\,500\,000\,000\,000\,000\,000}$$

Out[*]=

$$\frac{25\,242\,172}{26\,192\,926\,198\,805\,840\,899\,633\,571\,228\,338\,935\,011\,013\,\pi}$$

Out[*]=

$$\frac{801\,088\,317\,\pi}{154\,065\,991\,210\,937\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000}$$

Out[*]=

$$\frac{10\,240\,000\,000\,000\,000\,000\,000\,\pi}{31\,901\,702\,360\,570\,162\,399\,133}\,\text{mHz}/(\text{s J})$$

Out[*]=

1.1521 mHz / (s J)

Out[*]=

$$\frac{801\,088\,317\,\pi}{972\,240\,062\,374\,243\,164\,062\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000}$$

Out[*]=

$$\frac{1}{2\,819\,087\,600\,118\,795\,547\,470\,234\,375\,000\,000\,000\,000\,000}$$

Out[*]=

$$\frac{1}{446\,726\,628\,773\,276\,015\,625\,000\,000\,000\,000\,000}$$

Out[*]=

$$\frac{801\,088\,317\,\pi}{972\,240\,062\,374\,243\,164\,062\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000}$$

Out[*]=

$$\frac{\pi}{972\,240\,062\,374\,243\,164\,062\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000}$$

Out[*]=

$$\frac{78\,125\,000\,000\,000\,000\,000}{801\,088\,317}$$

Out[*]=

$$\frac{972\,240\,062\,374\,243\,164\,062\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000}{\pi}$$

Out[*]=

972 240 062 374 243 164 062 500 000 000 000 000 000 000 000 000 000 000

$$\begin{aligned}
 \ln[*] &:= \frac{972\,240\,062\,374\,243\,164\,062\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000}{6\,310\,543\,\alpha\delta\,c\,1.0\,c} \\
 c &\sqrt{\frac{972\,240\,062\,374\,243\,164\,062\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000}{6\,310\,543\,\alpha\delta\,c\,c}} \frac{\alpha}{2} = 6\,250\,000\,000\,\sqrt{\alpha} \\
 \frac{6\,250\,000\,000\,\sqrt{\alpha}}{12\,500\,000\,000} &= \frac{2}{\pi} \frac{4\pi}{\alpha\delta\,\alpha\gamma\,mP\,1P} = 972\,240\,062\,374\,243\,164\,062\,500\,000\,000\,000\,000\,000\,000\,000\,000\,000
 \end{aligned}$$

$$\text{Out[*]} = 869.258\,s^2/m^2$$

$$\text{Out[*]} = \text{True}$$

$$\text{Out[*]} = \frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241\,\pi}{12\,621\,086}}$$

$$\text{Out[*]} = 6\,250\,000\,000$$

$$\text{Out[*]} = \text{True}$$

$$\begin{aligned}
 \ln[*] &:= \left(\frac{\pi}{\Gamma} \frac{4\pi}{\alpha\delta\,\alpha\gamma\,mP\,1P} \right) = \frac{1P\,mP\,\alpha\gamma\,\alpha\delta}{4 \left(\frac{\alpha\gamma}{2\pi c} h \right)} \\
 \left(\frac{\pi}{\Gamma} \frac{4\pi}{\alpha\delta\,\alpha\gamma\,mP\,1P} \right) &= \frac{c\,\Gamma\,\pi\,\alpha\delta^2}{2\,h\,\alpha} \\
 \left(\frac{\pi}{\Gamma} \frac{4\pi}{\alpha\delta\,\alpha\gamma\,mP\,1P} \right) &= \frac{c\,\pi\,\alpha\delta^3\,\Gamma}{2\,h\,\alpha\gamma} \\
 \frac{\alpha\gamma}{\alpha\delta} &= \frac{\alpha\,h}{2\pi c} \\
 c &\sqrt{\frac{\frac{c\,\pi\,\alpha\delta^3\,\Gamma}{2\,h\,\alpha\gamma} \frac{\alpha\gamma}{\alpha\delta}}{6\,310\,543\,\alpha\delta\,c\,c\,2}} \\
 \frac{1}{2} c &\sqrt{\frac{\pi}{6\,310\,543}} \sqrt{\frac{\alpha\delta \frac{\alpha\,h}{2\pi c} \pi}{c\,h}} \\
 \frac{1}{2} c &\sqrt{\frac{\pi}{6\,310\,543}} \sqrt{\frac{\alpha\delta\,\Gamma}{c\,h}} \\
 6\,250\,000\,000 &\sqrt{\alpha}
 \end{aligned}$$

Out[⌘]=

True

Out[⌘]=

True

Out[⌘]=

True

Out[⌘]=

$$\frac{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}{1\,972\,044\,687\,500\,000\,000\,000\,000\,000}$$

Out[⌘]=

True

Out[⌘]=

$$\frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241\,\pi}{12\,621\,086}}$$

Out[⌘]=

$$\frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241}{12\,621\,086}} \pi$$

Out[⌘]=

$$\frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241\,\pi}{12\,621\,086}}$$

Out[⌘]=

$$\frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241\,\pi}{12\,621\,086}}$$

In[⌘]:=

$$\frac{1\text{P mP}}{2\,\Gamma} \alpha$$
$$\frac{1\text{P mP}}{2\,\Gamma\,100\,\sqrt[4]{4\,\pi}}$$
$$100\,\sqrt[4]{4\,\pi}$$

Out[⌘]=

$$\frac{1}{2}$$

Out[⌘]=

$$\frac{4\,930\,111\,718\,750\,000\,000\,000\,000\,\sqrt{2}}{4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi^{5/4}}$$

Out[⌘]=

$$100\,\sqrt{2}\,\pi^{1/4}$$

In[⌘]:=

$$\frac{1}{2}\,\mathbf{c}\,\sqrt{\frac{\pi}{6\,310\,543}}\,\sqrt{\frac{\alpha\delta\,\Gamma}{\mathbf{c}\,\mathbf{h}}}$$
$$6\,250\,000\,000\,\sqrt{\alpha}$$
$$\frac{1}{2}\,\mathbf{c}\,\sqrt{\frac{\pi}{6\,310\,543}}\,\sqrt{\frac{\alpha\delta\,\Gamma}{\mathbf{c}\,\mathbf{h}}} == 6\,250\,000\,000\,\sqrt{\alpha}$$
$$6\,310\,543$$

$$\frac{\Gamma}{1P \alpha} == mP$$

$$\frac{\Gamma}{mP \alpha} == 1P$$

Out[]=

$$\frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241 \pi}{12\,621\,086}}$$

Out[]=

$$\frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241 \pi}{12\,621\,086}}$$

Out[]=

True

Out[]=

6 310 543

Out[]=

267 029 439

Out[]=

986 022 343 750 000 000 000 000 000

Out[]=

1 972 044 687 500 000 000 000 000 000

Out[]=

68.518

Out[]=

$$\frac{9\,860\,223\,437\,500\,000\,000\,000\,000 \times 2^{3/4}}{4\,580\,703\,784\,999\,263\,461\,548\,761 \pi^{9/8}}$$

Out[]=

$$\frac{9\,860\,223\,437\,500\,000\,000\,000\,000 \times 2^{3/4}}{4\,580\,703\,784\,999\,263\,461\,548\,761 \pi^{9/8}}$$

Out[]=

1 mC / (kg s)

Out[]=

$$\frac{986\,022\,343\,750\,000\,000\,000\,000\,000}{4\,580\,703\,784\,999\,263\,461\,548\,761 \pi}$$

Out[]=

$$\frac{267\,029\,439}{2} \sqrt{\frac{64\,241\,241 \pi}{12\,621\,086}}$$

Out[]=

$$\frac{1}{50 \times 2^{1/4} \pi^{1/8}} \text{ mC / (kg s)}$$

Out[]=

1

Out[]=

$$50 \times 2^{1/4} \pi^{1/8} \text{ kg s / (mC)}$$

Out[]=

True

Out[]=

$$\frac{1}{100 \times 2^{1/4} \pi^{1/8}} \text{ m C / (kg s)}$$

Out[]=

True

Out[]=

$$6.30458 \times 10^9 \sqrt{\text{kg}} \sqrt{\text{Hz}} \sqrt{\text{s}} / (\text{C} \sqrt{\text{J}})$$

Out[]=

$$\frac{7 \sqrt{405397113703863}}{200000000000000 \pi^{3/4}} \text{ kg}^{3/2} \text{ s}^{3/2} \sqrt{\text{J}} / (\text{m}^2 \sqrt{\text{Hz}} \text{ C})$$

Out[]=

$$4.18547 \times 10^{-23}$$

Out[]=

$$\frac{7 \sqrt{405397113703863}}{200000000000000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

Out[]=

$$\frac{\sqrt{\frac{18931629}{21413747}}}{299792458000000000000000000000000 \times 2^{3/4} \pi^{1/8}} \sqrt{\text{s}} \sqrt{\text{J}} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

Out[]=

1

Out[]=

True

Out[]=

True

$$\text{In[]:=} \frac{1}{2\alpha}$$

Out[]=

$$\frac{986022343750000000000000000}{4580703784999263461548761 \pi}$$

$$\text{In[]:=} \text{Ezn[me, 1, 4] 1.0}$$

Out[]=

$$217.691 \text{ eV}$$

Out[1069]=

$$(-1)^{1/4}$$

Out[1070]=

$$-\frac{i \pi^{3/4}}{25\,000\,000\,000 \sqrt{2}} \text{ m}^3 / (\text{kg s}^2)$$

Out[1071]=

$$-6.674325731836412459517924452234 \times 10^{-11} i \text{ m}^3 / (\text{kg s}^2)$$

Out[1073]=

$$-1.3901788898528296790927272448668 \times 10^{51} i \text{ m} / \text{s}^2$$

$$f = \text{UnitConvert}\left[\sqrt{\frac{\text{Gr M}}{d^2} \frac{\alpha}{d \, 2 \, \pi}}, \text{"Hertz"}\right];$$

Print["f = ", SetPrecision[f, 32]]

In[]:= UnitSimplify[f]

Out[]:=

$$-\frac{1}{39\,822\,952\,954\,849} \times \frac{1\,056\,443\,862\,151\,977\,096\,474\,994\,945\,960\,914\,065\,435\,712\,847\,388\,905\,717\,918\,649\,165\,376\,000\,000\,000}{000\,000\,000\,000\,000\,000\,000\,000} i \sqrt{2} \pi^{1/4} \text{ per second}^2$$

In[]:= Clear[n]

$$\frac{\pi^{3/4}}{25\,000\,000\,000 \sqrt{2}} - \frac{i \pi^{3/4}}{25\,000\,000\,000 \sqrt{2}}$$

Out[]:=

$$\frac{\left(\frac{1}{25\,000\,000\,000} - \frac{i}{25\,000\,000\,000}\right) \pi^{3/4}}{\sqrt{2}}$$

In[]:=

► convert to exponential $\frac{\left(\frac{1}{25\,000\,000\,000} - \frac{i}{25\,000\,000\,000}\right) \pi^{3/4}}{\sqrt{2}}$

Out[]:=

$$\frac{\pi^{3/4} e^{\frac{1}{4} i (-\pi)}}{25\,000\,000\,000}$$

In[]:=

$$\text{Conjugate}\left[\frac{\left(\frac{1}{25\,000\,000\,000} - \frac{i}{25\,000\,000\,000}\right) \pi^{3/4}}{\sqrt{2}}\right]$$

Out[]:=

$$\frac{\left(\frac{1}{25\,000\,000\,000} + \frac{i}{25\,000\,000\,000}\right) \pi^{3/4}}{\sqrt{2}}$$

$$\text{In}[*]:= \frac{\pi^{3/4} e^{\frac{1}{4} i (-\pi)}}{25\,000\,000\,000}$$

$$\text{Out}[*]= \frac{e^{-\frac{i \pi}{4}} \pi^{3/4}}{25\,000\,000\,000}$$

$$\text{In}[*]:= \text{RootApproximant}\left[\frac{e^{-\frac{i \pi}{4}} \pi^{3/4}}{25\,000\,000\,000}\right]$$

$$\text{Out}[*]= \frac{275\,925\,716\,221\,129\,070\,306 - i \sqrt{76\,135\,000\,872\,130\,935\,862\,970\,482\,392\,803\,746\,594\,818}}{4\,134\,136\,200\,529\,702\,725\,388\,613\,373\,022}$$

$$\text{In}[*]:= \text{Gr} \frac{\text{mP}}{\text{1P}} == -i \, c^2$$

$$\text{G} \frac{\text{mP}}{\text{1P}} == c^2$$

$$\text{Out}[*]= \text{True}$$

$$\text{Out}[*]= \text{True}$$

$$\frac{1 + \sqrt{5}}{2.0}$$

$$M = 1.4 \text{ SunMass}$$

$$\text{length} = \text{Quantity}[1 * 10^2, \text{"Meters"}]$$

$$\text{Out}[*]= 1.61803$$

$$\frac{4 \pi}{(100 \sqrt{4 \pi})^2} * 10^{-7}$$

$$\text{Out}[*]= 6.67433 \times 10^{-11}$$

$$\text{In}[*]:= \text{1P}$$

$$\text{Out}[*]= \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz})$$

$$\text{In}[*]:= \text{Quantity}\left[1, \frac{\sqrt{\text{"Joules"}} \sqrt{\text{"Seconds"}}}{\sqrt{\text{"Hertz"}} \sqrt{\text{"Kilograms"}}}\right]$$

$$\text{Out}[*]= 1 \sqrt{s} \sqrt{J} / (\sqrt{kg} \sqrt{Hz})$$


```
Plot3D[Re[kappaField[x, y]], {x, -1, 1}, {y, -1, 1}, PlotRange -> All,
  AxesLabel -> {"x", "y", "Re[κ]"}, ColorFunction -> "SolarColors", Mesh -> Full]
```

$$\sqrt[4]{\left(1 - \frac{4 G m}{c^2 \left(\frac{m P 1 P}{m \alpha} \left(\sqrt{\frac{m \alpha \left(-\frac{1 P}{2} + R m \right)}{1 P m P}} \right)^2 \right)}\right)};$$

$$\sqrt{\frac{m \alpha \left(-1 P + R m \right)}{1 P m P}};$$

$$\left(1 - \frac{4 G M}{c^2 \left(-\frac{1 P}{2} + R \right)}\right)^{1/4};$$

$$\sqrt[4]{\left(1 - \frac{4 G m}{c^2 \left(\frac{m P 1 P}{m \alpha} \left(\sqrt{\frac{\left(-\frac{1 P}{2} + \text{Quantity}[R, \text{"Meters"} \right) - \frac{1 P}{2}} m \alpha}{m P 1 P} \right)^2 \right)}\right)};$$

$$\left(1 - \frac{4 G r m}{c^2 \left(1 P + \left(\frac{m P 1 P}{M \alpha} \left(\sqrt{\frac{r m \alpha}{m P 1 P}} \right)^2 \right) \right)}\right)^{1/4}$$

$$\text{kappa}[R_]:= \left(1 - \frac{4 G r m}{c^2 \left(1 P + \left(\frac{m P 1 P}{M \alpha} \left(\sqrt{\frac{R m m \alpha}{m P 1 P}} \right)^2 \right) \right)}\right)^{1/4} + \left(1 - \frac{4 G r m}{c^2 \left(1 P - \left(\frac{m P 1 P}{M \alpha} \left(\sqrt{\frac{R m m \alpha}{m P 1 P}} \right)^2 \right) \right)}\right)^{1/4}$$

$$\left(1 - \frac{4 G r m}{c^2 \left(1 P + \left(\frac{m P 1 P}{M \alpha} \left(\sqrt{\frac{r m \alpha}{m P 1 P}} \right)^2 \right) \right)}\right)^{1/4};$$

In[1042]:=

$$m = mp;$$

$$l = \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}};$$

$$\text{kappa}[R_] := \left(1 - \frac{4 G m}{c^2 (-1P + R m)} \right)^{1/4}$$

(*Convert to 2D radial field*)

$$\text{kappaField}[x_, y_] := \text{Module}[\{r = \text{Sqrt}[x^2 + y^2]\}, \text{kappa}[r]]$$

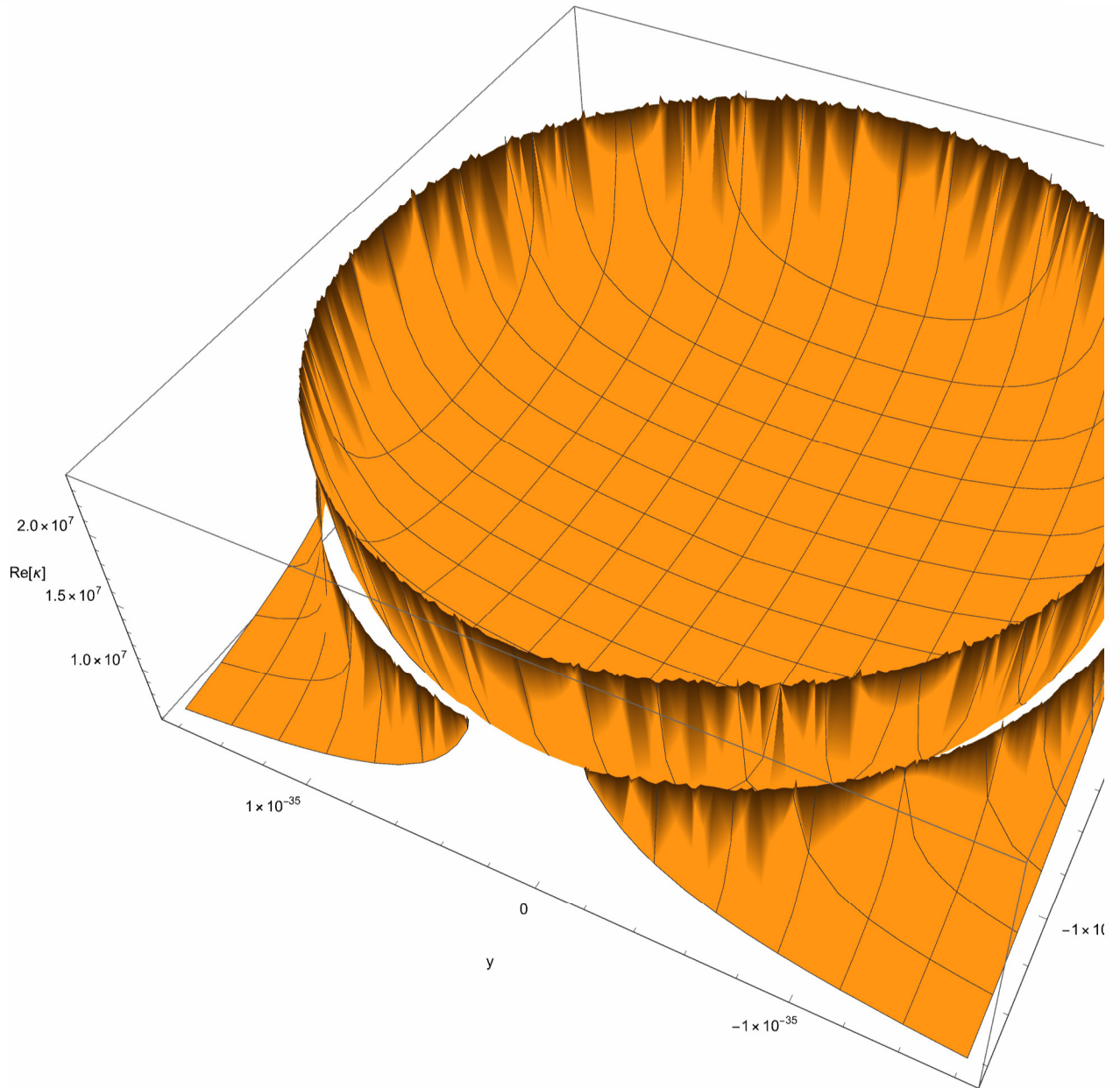
(*3D Plot of the real component of curvature*)

(*Define curvature function in radial distance r*)

$$\text{Plot3D}[\text{Re}[\text{kappaField}[x, y]], \{x, -1, 1\}, \{y, -1, 1\}, \text{PlotRange} \rightarrow \text{All}, \text{AxesLabel} \rightarrow \{ "x", "y", "Re[\kappa] " \}, \text{Mesh} \rightarrow \text{Full}]$$

$$\text{Clear}[m, M, r, R, q, Q]$$

Out[1046]=




```
(*kTable=Table[ $\sqrt[4]{1 - \frac{4G M}{c^2 \left( \frac{mP \text{ } 1P}{M \alpha} \left( \sqrt{\frac{(1P + \text{Quantity}[R, \text{"Meters"}]) M \alpha}{mP \text{ } 1P}} \right)^2}} \right)}, \{R, -50, 50\}]*)$ 
```

```
ListPlot[{Tooltip[{Re[#1], Im[#1]}] &} /@ kTable]
```

```
Clear[R]
```

```
GTable =
```

```
Table[ $\left( \frac{4 G M}{c^2 \left( \frac{mP \text{ } 1P}{m \alpha} \left( \sqrt{\frac{\left( -\frac{1P}{2} + \text{Quantity}[R, \text{"Meters"} \right) - \frac{1P}{2}} M \alpha}{mP \text{ } 1P} \right)^2} \right)} + \frac{4 Gr M}{c^2 \left( \frac{mP \text{ } 1P}{m \alpha} \left( \sqrt{\frac{\left( -\frac{1P}{2} + \text{Quantity}[R, \text{"Meters"} \right) - \frac{1P}{2}} M \alpha}{mP \text{ } 1P} \right)^2} \right)} \right), \{R, -21, 21\}]$ 
```

```
ListPlot[{Tooltip[{Re[#1], Im[#1]}] &} /@ GTable]
```

```
Clear[R]
```

Out[998]=

$$\frac{7 \sqrt{405397113703863}}{2000000000000000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{s} \sqrt{J} / (m \sqrt{\text{Hz}})$$

Out[999]=

$$\frac{\sqrt{\frac{18931629}{21413747}}}{299792458000 \times 2^{3/4} \pi^{1/8}} \sqrt{s} \sqrt{J} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

Out[1000]=

$$\frac{\sqrt{\frac{18931629}{21413747}}}{149896229000 \times 2^{3/4} \pi^{1/8}} \sqrt{s} \sqrt{J} / (\sqrt{\text{kg}} \sqrt{\text{Hz}})$$

Out[1001]=

$$\frac{267029439 \sqrt{\frac{64241241 \pi}{12621086}}}{12500000000}$$

Out[1002]=

$$\frac{\sqrt{\frac{18931629}{21413747}}}{299792458000} m$$

Out[1003]=

$$(-3)^{1/4}$$

Out[1004]=

$$-\frac{i \pi^{3/4}}{25000000000 \sqrt{6}} m^3 / (\text{kg } s^2)$$

Out[1005]=

True

Out[1006]=

$$\begin{aligned}
& \left\{ 1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \\
& \quad \left. \left. \left(-50 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right\}^{1/4}, \\
& \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \\
& \quad \left. \left. \left(-49 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right\}^{1/4}, \\
& \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \\
& \quad \left. \left. \left(-48 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right\}^{1/4}, \\
& \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \\
& \quad \left. \left. \left(-47 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right\}^{1/4}, \\
& \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \\
& \quad \left. \left. \left(-46 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right\}^{1/4}, \\
& \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.
\end{aligned}$$

$$\left(-45 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-44 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-43 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-42 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-41 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-40 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-39 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-38 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-37 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-36 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-35 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(-34 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-33 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-32 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-31 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-30 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-29 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-28 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-27 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-26 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-25 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-24 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(-23 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-22 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-21 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-20 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-19 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-18 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-17 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-16 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-15 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-14 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-13 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(-12 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-11 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-10 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-9 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-8 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$
$$\left(1 - \left(\sqrt{\frac{18931629}{21413747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$
$$\left. \left. \left(-7 + \frac{\sqrt{\frac{18931629}{21413747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-6 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-5 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-4 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-3 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.-2 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(-1 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$

$$(-3)^{1/4}, \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(1 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(2 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(3 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(4 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(5 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.6 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.7 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.8 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.9 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.10 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(11 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(12 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(13 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(14 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(15 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(16 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right) \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(17 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(18 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(19 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(20 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(21 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(22 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(23 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(24 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(25 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(26 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(27 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.28 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.29 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.30 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.31 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.32 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(33 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(34 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(35 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(36 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(37 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(38 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(39 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(40 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(41 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(42 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left. \left(43 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}}\right) \pi^{1/8}\right)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right.$$

$$\left(44 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \Bigg)^{1/4},$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right.$$

$$\left. \left. \left(45 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(46 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(47 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

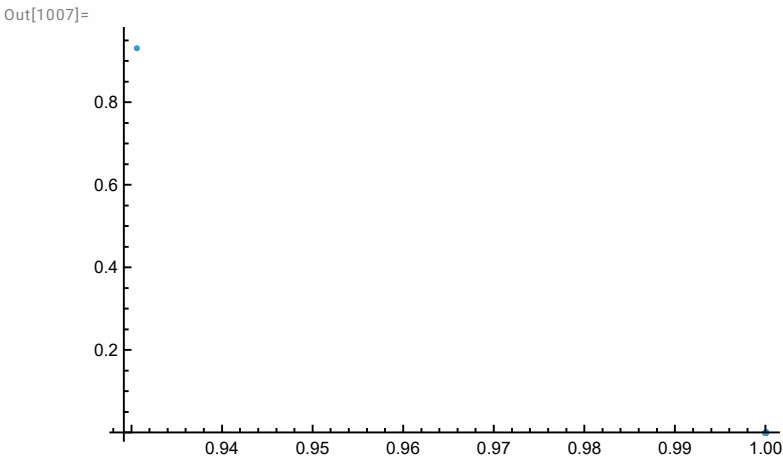
$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

$$\left. \left. \left(48 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left. \left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}} \right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \right. \right. \right.$$

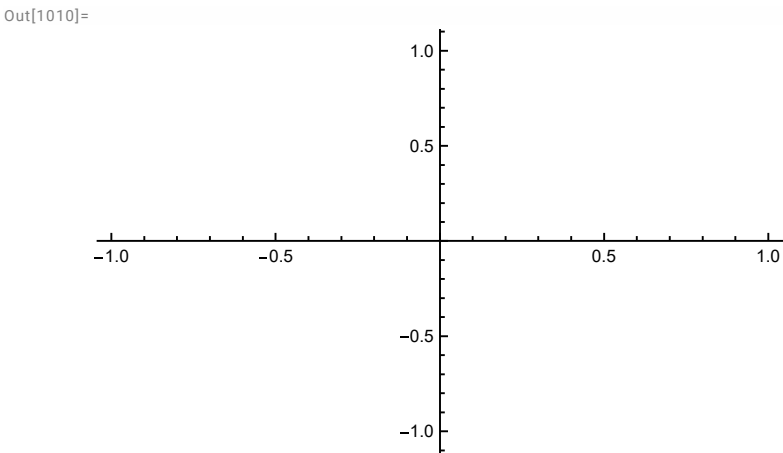
$$\left. \left. \left(49 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \right) \pi^{1/8} \right)^{1/4}, \right.$$

$$\left(1 - \left(\sqrt{\frac{18\,931\,629}{21\,413\,747}}\right) / \left(7\,494\,811\,450\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4}\right.\right. \\ \left.\left.50 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000 \times 2^{3/4} \pi^{1/8}} \pi^{1/8}\right)^{1/4} \right\}$$



Out[1009]=

$$\left\{ m \left(\frac{8\,000\,000\,000\,000\,000 \, i \, 2^{1/4} \pi^{7/8}}{63 \sqrt{135\,132\,371\,234\,621}} m \sqrt{\text{Hz}} / (\sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}}) \right) + \right. \\ \left. m \left(- \frac{8\,000\,000\,000\,000\,000 \times 2^{1/4} \pi^{7/8}}{21 \sqrt{405\,397\,113\,703\,863}} m \sqrt{\text{Hz}} / (\sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}}) \right) \right\}$$



In[338]:=

RootReduce[(−7)^{1/4}]

Out[338]=

$4 \sqrt[4]{-7} \, 1.15... + 1.15... \, i$

$$\ln[\cdot] := \frac{\left(\frac{1}{7\,948\,811\,450\,000\,000\,000\,000\,000\,000\,000} - \frac{i}{7\,948\,811\,450\,000\,000\,000\,000\,000\,000\,000} \right) \sqrt{\frac{18\,931\,629}{21\,413\,747}}}{2^{3/4} \left(50 + \frac{\sqrt{\frac{18\,931\,629}{21\,413\,747}}}{29\,979\,245\,800\,000\,000\,000\,000\,000\,000\,000\,000\,000} \cdot 2^{3/4} \pi^{1/8} \right)} \pi^{1/8}$$

[illegible]

$$\ln[*] := \mathbf{r} = \text{UnitConvert}\left[\frac{1\text{P}}{\Delta\alpha\left[\sqrt{\frac{\text{length M}\alpha}{\text{mP } 1\text{P}}}\right]}, \text{"Meters"}\right] 1.0$$

$$r / \left(\frac{m^P l^P}{M \alpha} n^2 \right) 1.0$$

$$G \frac{M}{\left(\frac{m_P l_P}{M \alpha} n^2 \right) c^2}$$

$$\text{Gr} = \frac{M}{\left(\frac{m_P l_P}{M \alpha} n^2 \right) c^2}$$

$$\text{Table} \left[\left(G \frac{M}{\left(\frac{m_P \text{LP}}{M \alpha} n^2 \right) c^2} + Gr \frac{M}{\left(\frac{m_P \text{LP}}{M \alpha} n^2 \right) c^2} \right) \left(\frac{\text{LP } m_P}{M r \alpha} \right), \left\{ n, \frac{1}{\sqrt{\frac{\text{length } M \alpha}{m_P \text{LP}}}}, \frac{1}{\sqrt{\frac{\text{length } M \alpha}{m_P \text{LP}}}} + 50 \right\} \right]$$

$$\frac{mP \text{ lP}}{M \alpha} \left(\sqrt{\frac{\text{length } M \alpha}{mP \text{ lP}}} \right)^2$$

$$\text{Table} \left[\left(\text{G} \frac{\text{M}}{\left(\frac{\text{mP}}{\text{M}} \frac{\text{1P}}{\alpha} \text{n}^2 \right) \text{c}^2} + \text{Gr} \frac{\text{M}}{\left(\frac{\text{mP}}{\text{M}} \frac{\text{1P}}{\alpha} \text{n}^2 \right) \text{c}^2} \right) \left(\frac{\text{1P mP}}{\text{M} \frac{\text{mP}}{\text{M}} \frac{\text{1P}}{\alpha} \left(\sqrt{\frac{\text{length M} \alpha}{\text{mP 1P}}} \right)^2} \right) \right],$$

$$\left\{ n, \frac{1}{\sqrt{\frac{\text{length } M \alpha}{\text{mP } \text{IP}}}}, \frac{1}{\sqrt{\frac{\text{length } M \alpha}{\text{mP } \text{IP}}}} + 50 \right\}$$

$Out[\bullet]=$ 24.5647 m

Out[•]= 2.45647

Out[•]= 206.727

$Out[\bullet]=$

0. - 10.1791 i

Out[*]=

```
{4.85991 × 1073 - 2.39298 × 1072 i, 84.1562 - 4.14379 i, 21.0391 - 1.03595 i,
9.35069 - 0.460422 i, 5.25976 - 0.258987 i, 3.36625 - 0.165752 i, 2.33767 - 0.115105 i,
1.71747 - 0.0845672 i, 1.31494 - 0.0647468 i, 1.03897 - 0.051158 i, 0.841562 - 0.0414379 i,
0.695506 - 0.0342462 i, 0.584418 - 0.0287763 i, 0.497966 - 0.0245195 i,
0.429368 - 0.0211418 i, 0.374028 - 0.0184169 i, 0.328735 - 0.0161867 i,
0.291198 - 0.0143384 i, 0.259741 - 0.0127895 i, 0.23312 - 0.0114787 i, 0.210391 - 0.0103595 i,
0.19083 - 0.00939636 i, 0.173876 - 0.00856156 i, 0.159085 - 0.00783326 i,
0.146105 - 0.00719409 i, 0.13465 - 0.00663007 i, 0.124491 - 0.00612987 i,
0.115441 - 0.00568422 i, 0.107342 - 0.00528545 i, 0.100067 - 0.00492722 i,
0.0935069 - 0.00460422 i, 0.0875715 - 0.00431196 i, 0.0821838 - 0.00404667 i,
0.0772784 - 0.00380514 i, 0.0727995 - 0.0035846 i, 0.068699 - 0.00338269 i,
0.0649354 - 0.00319737 i, 0.0614728 - 0.00302688 i, 0.0582799 - 0.00286966 i,
0.0553295 - 0.00272439 i, 0.0525976 - 0.00258987 i, 0.0500632 - 0.00246508 i,
0.0477076 - 0.00234909 i, 0.0455145 - 0.0022411 i, 0.0434691 - 0.00214039 i,
0.0415586 - 0.00204632 i, 0.0397714 - 0.00195831 i, 0.038097 - 0.00187587 i,
0.0365261 - 0.00179852 i, 0.0350505 - 0.00172586 i, 0.0336625 - 0.00165752 i}
```

Out[*]=

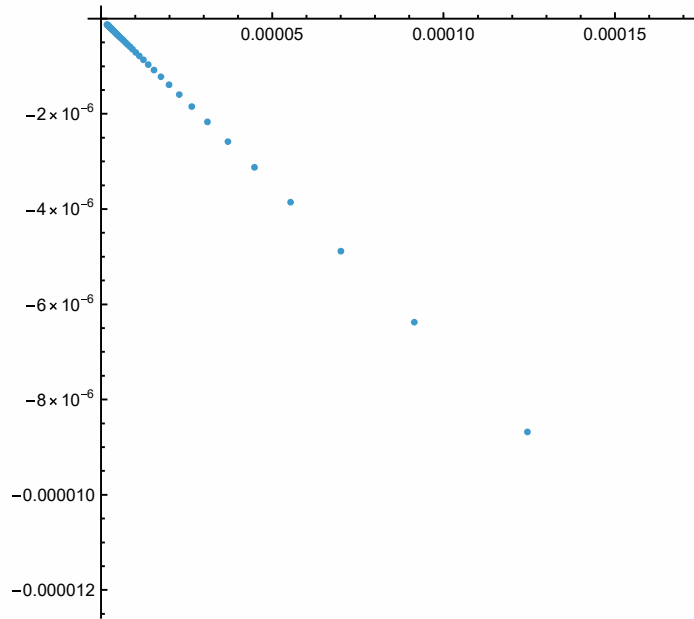
10. s J / (kg mHz)

Out[*]=

```
{8.71174 × 1071 - 4.2896 × 1070 i, 1.50856 - 0.0742805 i, 0.37714 - 0.0185701 i,
0.167618 - 0.00825339 i, 0.0942851 - 0.00464253 i, 0.0603425 - 0.00297122 i,
0.0419045 - 0.00206335 i, 0.030787 - 0.00151593 i, 0.0235713 - 0.00116063 i,
0.0186242 - 0.000917044 i, 0.0150856 - 0.000742805 i, 0.0124675 - 0.000613889 i,
0.0104761 - 0.000515837 i, 0.0089264 - 0.00043953 i, 0.00769674 - 0.000378982 i,
0.00670472 - 0.000330136 i, 0.00589282 - 0.000290158 i, 0.00521994 - 0.000257026 i,
0.00465605 - 0.000229261 i, 0.00417884 - 0.000205763 i, 0.0037714 - 0.000185701 i,
0.00342078 - 0.000168437 i, 0.00311686 - 0.000153472 i, 0.00285172 - 0.000140417 i,
0.00261903 - 0.000128959 i, 0.0024137 - 0.000118849 i, 0.0022316 - 0.000109882 i,
0.00206936 - 0.000101894 i, 0.00192419 - 0.0000947456 i, 0.00179377 - 0.0000883241 i,
0.00167618 - 0.0000825339 i, 0.00156978 - 0.000077295 i, 0.0014732 - 0.0000725396 i,
0.00138527 - 0.0000682099 i, 0.00130498 - 0.0000642565 i, 0.00123148 - 0.0000606372 i,
0.00116401 - 0.0000573152 i, 0.00110194 - 0.000054259 i, 0.00104471 - 0.0000514408 i,
0.000991822 - 0.0000488366 i, 0.000942851 - 0.0000464253 i, 0.000897419 - 0.0000441883 i,
0.000855194 - 0.0000421092 i, 0.00081588 - 0.0000401734 i, 0.000779216 - 0.000038368 i,
0.000744969 - 0.0000366817 i, 0.000712931 - 0.0000351042 i, 0.000682916 - 0.0000336263 i,
0.000654758 - 0.0000322398 i, 0.000628306 - 0.0000309373 i, 0.000603425 - 0.0000297122 i}
```

```
In[*]:= ListPlot[ (Tooltip[{Re[#1], Im[#1]}] &) /@%928, AspectRatio -> 1]
```

```
Out[*]=
```



```
In[*]:= Clear[n]
```

$$\left(\frac{m_P l_P}{M \alpha} n^2 \right) \frac{\Gamma}{m_P r} \sqrt{\frac{\text{length } M \alpha}{m_P l_P}} 1.0$$

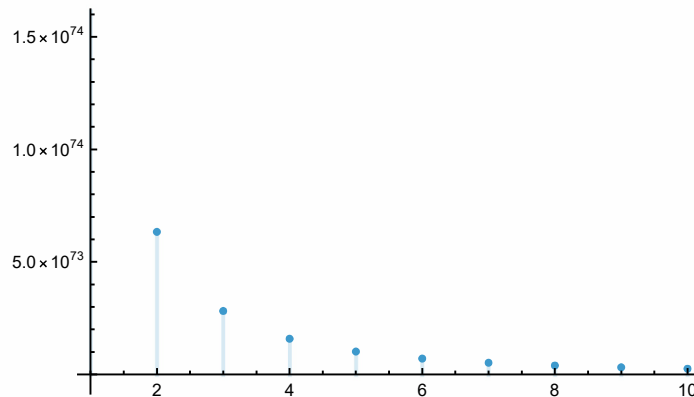
$$\text{DiscretePlot} \left[G \frac{M}{\left(\frac{m_P l_P}{M \alpha} n^2 \right) c^2} + G r \frac{M}{\left(\frac{m_P l_P}{M \alpha} n^2 \right) c^2}, \{n, 1, 10\} \right]$$

```
Clear[n]
```

```
Out[*]=
```

$$n^2 \left(6.3182 \times 10^{-74} \text{ s J / (kg mHz)} \right)$$

```
Out[*]=
```



$$\text{In}[*]:= G \frac{M}{d c^2} + G r \frac{M}{d c^2}$$

$$r = \text{UnitConvert} \left[\frac{1P}{\Delta \alpha \left[\sqrt{\frac{\text{length } M \alpha}{mP 1P}} \right]}, \text{"Meters"} \right] 1.0$$

$$r / \left(\frac{mP 1P}{M \alpha} n^2 \right) 1.0$$

$$\left(\frac{mP 1P}{M \alpha} n^2 \right) / r 1.0$$

$$\sqrt{\left(\frac{mP 1P}{M \alpha} n^2 \right) - r}$$

$$G \frac{M}{r c^2} + G r \frac{M}{r c^2}$$

$$G \frac{M}{\left(\frac{mP 1P}{M \alpha} n^2 \right) c^2} + G r \frac{M}{\left(\frac{mP 1P}{M \alpha} n^2 \right) c^2}$$

Out[*]=

$$206.727 - 14.4129 i$$

Out[*]=

$$24.5647 \text{ m}$$

Out[*]=

$$2.45647$$

Out[*]=

$$0.407088$$

Out[*]=

$$(0. + 3.81637 i) \sqrt{m}$$

Out[*]=

$$84.1562 - 5.86733 i$$

Out[*]=

$$206.727 - 14.4129 i$$

$$\text{In}[*]:= \sqrt[4]{\left(1 - \frac{G M}{c^2 d}\right)}$$

$$\sqrt[4]{\left(1 - \frac{G M}{c^2 \left(\frac{mP \text{ lP}}{M \alpha} n^2\right)}\right)}$$

$$k == \sqrt[4]{\left(1 - \frac{G M}{c^2 \left(\frac{mP \text{ lP}}{M \alpha} \left(\sqrt{\frac{\text{length } M \alpha}{mP \text{ lP}}}\right)^2\right)}\right)}$$

$$\frac{mP \text{ lP}}{M \alpha} n^2$$

Out[*]=

$$1.48919 + 1.48919 i$$

Out[*]=

$$1.48919 + 1.48919 i$$

Out[*]=

True

Out[*]=

$$100. \text{ s J / (kg m Hz)}$$

```

In[*]:= Clear[m, M, n, d];
SunMass = Quantity[1988400 * 1024, "Kilograms"];
M = 1.4 SunMass;
length = Quantity[1 * 10, "Meters"]
length = Quantity[1 * 102, "Meters"]
Rs =  $\frac{2 G M}{c^2}$ ;
n =  $\sqrt{\frac{\text{length } M \alpha}{mP \text{ lP}}}$ 
d = UnitConvert[ $\frac{mP \text{ lP}}{M \alpha} n^2$ , "Meters"]

k = SetPrecision[ $\sqrt[4]{\left(1 - \frac{G M}{c^2 d}\right)}$ , 32]

Gr = SetPrecision[ $\frac{G}{k^2}$ , 32]

g =  $\frac{Gr M}{d^2}$ 

ScientificForm[Rs]
ScientificForm[n]
    
```

```

ScientificForm[d]
ScientificForm[k]
ScientificForm[Gr]
ScientificForm[g]
ScientificForm[M]

```

$$\text{Gr} \frac{m_P}{l_P} == \frac{G}{k^2} \frac{m_P}{l_P}$$

$$G \frac{m_P}{l_P}$$

$$\text{Gr} \frac{M}{d}$$

$$G \frac{M}{d}$$

$$\text{Gr} \frac{M}{d c^2}$$

$$G \frac{M}{d c^2}$$

$$\sqrt{\frac{G}{\text{Gr}}} == k$$

$$G \frac{M}{d c^2} + \text{Gr} \frac{M}{d c^2}$$

Out[]=

10 m

Out[]=

100 m

Out[]=

2.40309×10^{36}

Out[]=

100. m

Out[]=

$1.4891935511070053532733936663135 + 1.4891935511070053532733936663135 \, i$

Out[]=

$(0. \times 10^{-43} - 1.5047872702442726375301440984877 \times 10^{-11} \, i) \, \text{m}^3 / (\text{kg s}^2)$

Out[]=

$(0. - 4.18897 \times 10^{15} \, i) \, \text{m/s}^2$

Out[]//ScientificForm=

$4.13455 \times 10^3 \, \text{m}$

Out[]//ScientificForm=

2.40309×10^{36}

Out[]//ScientificForm=

$1. \times 10^2 \, \text{m}$

```
Out[ ]//ScientificForm=
1.4891935511070053532733936663135 + 1.4891935511070053532733936663135 i
```

```
Out[ ]//ScientificForm=
(0. × 10-42 - (1.5047872702442726375301440984877 × 10-11) i) m3 / (kg s2)
```

```
Out[ ]//ScientificForm=
(0. - (4.18897 × 1015) i) m/s2
```

```
Out[ ]//ScientificForm=
2.78376 × 1030 kg
```

```
Out[ ]=
True
```

```
Out[ ]=
89 875 517 873 681 764 m2/s2
```

```
Out[ ]=
(0. - 4.18897 × 1017 i) m2/s2
```

```
Out[ ]=
1.85797 × 1018 m2/s2
```

```
Out[ ]=
0. - 4.66085 i
```

```
Out[ ]=
20.6727
```

```
Out[ ]=
True
```

```
Out[ ]=
20.6727 - 4.66085 i
```

```
In[ ]:=  convert to exponential 206.727 - 14.4129 i
```

```
Out[ ]=
207.229 ei (-0.0696068)
```

```
In[ ]:= Abs[207.229 ei (-0.0696068)]
```

```
Out[ ]=
207.229
```

```
In[ ]:=  convert to exponential 2.06727 - 2.00106 i
```

```
Out[ ]=
2.87713 ei (-0.769124)
```

```
In[ ]:= Abs[2.87713 ei (-0.769124)]
```

`In[ ]:=`
$$\frac{\text{mP } 1\text{P } n^2}{\alpha \text{ m}}$$
$$\sqrt[4]{1 - \frac{4 \text{ G m}}{c^2 \left(\frac{\text{mP } 1\text{P } n^2}{\alpha \text{ m}} \right)}}$$
$$\left(1 - \frac{4 \text{ G m}^2 \alpha}{c^2 1\text{P mP } n^2} \right)^{1/4}$$

`Out[ ]=`

$$\frac{6\,774\,777\,596\,359\,697\,247\,579\,907\,171\,385\,582\,163\,155\,329\,482\,752}{24\,945\,585\,106\,892\,044\,818\,102\,365\,167\,609\,405\,517\,578\,125 \pi^2} \text{ s J / (m Hz)}$$

m

`Out[ ]=`

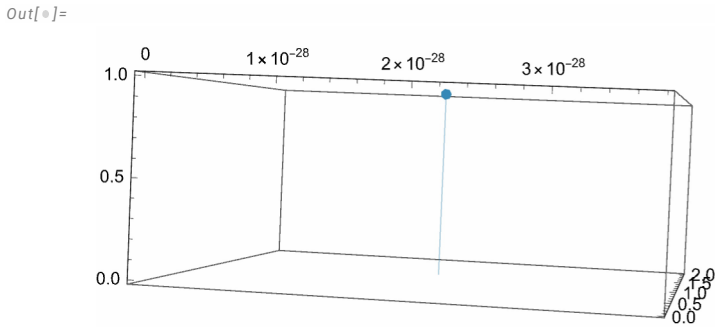
$$\left(1 + m^2 \left(- \frac{1\,114\,136\,270\,197\,730\,015\,625 \pi^{11/4}}{169\,965\,620\,337\,472\,084\,547\,284\,711\,115\,721\,485\,309\,240\,906\,063\,282\,176 \sqrt{2}} \text{ m}^2 \text{Hz / (kg s J)} \right) \right)^{1/4}$$

`Out[ ]=`

$$\left(1 + m^2 \left(- \frac{1\,114\,136\,270\,197\,730\,015\,625 \pi^{11/4}}{169\,965\,620\,337\,472\,084\,547\,284\,711\,115\,721\,485\,309\,240\,906\,063\,282\,176 \sqrt{2}} \text{ m}^2 \text{Hz / (kg s J)} \right) \right)^{1/4}$$

`In[ ]:=` `DiscretePlot3D`
$$\left[\left(1 - \text{QuantityMagnitude} \left[\frac{4 \text{ G m}^2 \alpha}{c^2 1\text{P mP } n^2} \right] \right)^{1/4}, \right.$$

$$\left. \{m, \text{QuantityMagnitude}[m\mu], \text{QuantityMagnitude}[2m\mu]\}, \{n, 1, 1\} \right]$$

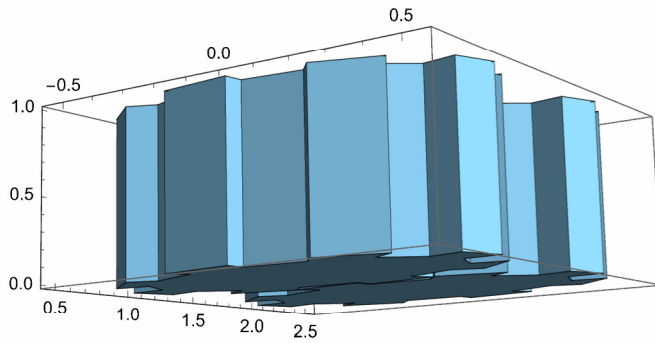


```

In[ ]:= DiscretePlot3D[ $\left(1 - \text{QuantityMagnitude}\left[\frac{4 G m^2 \alpha}{c^2 l_P m_P n^2}\right]\right)^{1/4}$ ,
  {m, QuantityMagnitude[me], QuantityMagnitude[2 me]}, {n, 1, 2},
  ExtentSize → Full, ExtentElementFunction → "SquareWaveCube"]

```

Out[]=

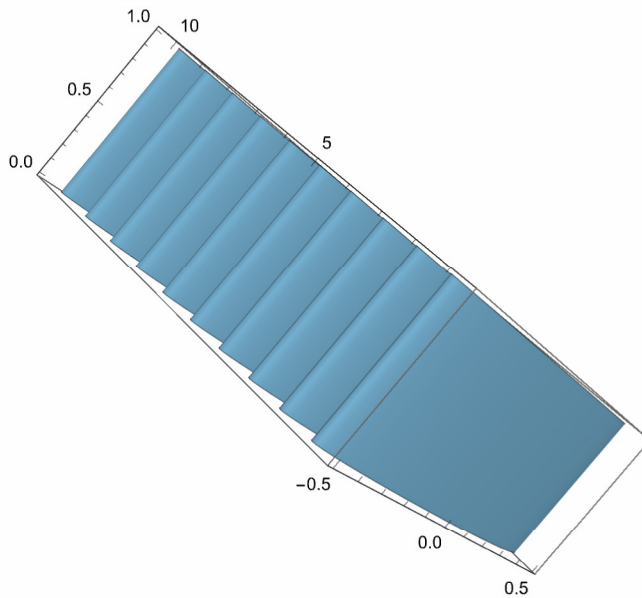


```

In[ ]:= DiscretePlot3D[ $\left(1 - \text{QuantityMagnitude}\left[\frac{4 G m^2 \alpha}{c^2 l_P m_P n^2}\right]\right)^{1/4}$ ,
  {m, QuantityMagnitude[me], QuantityMagnitude[me 1836]}],
  {n, 1, 10}, ExtentSize → Full, ExtentElementFunction → "Cylinder"]

```

Out[]=

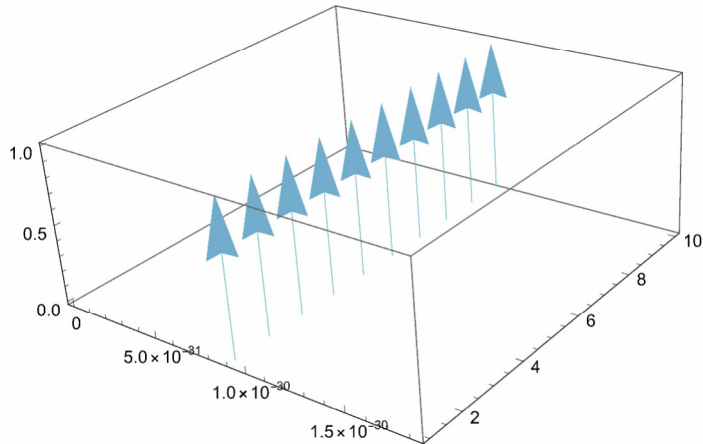


```

In[ ]:= DiscretePlot3D[ $\left(1 - \text{QuantityMagnitude}\left[\frac{4 G m^2 \alpha}{c^2 l_P m_P n^2}\right]\right)^{1/4}$ ,
  {m, QuantityMagnitude[me], QuantityMagnitude[me 1836]},
  {n, 1, 10}, ExtentSize → Full, ExtentElementFunction → "Arrow3D"]

```

Out[]=

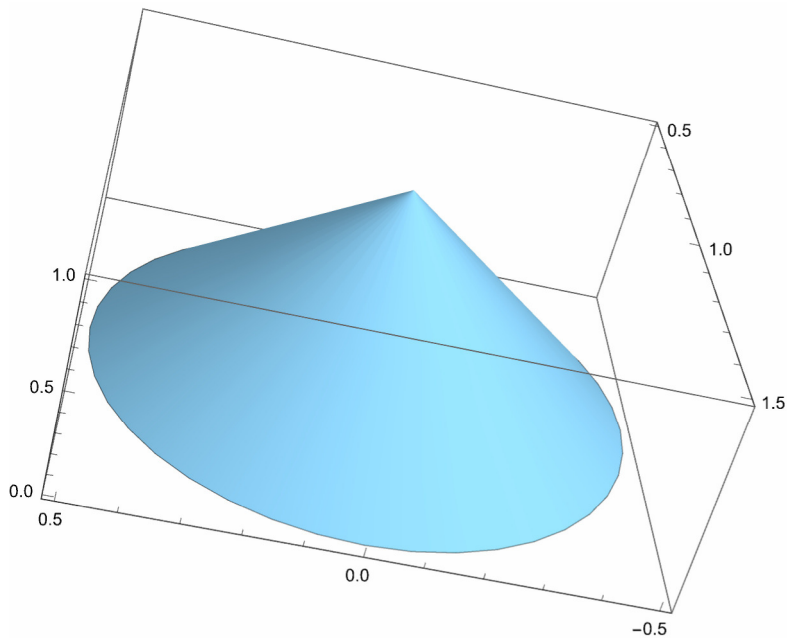


```

In[ ]:= DiscretePlot3D[ $\left(1 - \text{QuantityMagnitude}\left[\frac{4 G m^2 \alpha}{c^2 l_P m_P n^2}\right]\right)^{1/4}$ ,
  {m, QuantityMagnitude[me], QuantityMagnitude[100 me]},
  {n, 1, 1}, ExtentSize → Full, ExtentElementFunction → "Cone"]

```

Out[]=



```
In[ ]:= m = mP
```

```
n = 10
```

$$\left(1 - \frac{4 G m^2 \alpha}{c^2 l_P m_P n^2}\right)^{1/4} 1.0$$

```
Out[ ]=
```

$$\frac{7 \sqrt{405397113703863}}{2000000000000000 \times 2^{1/4} \pi^{7/8}} \sqrt{\text{kg}} \sqrt{\text{s}} \sqrt{\text{J}} / (\text{m} \sqrt{\text{Hz}})$$

```
Out[ ]=
```

```
10
```

```
Out[ ]=
```

```
0.999927
```